Behavioural Effects of High Doses of Psilocybin

in Female Rats

by

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Abstract

Depression and anxiety are two of the most predominant mental disorders and leading causes of disability. The World Health Organization estimated that 3.8% of the world's population suffers from depression, and between 0.4% and 3.6% have an anxiety spectrum diagnosis. Psychedelics, particularly psilocybin, have been extensively studied as new and promising treatments, showing long-lasting positive results when combined with psychotherapy for patients with depression and anxiety, among other disorders. However, pre-clinical trials investigating the precise neurobiological mechanisms underlying these results are still scarce, especially with female subjects. Additionally, emerging evidence suggests that hormonal fluctuations may influence the drug's efficacy, potentially providing a protective effect.

This study utilised adult female rats to examine the behavioural effects of high doses of psilocybin on locomotor activity and on depression and anxiety-like measures. Symptomatology was induced using either early maternal separation or a genetic reduction in the serotonin transporter. The open field (OFT) and affective disorders test (ADT) were employed to assess the effects of doses of 4, 8, or 16 mg/kg of psilocybin. Additionally, the effects of the phase of the oestrous cycle when animals were injected were evaluated. Results showed a significant effect of psilocybin on locomotion activity in the OFT 3 hours and 24 hours post-injection but not after 8 days. No robust effects on depression and anxiety-like measures were observed at any of the doses tested. However, trends were observed in both depression and anxiety-like measures with doses of 4 and 8mg/kg. Moreover, a preliminary evaluation in small groups revealed an interaction between treatment and the phase of the oestrous cycle, suggesting that female hormones might have some protective effect against the effects of the drug in depression-like measures.

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Abbreviations List

5D-ASC - Five Dimensions of Altered States of Consciousness

5HT - 5-hydroxytryptamine (Serotonin)

5HT2A - Serotonin Receptor (2A)

ACTH - Adrenocorticotropic Hormone

ADT - Affective Disorders Test

ANOVA: Analysis of Variance

APA - American Psychiatric Association

BDI - Bech Depression Inventory

BDN - Brain Default Network

CBT - Cognitive Behavioural Therapy

DM - Diestrus/Metoestrus

DSM - V- Diagnostic and Statistical Manual of Mental Disorders – Fifth edition

EMS - Early Maternal Separation

EPM - Elevated Plus Maze

FC - Functional Connectivity

FMRI - Functional Magnetic Resonance Imaging

FSL - Flinders Sensitive Line

FST - Forced Swim Test

GAD - Generalized Anxiety Disorder

GBD - Global Burden Disease Study

HAM - D -Hamilton Depression Rating Scale

HPA - Hypothalamic Pituitary Adrenal

HPG - Hypothalamic Pituitary Gonadal

I.P. - Intraperitoneal

LSD - Lysergic Acid Diethylamide

MDD - Major Depressive Disorder

MDMA - 3,4-Methylenedioxymethamphetamine

mSAT - Modified Successive Alleys Test

OBN - Oceanic Boundlessness

OFT - Open Field Test

OTT - One Trial Tolerance

PE - Proestrus/Oestrous

PND - Post Natal Day

POMS - Profile of Mood States

PPI - Prepulse Inhibition

QIDS - Quick Inventory of Depressive Symptoms

SAT - Successive alleys Test

SEM: Standard Error of the Mean

SERT KO - Serotonin Transporter knockout

SHRP - Stress Hyporesponsive Period

SSRI - Selective Serotonin Reuptake Inhibitor

USV - Ultrasound Vocalization

WHO - World Health Organization

WT – Wild Type

Chapter 1 – General Introduction

> "Life is fraught with sorrows: no matter what we do, we will in the end die; we are, each of us, held in the solitude of an autonomous body; time passes, and what has been will never be again. Pain is the first experience of world helplessness, and it never leaves us. We are angry about being ripped from the comfortable womb, and as soon as that anger fades, distress comes to take its place. Even those people whose faith promises them that this will all be different in the next world cannot help experiencing anguish in this one; Christ himself was the man of sorrows."
>
> (Solomon A. The Noonday Demon, 2001. Chap 1)

Emotional pain, helplessness, sorrow, and distress have been part of the human experience since we began documenting our thoughts and emotions. While such feelings are present in the regular human experience, they differ significantly from pathological conditions. Although on a continuum with common emotional experiences, pathologies are distinguished by their severity and impact on daily functioning. Specifically, they manifest as persistent, intense disturbances in emotional, cognitive, or behavioural patterns, deviating from what is typically considered normal within a given cultural or societal context.

Among these dysfunctional patterns, mental disorders, anxiety and depression are the most prevalent. Numerous treatments have been developed to combat these disorders, with psychedelics such as psilocybin emerging as a promising option. Clinical trials with psilocybin have shown positive and long-lasting effects in alleviating symptoms of certain mental disorders, including anxiety and depression, suggesting its potential as a novel treatment option.

Despite these promising results, many underlying biological mechanisms remain unclear. Controlled laboratory environments are crucial for elucidating these mechanisms and advancing our understanding of psilocybin's therapeutic potential.

The primary objective of this thesis is to expand the body of knowledge on the neurobiology of psilocybin through preclinical research. The following chapters will present three preclinical behavioural studies conducted on female rats using relatively high doses of psilocybin. However, before exploring these findings, this chapter will discuss anxiety and depression and examine their profound effects on society and individuals. Additionally, it will provide an overview of current research on psychedelics, specifically focusing on psilocybin, and highlight areas that require further investigation in preclinical research related to mental disorders. This foundation will set the stage for a deeper understanding of psilocybin's mechanisms and its potential implications, emphasising the necessity for ongoing research in this promising field.

1.1- Background on Anxiety and Depression

1.1.1- Anxiety

Etymologically, the term anxiety originated from the Latin word anxieties (nominative anxiety), “anguish, solicitude,” a noun or quality from anxious “uneasy, troubled in mind”. In its definition from the XVI century, anxiety was defined as: "apprehension caused by danger, misfortune, or error, uneasiness of mind respecting some uncertainty, a restless dread of some evil". (*Online Etymology Dictionary*, "Anxiety," n.d.)Feelings of anxiety can be understood as adaptive behaviours that evolved throughout our evolutionary history, playing a crucial role in our vigilance and survival. On the other hand, anxiety disorders are groups of non-adaptive behaviours that jeopardise overall well-being and general health. (Campos et. al., 2013; Hofmann et al., 2002). What differentiates a natural and adaptive behaviour from a dysfunctional pathology is the degree to which this feeling affects daily life.

At the heart of the thoughts that pervade states of anxiety lies the underpinning emotion of fear. This foundational element is a crucial constituent within this category of mental health conditions and can be explained as an emotional reaction to actual or perceived imminent threats. In contrast, anxiety, in a more intricate manner, is typified by the anticipation of potential future threats.

Since the early 1900s, the term Anxiety has been employed to define various psychological conditions, and they have become a prominent part of psychological and psychiatric manuals. Currently, the term “anxiety disorders" encompasses a range of specific conditions, sharing features of excessive fear, anxiety and related behavioural disturbances, leading to physical symptoms such as sleep disturbances, muscle tension, restlessness, shortness of breath, and increased heart rate (Association, 2013).

Among the different conditions classified as anxiety disorders, it is possible to include separation anxiety disorder, selective mutism, specific phobias, social anxiety disorder, panic disorder, agoraphobia, substance/medication-induced anxiety, and generalised anxiety disorder (Association, 2013).

For the scope of this work, we will focus on generalised anxiety disorder (GAD), as diagnosed in the fifth edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM–V), which is characterised as excessive preoccupation and worry about some events or activities with difficulty to control the worry. These symptoms are accompanied by at least three of the following physical or cognitive symptoms of: restlessness or feeling keyed up or on edge, feeling fatigued, irritability, difficulty concentrating or mind going blank and muscle tension (Association, 2013).

1.1.2- Depression

Depression, meaning "dejection, state of sadness, a sinking of the spirits," is derived directly from the Medieval Latin depressionem (nominative depressio), a noun of action from the past participle stem of Latin deprimere, to press down, depress" (*Online Etymology Dictionary*, "Depression," n.d.)

Similar to anxiety, the term 'depression' has been used for a long time to define a group of mental disorders that share similarities but also present specific characteristics. Conditions such as disruptive mood dysregulation disorder, major depressive disorder (including major depressive episodes), persistent depressive disorder, premenstrual dysphoric disorder, substance/medication-induced depressive disorder, depressive disorder due to another medical condition, other specified depressive disorders, and unspecified depressive disorder are listed in the most recent editions of the psychological and psychiatric manuals. While these disorders share cognitive and physical symptoms, each presents unique characteristics used in the differential diagnosis (Association, 2013b).

For the scope of this work, the focus will be on Major Depressive Disorder (MDD), which is the most common form of depressive disorder. This condition is characterised by at least 1 of 2 core symptoms: Depressed mood (sadness, hopelessness) almost all day and/or anhedonia (loss of interest or pleasure in activities) most of the day, nearly every day, together with additional symptoms (must total at least five including core symptom (s): Sleep disturbance (Hypo- or hypersomnia), weight loss or weight gain, psychomotor agitation or retardation, loss of energy, feelings of worthlessness or guilt inability to focus, indecisiveness, thoughts of death or suicide (Association, 2013b).

While scientific studies provide less substantiation for the connection between fear and depression compared to anxiety, the emotional component of fear appears to be linked to both conditions. Research has shown, for example, a direct connection between fear and depression in the context of the COVID-19 pandemic, indicating that fear plays a central role in the development of depressive conditions (Rodríguez-Hidalgo, 2020; Pak et al., 2021).

Beyond the fear component, anxiety and depression disorders share several similarities. Understanding these similarities is crucial for comprehending both groups of disorders and developing potential new treatments. Before exploring this topic, it is essential to understand the impacts of these disorders on society.

1.1.3- Mental Disorders – Prevalence and Economic Impact

According to a 2022 report by the World Health Organization (WHO, 2022), in 2019, one in every eight people around the world was living with some form of mental disorder, which corresponds to 970 million people or 13% of the global population. Specifically addressing anxiety and depression disorders, the numbers are notably high, representing 31% (301 million people) and 28.9% (280 million) of this total, respectively.

Additionally, forecasts that consider the impact of the COVID-19 pandemic anticipate a significant increase in mental health issues: anxiety disorders are expected to rise by 28% and depressive disorders by 26% worldwide. (WHO, 2022).

The impact of mental disorders extends beyond patients and their families, encompassing various social and economic perspectives. The global economic burden of mental health issues has been estimated to exceed US$2.5 trillion in 2010 and is projected to be around six trillion by 2030 (WHO, 2022). This staggering impact justifies the necessity for investments from both the public and private sectors. Even though the current levels of investment are increasing, they can be considered low and inefficient.

The WHO Mental Report of 2022 (WHO, 2022) attests that over 86% of the 196 country members have policies or plans addressing mental health concerns. However, only 31% of them reported plans that were being implemented, and the disparity across income groups is considerable, varying from 32% for upper-middleincome countries to just 3% for low-income countries. Regarding the budget distribution, the report points out that just 2% of the health budget is destined for treating and preventing mental health disorders. This value can increase to 10% in high-income countries but decreases to only 1% for low-income countries.

This scenario perpetuates existing social inequalities, leaving vulnerable individuals even more marginalised and without the support they desperately need.

These figures underscore the imperative need for the development of innovative treatment alternatives that can offer support to individuals suffering from mental health issues while also addressing the financial constraints faced by governments with diminishing budgets in this critical domain.

1.1.4- Connections between Anxiety and Depression

While each of these pathologies is individually well-defined in psychiatric manuals, there is a growing understanding of their interconnectedness. Factors such as comorbidity rates, sex differences in terms of prevalence, pathogenesis and options of effective treatments approximate these two disorders originally understood as wholly separated.

Generally, comorbidity rates between MDD and Anxiety Disorders can be as high as 60% (Fava et al., 2000). Amongst youth, for example, anxiety rates of patients previously diagnosed with depression range from 15% to 75% (Cummings et al., 2014). On the other hand, in anxiety patients, it is estimated that depression co-occurs in 10% to 15% of the cases. (Cummings et al. 2014). This scenario again makes evident the intricate connection between these groups of disorders.

There are several key points to be observed regarding anxiety disorders in youth and their relationship with depression. Firstly, these youth may exhibit depressive symptoms that fall below the threshold for diagnosis, as traditional diagnostic interviews may not fully capture them. Secondly, existing studies often group together diverse anxiety disorders, such as panic disorder and social phobia, making it challenging to discern the specific connections between individual anxiety disorders and depression. Thirdly, anxiety tends to manifest at an earlier age than depression and is generally more prevalent in childhood, while depression becomes more common in adolescence. Therefore, the coexistence of these conditions may vary depending on the individual's age. Lastly, many studies do not distinguish between concurrent comorbidity (i.e., disorders co-occurring) and sequential comorbidity (i.e., one disorder preceding the other), contributing to potential misinterpretations of their relationship.

A second similarity between these two conditions is the consensus that anxiety and depression disorders are approximately twice as prevalent in women compared to men (Altemus et al., 2014). Multiple factors can explain these differences, including hormonal changes during puberty, pregnancy and postpartum, societal expectations, and biological differences between genders. It is crucial to address these gender-specific risk factors and their impact on mental health to ensure appropriate support and intervention for individuals experiencing anxiety and depression.

In terms of pathogenesis, the theory of Stress-Related Disorders reinforces the connectivity between depression and anxiety. It postulates that external factors, such as life events, trauma, or chronic stress, can trigger and exacerbate both these mental health conditions. Moreover, the genetic evidence supporting this theory underscores the role of individual susceptibility to stress-related disorders, with certain genetic variations increasing vulnerability to both conditions when combined with adverse environments or stressors. (Smoller, 2016).

The convergence of depression and anxiety under the umbrella of stress-related disorders challenges the traditional view of categorising mental health conditions into isolated groups. Instead, it highlights the need for a more comprehensive and integrated approach to understanding and treating these disorders. By embracing the connection between depression and anxiety disorders through the lens of the Stress-Related Disorders theory, we can advance our understanding of the intricate interplay between genetic predisposition and environmental factors in the development of these mental health conditions. Additionally, recognising their shared pathogenesis encourages mental health professionals to address common underlying mechanisms with greater precision when designing therapeutic interventions, resulting in a more effective and targeted treatment approach.

Lastly, treatment is another piece of evidence connecting these two groups of disorders. The approaches with better outcomes for patients with both disorders are similar: a combination of psychotherapy and pharmacotherapy. Convincing evidence points to the efficacy of psychotherapies, including cognitive behavioural therapy (CBT) for patients with both disorders (Higa-McMillan et al., 2016; Zakhour et al., 2020). Additionally, the front-line pharmacological treatments for both anxiety and depression disorders are selective serotonin reuptake inhibitors (SSRIs) (Ravindran & Stein, 2010; Cordner, Z.A et al., 2021).

Considering the interconnectedness of these two groups of disorders, this broader perspective may lead to identifying novel therapeutic targets and more personalised treatments, ultimately improving the outcomes and well-being of individuals affected by these debilitating conditions. However, further research and interdisciplinary collaboration are essential to fully comprehend the complexities of these conditions and develop more effective interventions. This need is particularly urgent given that many current treatment options were introduced several decades ago, prior to significant advances in neuroscience.

1.2- Treatments

1.2.1- Current Pharmacological Treatments

As already mentioned, psychotherapy and pharmacotherapy are the frontline treatments and have comparable efficiency for both disorders, depression (Cuijpers et al., 2013; Spielmans et al., 2011) and anxiety (Cuijpers et al., 2013; Mitte, 2005). However, pharmacotherapy is the predominant path chosen by the majority of patients (Ahrnsbrak & Stagnitti, 2020; Hockenberry et al., 2019).

In their meta-analysis, Cuijpers and colleagues (2011) found that diverse kinds of psychotherapy are effective for the treatment of depression, including cognitive behaviour therapy, interpersonal psychotherapy, problem-solving therapy, non-directive supportive therapy, and behavioural activation therapy, which have slight differences in terms of efficacy. However, according to this study, the best approach in terms of results seems to be combining psychotherapy and pharmacotherapy (Cuijpers et al., 2011).

Pharmacological treatments for anxiety and depression disorders made significant progress in the second half of the last century. Several classes of drugs were developed, targeting, for example, monoamine receptors or transporters, such as norepinephrine, dopamine, and serotonin, with particular emphasis on serotonin (Garakani A., 2020; Tamminga et al., 2002). The specific mechanism of action and the expected results vary depending on the specific drug.

In general, the expected results from treatments are related to the reduction of symptoms. In patients with depression, the aim is to improve mood, increase motivation, restore overall functioning and improve sleeping. For patients with anxiety, the main goal is to help patients feel calmer, more relaxed, and better able to manage their anxiety thoughts and worries (Popovic et al., 2015).

However, the results are not always the remission of symptoms. It is important to note that the onset of effects for some drugs may require time to be seen, and, in some instances, this period can extend up to six weeks. Additionally, typical side effects such as dry mouth, gastrointestinal symptoms, insomnia, and sexual dysfunction, among others, can significantly impact daily life. These side effects and the delayed onset of therapeutic effects can complicate finding the appropriate medication type and adjusted dosage for each patient. Consequently, multiple adjustments in the doses and even in the drug or class of drug may be necessary before a patient achieves a favourable balance between the expected therapeutic results and manageable side effects. All these factors (e.g., dose adjustment, side effects and latency to onset of effects) can be cited as reasons for a significant proportion of patients failing to achieve the benefits of prescribed drugs and, consequently, remission of symptoms.

According to Halaris (2021), it is estimated that around one-third of patients with depression do not experience even partial response, and remission is attained for less than 30% of patients, even with adjusted doses.

In essence, despite decades of ongoing research and the development of new treatments for anxiety and depression, the current pharmacological scenario remains unsatisfactory for many patients. There is a clear need for new and more effective drug options, particularly for individuals resistant to existing treatments. Notably, psychedelics have recently gained considerable attention as a promising alternative. However, more dedicated studies and research efforts are essential to fully explore and understand their potential as a viable therapeutic option for those facing resistance to current medications.

1.2.2- Psychedelics

Psychedelics are a class of psychoactive substances that produce changes in perception, mood, and cognitive processes. Throughout millennia, the naturally occurring forms of these compounds have been employed predominantly in spiritual and mystical contexts. More recently in history, together with their synthetic variations, these substances have ignited a growing interest in the scientific community driven by their potential for treating various mental conditions.

The meaning of the word psychedelic, coming from its Greek origin, means "to reveal the psychē" (Carhart-Harris & Goodwin, 2017). It sounds audacious, but the reports from patients about the effects of this class of substances support the idea that psychedelics are a window to the mind's secret world.

To understand what psychedelics are, it is essential to say that there are multiple approaches to defining them. Some focus on chemical composition: a compound with appreciable serotonin 2A receptor agonist properties that can alter consciousness in a marked and unique way (Carhart-Harris et al., 2017). Other definitions emphasise the resulting effects: "a drug which, without causing physical addiction, craving major physiological disturbances, delirium, disorientation, or amnesia, more or less reliably produces thought, mood and perceptual changes otherwise rarely experienced except in dreams, contemplative and religious exaltation, flashes of vivid involuntary memory, and acute psychosis." (Grispoon and Balakar, 1997). Despite the differences, both approaches highlight the potential of psychedelics to experience the conscience distinctly.

Research in the 1960s established the classification of "classical psychedelics" as those having serotonergic effects, including psilocybin and LSD, and "dissociative anaesthetics" having an interaction with the glutamatergic system, such as ketamine and phencyclidine (Wojtas et al., 2022). This classification paved the way for further studies into the mechanisms of action of these substances in the brain.

Interestingly, serotonergic pathways play crucial roles in the regulation of anxiety and depression disorders. (Gordon & Hen, 2007). More recently, however, discoveries suggest that both classes of psychedelics (i.e., classical and dissociative anaesthetics) act simultaneously on multiple neurotransmitter systems. (Wojtas et al, 2022).

It would be inaccurate to say that psychedelics are a recent discovery or even that the studies of their effects on mental health represent a novel research avenue. Some studies about peyote cactus, for instance, claim that native North Americans explored its psychotropic properties more than 5000 years ago (El-Seedi et al., 2005). Throughout a significant part of history, humanity has been endeavouring to comprehend the effects of hallucinogenic compounds found in plants or fungi and their potential applications for diverse purposes, including religious, recreational, and medical contexts.

The scientific interest in these substances also has a long history. Some particularly significant points are the work of Arthur Heffter in the 1800s, who isolated and identified the effects of Mescaline (Bogenschutz & Johnson, 2016) and the work of Albert Hoffmann and his team, who in 1943 discovered the psychoactive effects of the Lysergic acid diethylamide (LSD). Hoffman's name is also connected with the beginning of the studies with psilocybin (Bogenschutz & Johnson, 2016), which will be discussed in the next section. Thanks to their efforts, there has been substantial progress in the research into these substances, positioning this field (i.e., the use of psychedelics for mental disorders) as a new frontier with promising potential in psychopharmacology.

The initial studies conducted on psychedelics in the last century and the resulting discussions not only attracted the attention of researchers, especially in the USA, but also captured the interest of individuals seeking recreational experiences with these substances. This increased widespread interest eventually drew the attention of the US government and public opinion. This resulted in the passing of the Controlled Substance Act in 1971, which made the possession and use of psychedelic substances illegal. (Geiger, 2018). The consequence of this legislation was a complete cessation of research on psychedelics within the United States and worldwide.

Presently, psilocybin and other psychedelics continue to be prohibited globally. Nonetheless, there has been a notable resurgence in research in this field since the late 1990s, following a pause of nearly three decades. The scientific interest in psychedelics has grown in parallel with the advancement of new laboratory methods and techniques. The potential of these substances as viable treatments for mental disorders has captured the global research community's focus, given that they can offer a response to a longstanding hiatus in terms of innovative pharmacological options available for mental disorders since the release of SSRIs onto the market in the late eighties. The current resurgence of enthusiasm regarding psychedelics' potential for treating mental disorders marks a significant development, with psilocybin as one of the most studied substances.

1.3- Psilocybin

1.3.1- History

The word psilocybin originates from the Greek words *psilos*, meaning "bare" or "naked," and *kybē*, meaning "head" (Online Etymology Dictionary, "Psilocybin," n.d.). The substance was first studied in 1958 by Albert Hoffman, a renowned pioneer of psychedelic science.. He started studying the compounds found in the Psilocybe mexicana mushroom (Figure 1.1), which is native to regions of Mexico and South/Central America. He aimed to isolate the mushroom's active compound through experiments conducted on laboratory rats. However, it was not until he fearlessly decided to subject himself to a self-experimentation that Hoffman and his team successfully identified two active compounds, psilocybin and psilocin, along with their corresponding molecular structures. (Hasselgrave, 2020)

Figure 1.1- Psilocybe mexicana - Rockefeller, A. (2019) Source: iNaturalist

Figure 1.2– Psilocybe weraroa - Townsend, D. (2017) Source: *iNaturalist*

Figure 1.3- C - *Psilocibe aucklandiae* - Elle (2023) Source: *iNaturalist*

Note: Figures 1.2 and 1.3 species endemic to New Zealand

Due to the similarity between the molecules of psilocybin, psilocin, and LSD, together with previous studies associating the effects of psychedelics with positive changes in self-perception, the interest in the effects of these drugs on mental disorders rose exponentially in the following decades. A considerable number of papers from the 1950s and 1960s can be found on the use of psilocybin in psychiatry, although their methods should be considered cautiously.

Fasting forward to recent times, the last decades have witnessed a rise in the number of studies exploring the effects of psilocybin (and other psychedelics) on psychiatric disorders such as eating disorders, obsessive-compulsive disorder, depression, anxiety, and substance abuse. This surge is considerable in pre-clinical research but more notable in clinical trials (Nichols, 2020). As a result, a cohesive body of knowledge has emerged, encompassing various facets of psilocybin, including its molecular structure and therapeutic effects, and this time, differently from last century's studies, with more advanced and rigorous methods.

1.3.2- Molecular Structure of Psilocybin

To understand this substance's effects on the brain, it is first essential to understand its molecular structure and mechanism of action after consumption.

Psilocybin is a secondary metabolite produced by over 140 mushroom species that grow in diverse parts of the world (Guzmán, 2005). Its chemical composition consists of the molecular structure psilocybin 4-phosphoryloxy-N, N-dimethyltryptamine (Tylš et al., 2016). After consumption, psilocybin undergoes a swift enzymatic dephosphorylation process, resulting in the formation of psilocin. In essence, psilocybin serves as a prodrug, acting as a precursor for the active compound psilocin, which mediates the psychoactive effects associated with psilocybin-containing mushrooms. (Nichols, 2020). Psilocin (4-hydroxy-N, N-dimethyltryptamine) is a serotonin (5-HT) 2A receptor agonist.

Extensive research has established a widespread consensus that the agonist activity at the 5-HT2A receptor plays a pivotal role in mediating the effects of psilocin and other psychedelic compounds (Nichols, 2020). In fact, the molecular structure of psilocin shows a strong resemblance to serotonin, as demonstrated in Figure 1.4.

Figure 1.4 - Molecular structures of serotonin, *p*silocybin and *p*silocin - Adapted from Hesselgrave (2020)

1.3.3- Psilocybin Clinical Trials

As previously stated, the comparison between psychedelic research in general (including psilocybin) conducted in the 1960s and contemporary investigations highlights a significant evolution in the methods and techniques employed to investigate the effects of this substance in the brain. For example, the advancement and refinement of psychological assessment batteries produced more precise tools encompassing a broader spectrum of psychological disorders. Concurrently, neuroimaging techniques have played a pivotal role in enhancing the comprehension of the neurological pathways affected by these substances and their effects. Additionally, laboratory techniques have shed light on the effects of this drug on the body and behaviour in a more precise way. All these innovations, applicable to clinical and preclinical trials, have culminated in a more robust body of knowledge, elevating the methodological rigour and overall quality of research.

Clinical trials throughout the time consistently demonstrated positive results of the use of psilocybin for conditions such as anxiety (Grob, 2011; Muttoni et al., 2019), depression (Carhart-Harris & Goodwin, 2017; Vargas et al., 2020), post-traumatic stress disorder (Khan et al. et al. 2022), substance abuse (Johnson et al., 2014), and obsessive-compulsive disorder (Moreno, 2016; Bouso et al., 2021).

The vast majority of trials have been done in patients with depression and anxiety, using doses ranging from 0.2 mg to 30 mg (Bouso et al., 2021) that are administered once or twice within a psychotherapeutic context where psychological support is offered before, during and after the psychedelic session(s). Standard psychological batteries such as the Quick Inventory of Depressive Symptoms (QIDS), Hamilton Depression Rating Scale (HAM-D), Beck Depression Inventory (BDI) for depression, and Spielberger State-Trait Anxiety Inventory (STAI) for anxiety (Santos & Gama, 2021) are typically used to measure the effects of the treatment offering data about the psychological states of participants before and after the sessions and typically demonstrating the rapid onset and long-lasting effects of treatment.

A particularly interesting study on depression was conducted by Carhart-Harris and colleagues (2017). Twelve patients with moderate to severe major depressive disorder (scoring 17+ on the 21-item HAM-D) with no improvement from treatments with two different classes of antidepressants were included. After a series of biological and psychological tests, the patients received two doses of psilocybin (10 mg and 25 mg) in two separate sessions one week apart. They also received psychiatric support during and after the sessions. Based on comparisons of HAM-D and BDI scores, the response rate to psilocybin was 67% (n=8) one week after treatment, with seven of these eight patients meeting the criteria for remission. Furthermore, 58% (n=7) of the patients maintained their response for three months, and 42% (n=5) remained in remission in the same period.

Effects of psilocybin on GAD were investigated, for instance, by Grob (2011) in a double-blind placebo-controlled cross-over trial with 12 subjects (1 male, 11 females) with advanced-stage cancer. Oral doses of psilocybin (0.2 mg/kg) or niacin (250 mg as a control) in two separate dosing sessions, with subjects acting as their own control. Results from Beck BDI, Profile of Mood States (POMS), and State-Trait Anxiety STAI batteries were administered regularly for up to 6 months. The results demonstrated significant decreases observed in STAI scores at one month and three months follow-up and in BDI scores at six months follow-up. POMS identified mood improvement after treatment that approached but did not reach significance.

Another study conducted by Ross and colleagues (2016) found that 0.3 mg/kg of psilocybin mediated improvements in psychiatric and existential distress in patients with cancer. In an evaluation made 6.5 months after administration, Psilocybin was linked to lasting anxiolytic and antidepressant effects, with about 60–80% of participants showing clinically significant reductions in depression or anxiety. In a second evaluation conducted 4.5 years later, 60–80% of the 16 participants who were still alive continued to meet the criteria for clinically significant antidepressant or anxiolytic responses (Agin-Liebes et al., 2020).

From a psychological perspective, the use of psychedelics induces changes in personality traits such as openness to experience, neuroticism, mindfulness, and optimism. (Carhart-Harris et al., 2016). These effects, monitored and developed in a psychotherapeutic setting, are thought to be at the heart of the therapeutic effects of these drugs on disorders such as anxiety and depression.

Additionally, from the patient's point of view, the reports after drug administration provide valuable insights into the cognitive processes occurring during the treatment. In the therapeutic setting, following the ingestion of the drug, alterations in perceptions of self, deep consciousness, fostering feelings of forgiveness, self-compassion, and love, increasing mindfulness, and achieving better control over choices and behaviour are typically reported. In other words, psilocybin-assisted therapy seems to induce a wide range of experiences that are remarkably diverse yet consistently tailored to the unique necessities of each individual (Bogenschutz et al., 2018). This disruption in the sense of self is commonly referred to as "ego dissolution" and is frequently reported as an outcome of psilocybin treatment. It can be understood as a reduction in self-referential awareness present in ordinary waking consciousness (Nour & Carhart-Harris, 2017) and is deeply associated with long-lasting positive results in alleviating anxiety and depression-like symptoms.

The most common effects attributed to the ingestion of psilocybin include changes in perception, such as hallucinations or synaesthesia and alterations in cognition, mood, and sense of self. Together these effects can be understood as altered state of conscience that were described, in 5 dimensions by Dittrich (1998): the oceanic boundlessness (OBN) - a pleasurable state of self-dissolution which includes items that measure changes in ego-boundaries, a sense of unity and transcendence of time and space, spiritual experiences, intuitive understanding, and shifts in emotions ranging from elevated mood to bliss and ecstasy; the anxious ego-dissolution (AED) - an anxious state of ego-dissolution characterized by a fragmented sense of self, detachment from the world, loss of autonomy, cognitive impairment, anxiety or panic, and motor disturbances; the visionary restructuralization (VIR) - consists of items assessing various perceptual phenomena such as hallucinations; the acoustic alterations (AA) - involving distortions, illusions, and hallucinations in terms of auditory information and the vigilance reduction (VR) which depicts dreaminess, reduced alertness, and drowsiness illusions, synaesthesia, altered meaning of precepts, vivid memories, and enhanced imagination.

These five groups of symptoms can be measured by the (Abnormal Mental States) APZ-OAV questionnaire and its renewed version of the Five Dimensions of Altered States of Consciousness (5D-ASC) (Dittrich, 1998; Preller & Vollenweider, 2018).

As already mentioned, the duration of the effects achieved with psilocybin-assisted therapy can last from a few weeks (Santos et al., 2021) to up to 12 months (Griffiths, et al., 2016; Gukasyan N. et al., 2022) for both anxiety and depression. A randomised controlled trial with sixteen participants with cancer-related psychiatric and existential distress even found significant antidepressant or anxiolytic responses 4.5 years after a single dose of psilocybin in 60–80% of the patients of the study (Agin-Liebes et al., 2020).

Reports of side effects are not frequent but include temporary anxiety and distress, sensorial illusions, physical symptoms such as headache, nausea and vomiting, psychotic-like behaviour and a mild increase in blood pressure and heart rate (Bouso, 2021).

However, the psychedelic experience can sometimes be seen as an adverse effect of this therapy, often referred to as a 'bad trip, this overwhelming distress during the drug action can develop into more serious conditions that, although transient, should be taken into consideration, for instance, prolonged psychosis triggered by hallucinogens. The overall discomfort experienced during a psychedelic experience can vary greatly depending on individual factors, the specific hallucinogen used, dosage and set and setting conditions. However, they can be significantly minimised if all the safeguards and guidelines are followed (Johnson et al., 2008). In this context, set refers to the mindset, expectations, and psychological preparedness of the individual undergoing the psychedelic experience. Setting refers to the physical and social environment in which the experience occurs. Both are considered crucial factors in determining the quality and outcome of a psychedelic experience.

The role of the psychedelic experience in clinical outcomes is important and continues to fuel scholarly debate. Studies such as Roseman et al. (2018) emphasise its significance, showing that OBN scores during psilocybin sessions strongly correlate with subsequent reductions in depressive symptoms. This finding suggests that these experiences might be pivotal in therapeutic effects. Conversely, Hesselgrave (2020), for example, advocates for a cautious approach, suggesting the use of the 5-HT2A antagonist Ketanserin in trials with psilocybin to mitigate the psychedelic experience, aiming to reduce the risks associated with altered states of consciousness while still achieving therapeutic benefits in mental disorders. A prevailing consensus in this discussion is the imperative for additional research to gain a comprehensive understanding of the role played by mystical experiences in the therapeutic outcomes associated with psilocybin.

Regarding brain activity, functional neuroimaging tests in patients have demonstrated significant brain connectivity alterations after psilocybin administration. Girn and Cols (2023) demonstrated that this substance induces a global functional integration, evidenced by the diminished functional segregation of large-scale brain networks and amplified global functional connectivity. Additionally, there is an increase in the diversity of functional connectivity (FC) states. By promoting new connectivity patterns, breaking down rigid thought patterns, and encouraging exploration of different states of consciousness, the treatment allows individuals to confront and process deep-seated emotions and unresolved issues. Additionally, the reduction in self-referential thinking can help them break free from rumination and negative thought patterns at the core of mental disorders.

In the same direction as these findings, there is a growing interest in investigating the effect of psilocybin in the Default Mode Network (DMN). The DMN comprises a network of interconnected brain regions, including the medial prefrontal cortex, the posterior cingulate cortex, and the lateral inferior parietal cortex. These regions exhibit higher activity levels during internally oriented activities such as predictions, judgments, and contemplations. These properties support the attribution of this region as the biological system underlying the psychological concept of ego. This region, which works like a convergence zone, receives more blood flow and consumes more energy than any other region in the brain. (Carhart-Harris et al 2014). Psilocybin markedly decreases FC within the DMN, which is correlated with positive states of ego dissolution, such as Oceanic Self-Boundlessness measured by the 5D-ASC. (Gattuso et. al., 2023)

In conclusion, the results from human trials demonstrate the drug's general effectiveness and safety for treating mental disorders. However, these clinical trials involved small and carefully selected groups of patients, necessitating further research with larger and more diverse populations. Additionally, several questions regarding dosage, timing of administration and sex differences remain unresolved. Animal studies in the laboratory setting are better suited to address these issues comprehensively.

1.3.4- Mechanism of Action

A multifaceted approach is essential to comprehend the mechanisms underlying the effects of psilocybin. This approach must encompass neuroendocrine, neuroanatomic, neurofunctional and psychological perspectives.

Psilocybin acts as a modulator of serotonergic neurotransmission, especially in the limbic system and related areas in the prefrontal cortex (Grim et al., ., 2018). According to Nutt (2020), "Activation of 5-HT2A receptors appears to increase the excitability of the host neuron, causing a spike to wave decoherence and associated dysregulation of spontaneous activity in cortical populations. This dysregulation reliably manifests as increased entropy in ongoing activity recorded via local field or scalp potentials, as well as related changes in the power spectrum, e.g., marked decreases in alpha oscillations".

However, merely binding to the 5-HT2A receptor is not enough to explain the psychedelic effects. There are non-psychedelic 5-HT2A receptors and 5-HT2A agonists, such as lisuride and ergotamine, which do not induce hallucinogenic effects (López-Giménez, 2018). The critical factor appears to be the specific signalling pathway activated by the ligand binding to the 5-HT2A receptor.

Psychedelic and non-psychedelic 5-HT2A agonists differentiate in their interaction with intracellular signalling pathways through a process known as biased agonism or functional selectivity. This allows different ligands to activate specific signalling cascades selectively. Research indicates that psychedelic effects are predominantly mediated by coupling the 5-HT2A receptor to Gq proteins, which increases intracellular calcium and activates phospholipase C (Wallach et al., 2023). In contrast, non-psychedelic agonists tend to preferentially activate β-arrestin-mediated pathways, leading to receptor internalisation and the initiation of alternative signalling cascades. Moreover, a certain threshold of Gq protein activation is necessary to produce psychedelic-like effects; below this level, even if the 5-HT2A receptor is activated, hallucinogenic effects are not observed. The duration and localisation of signalling also vary between psychedelic and non-psychedelic agonists, further contributing to their distinct pharmacological impacts. (Wallach, et al., 2023).

Additionally, findings by Vollenweider (1998) suggest that the interaction with 5-HT1A receptors may play a role in modulating dopamine release in the basal ganglia, influencing the effects of psilocybin.

After administration of psilocybin, anatomic alterations in the brain can be observed. In a study conducted by Shao and colleagues in 2021 using chronic two-photon microscopy to image dendritic spines, they demonstrated that this drug-induced synaptic rewiring in the layer five pyramidal neurons of the medial prefrontal cortex in mice, resulting in a remarkable 10% increase in both spine size and density.

According to Heck and Santos (2023), spine remodelling aligns with learning processes and the stability of spines over time supports the retention of lifelong memories. This suggests that increases in spine density and size, such as those induced by psilocybin, could enhance synaptic efficiency and plasticity, thereby facilitating learning and memory reprocessing. Furthermore, learning involves transient increases in spine density and the formation of new spines in clusters, which are essential for rewiring neural circuits and enhancing cognitive processes (Ma & Zuo, 2022). Therefore, the increase in spine density observed in psilocybin-treated mice could reflect a similar enhancement in neural connectivity and plasticity. This offers a mechanistic basis for the cognitive and emotional benefits observed in human studies following psilocybin administration.

Although direct measurement of dendritic spines in humans is not currently feasible, neuroimaging techniques such as functional MRI (fMRI) have provided indirect evidence of psilocybin's impact on brain connectivity and function. While direct evidence of increased dendritic spines in humans' post-psilocybin administration remains unavailable, the compelling results from animal studies, combined with the observed therapeutic benefits in human trials, suggest that psilocybin may similarly promote neuroplastic changes in the human brain. This hypothesis warrants further exploration in future research.

In terms of neurophysiology, there is evidence that psilocybin modulates the activity and connectivity within an extended visual–limbic–prefrontal neuronal network during the processing of a perceived threat. Additionally, within the limbic system, the amygdala has a central role in mediating the effects of this drug, Psilocybin reduces amygdala reactivity (Kraehenmann et al., 2015). Indeed, the amygdala-mediated processes of threat perception and fear response are central to the symptoms commonly observed in depression and anxiety disorders.

Collectively, all these perspectives offer a comprehensive, although not complete, understanding of the psilocybin experience and its impact on the human brain.

They also pave the way for discovering this drug's medical potential and point to avenues for further investigation.

Unlike the typical course of pharmacological research, where laboratories usually establish a solid and consistent body of knowledge with animal research before initiating clinical trials with humans, experiments involving psilocybin exhibit a more extensive and coherent collection of studies from clinical trials when contrasted with laboratory tests conducted on animals. This fact can be attributed to the compound's relative safety, substantiated by its extensive historical usage for recreational and ritualistic purposes, coupled with research predominantly carried out on humans during the 1950s and 1960s (Geiger et. al., 2018), with considerable safety for participants. The following sessions will present the main findings of preclinical trials with this drug.

1.3.5- Preclinical Trials

As stated above, pre-clinical trials using psilocybin are still incipient. There are a limited number of studies, mostly with conflicting results, and *p*redominantly carried out with male subjects, making it difficult to generalise the results and address the questions that remain in this field.

In this context, animal experiments involve understanding and comparing the results of various factors, including models, strains, and doses, to create a more comprehensive and realistic exploration of these complex biological processes. In terms of behavioural assessments, the most employed test for evaluating depression-like symptoms is the forced swim test (FST). On the other hand, the elevated plus maze (EPM), as well as assessments of locomotion and exploratory behaviours such as the open field test (OFT) (Jones et al., 2020) for measuring anxiety-like symptomatology. The doses used in the assays range from 0.01 mg/kg up to 20 mg/kg, and the strains of rats most used are Flinders Sensitive Line (FSL), Sprague Dawley, Wistar (with Serotonin transporter knockout (SERT KO) and Kyoto variations). FSL and both Wistar strains are well-established genetic animal models for depression and anxiety. These genetically modified strains exhibit heightened susceptibility to stress, along with anxiety-like behaviours (Hibicke et al., 2020).

As mentioned before, results from preclinical trials present conflicting results. For example, Hibicke et al. (2020) found a persistent antidepressant effect with 1 mg/kg in male Wistar Kyoto rats subjected to the FST. On the other hand, Jefsen (2019), using male animals of the FSL rat model, did not find significant effects on immobility in the FST using 0.5 and 2 mg/kg doses. Risca (2021), using male Sprague Dawley rats and administering chronic intermittent low and moderate doses of psilocybin (0.5 and 2 mg/kg), also did not find significant changes in immobility in the FST or significant reductions in anxiety-like behaviours in the OFT and Light Dark Box.

Jones et al. (2020) found anxiolytic responses in mice treated with an Intraperitoneal (IP) injection of 0.3 mg/kg psilocybin in the OFT in acute effects but inversely found anxiogenic responses when animals treated with the same dose of psilocybin had been previously chronically treated with corticosterone (80 μg/mL in drinking water for 21 days) as an animal model to induce stress. They suggested that chronic corticosterone exposure may interact with or counteract the anxiolytic properties of psilocybin.

The results for the EPM testing anxiety-like behaviours are equally inconclusive. Hibicke (2020) found persistent anxiolytic effects using 1 mg/kg psilocybin (more time in the open arms and significantly less time in the closed arms), and Horsley (2018) found only a slight reduction in the number of entries into the open arms. The test happened 48 hours after drug treatment with 0.05 or 0.075 mg/kg in male Wistar rats.

Studies indicate that the effects of psilocybin on locomotor activity are dose-dependent and may vary by sex. Most research suggests an impairment in locomotor activity following psilocybin administration at doses exceeding 1 mg/kg (Wojtas & Gołembiowska, 2024). This effect appears more pronounced in males than in females but may be influenced by the oestrous cycle phase, as noted by Tylš et al. (2016). However, evidence also shows no effects on locomotor activity (Jefsen et al., 2019). The presence of positive and negative effects may be influenced by various factors, including the type of behaviour being measured and the specific animal model used.

An important point to note is that most studies have used male subjects (Lovick, 2021). Consequently, the findings on sex differences and even on the effects of the drug on females remain virtually unknown due to the limited number of studies available. Interestingly, it is well-established in the literature that anxiety is more prevalent in women and that women are diagnosed with depression at twice the rate of men (Altemus et al., 2014).

This gap in psilocybin research raises questions about whether female hormones can interfere with treatment outcomes. Several lines of evidence suggest interactions between serotonin-like drugs and female hormones, highlighting the need for deeper investigation. Understanding the modulatory effect of hormones in psilocybin administration has significant therapeutic implications, as the magnitude of treatment effects may vary depending on the hormonal cycle phase. Administering psilocybin during a phase when hormones may have protective effects might reduce the drug's efficacy, considering that the treatment is usually administered with a single dose or two.

The oestrous cycle influences hormonal levels and stress responses due to the close interplay between the hypothalamic-pituitary-adrenal (HPA) axis and the hypothalamic-pituitary-gonadal (HPG) axis (Toufexis, 2014). The HPA axis regulates stress responses, while the HPG axis regulates the oestrous/menstrual cycle. Oestrogen levels fluctuate during cycle phases and influence stress responses and mood regulation. Additionally, oestrogen affects binding at serotonin receptor sites (Toufexis, 2014).

The oestrous cycle phase significantly impacts the results of tests based on unconditioned threatening stimuli, such as the EPM. Several studies measuring anxiety levels found that females in the proestrus phase exhibited reduced anxiety levels. However, some other studies found no difference when comparing cycle phases. (Lovick et al., 2021)

In psychedelic research, the limited studies exploring this relationship have found varying drug effects depending on the oestrous phase. For instance, using doses of 2.5, 5, and 10 mg/kg injected subcutaneously of the synthetic hallucinogen MDMA (3,4-Methylenedioxymethamphetamine), it was found that female rats in the dioestrus and metoestrus phases (MD) being more sensitive to the drug, presenting higher deficits in prepulse inhibition (PPI) than females in the proestrus and oestrus phases (PE) (Bubeníková, 2005). Similarly, Páleníček (2010) tested LSD (5, 50, and 200 µg/kg) and found mainly inhibitory locomotor effects in males and MD female rats. Both female groups were less sensitive to the hypo locomotor effects of the drug and exhibited less pronounced thigmotaxis than males. Furthermore, in the same study, PE females had increased rearing after a 5µg/kg dose of LSD. While LSD disrupted PPI in male and MD female rats, PE females were protected from this disruptive effect, suggesting lower sensitivity to LSD's behavioural effects for females in PE phases.

Moreover, Tylš et al. (2016) tested psilocybin doses of 0.25 mg/kg, 1 mg/kg, and 4 mg/kg in male and female rats. They found that female subjects were impacted to a lesser degree than males in terms of inhibition of locomotor activity and increased sniffing behaviour at the lowest dose, which was not observed in males. It was also found that female rats, especially during the proestrus and oestrus phases (PE), are protected from some effects of hallucinogens. They hypothesised that high levels of female hormones during these phases might protect females from the effects of these substances. One possible explanation is the increased expression of 5-HT2A receptors in females due to oestrogen effects (Summer & Fink, 1995). Additionally, during PE phases, females are protected from the effects of 5-HT1A-mediated changes and exhibit a higher activity in the dopaminergic system, which may counteract inhibitory effects on locomotion.

The inconsistent results in behavioural tests involving psilocybin underscore the need for further laboratory research to understand its effects on the brain. Several aspects are crucial in this pursuit. First, testing high doses is necessary to establish a more precise dose-response curve. Second, employing different behavioural assays with innovative approaches beyond the Elevated Plus EPM, FST and PPI will offer new perspectives on the effects of psilocybin. Additionally, designing studies with extended periods of tests would enhance understanding of the long-term effects of treatment. Third, including female subjects is essential to assess specific drug effects and the impact of the oestrous cycle, complementing existing data from male subjects. Addressing these aspects will provide a more comprehensive understanding of psilocybin's potential and limitations.

This thesis will, therefore, investigate the effects of high doses of psilocybin in female rats using a combination of behavioural and genetic techniques. The next chapter outlines the general methods used across all experiments. Chapter 3 will present data on the effects of 4 and 8 mg/kg doses on locomotor activity at three time points. Chapter 4 will assess the impact of 8 and 16 mg/kg doses on anxiety and depression-like symptoms, using an environmental challenge designed to induce these symptoms. Chapter 5 will explore the effects of 4 and 8 mg/kg doses of psilocybin, using a genetic model to induce the symptomatology. This chapter will also present the interaction of this treatment with the oestrous cycle in a dedicated subchapter. The final sections summarise the findings, discuss limitations, and propose future research directions. The main objective of this work is to offer answers to questions that are critical for our understanding of the effect of this drug and lay the groundwork for further studies.

Chapter 2- General Methods

The three experiments of this thesis test the effects of high doses of psilocybin in adult female rats. This section will present the general procedures used in all three experiments. Specific methods, objectives, and theoretical frameworks will be described in their corresponding chapters, with results and a discussion.

2.1- Ethics Statement

A total of 214 adult female rats were used in the experiments. They were born, housed, and tested at the Victoria University of Wellington laboratory in accordance with the Animal Welfare Act 1999 (Animal Welfare Act 1999, Part 1) and under the Animal Ethics Committee applications #29112, approved on 03/12/2020 (for Sprague Dawley rats used in experiments 1 and 3) and #29878, approved in 20/09/2021 (for Wistar rats used in experiment 2).

2.2- Animals

Litters were weaned at PND (Post Natal Day) 21. After reaching maturity, the females were housed in standard polycarbonate cages in groups of two, three or four animals and kept together until the end of the experiment. At least five days were allowed between this separation and the start of the experiments.

The rats were kept in a room with a 12-hour inverted light-dark cycle (lights off at 7:00 a.m.) to ensure testing occurred during the active period. The room has a controlled temperature (19-21°C) and humidity (55%). A technician changed cages, water bottles, and food containers once a week. The animals had access to food and water ad libitum, except when subject to food deprivation for experiments two and three.

All the manipulations (tests, injections, feeding, weighing, cleaning of cages) were done during the dark time of the cycle (7am—7pm), starting at least one hour after the lights were off and finishing one hour before the lights were on.

At the end of tests with each cohort, the animals were euthanised in their home cages via carbon dioxide ($CO2$) inhalation followed by cervical dislocation.

2.3- Drug

The psilocybin used in the experiments (BDG Synthesis, Wellington, New Zealand) was prepared the day before the injections and kept in the freezer (-18°C). For the preparation, the pure powder was dissolved in 0.9% sterile saline at a concentration of 4 mg/ml for the first and third experiments and 8 mg/ml for the second experiment. A sterile solution of 0.9% saline was used for the control treatment. Two hours before the injections, both solutions were thawed at room temperature and homogenised using a vortex agitator.

2.4- Treatments

Within each cohort, animals were randomly allocated to one of the treatment conditions: high or low dose of psilocybin or saline. For all experiments, a single dose of psilocybin or saline was administered intraperitoneally (i.p.).

For experiment one, injections happened between 9:00 a.m. and 1:00 *p*.m. and in two days for each cohort (conditions 1 and 2 on the first day and condition 3 on the second day). For experiments 2 and 3, injections were always administered between 10:00 a.m. and 2:00 p.m. on a single day for each cohort.

2.5- Handling

Before the beginning of tests, each animal received a 2-minute handling session for three consecutive days for experiments one and two days for experiments 2 and 3. The sessions happened in the housing room before feeding time (at least 30 minutes) to habituate the subjects to human contact. For the first experiment, handling started three days before the injection. The group was tested 8 days after the injection, and an additional handling session was performed one day before the test. For the second experiment, handling happened 2 days before the habituation sessions started. For the last experiment, the handling sessions started two days before the beginning of vaginal sample collections.

2.6- Food Restriction

For experiments two and three, using ADT, the animals were kept on a food restriction regimen that began one week before the experiment and lasted for the duration of the experiment.

Animals were maintained at 85-90% of their free-feeding weight (weight on the first day of experiment preparation) to establish food-directed motivation (Woods & Begg, 2015). To maintain this weight, 10-12g of regular chow was provided daily by the experimenter per animal. All animals were given the exact amount of food to prevent differential motivation levels, and the feeding time was always between 4 and 5:30 *p*.m. and at least 20 minutes after the last testing session to maintain temporal variability between feeding and testing time. Throughout the entire period of the experiments, the animals were weighed every two days at the end of the day before the feeding.

2.7- Sanitisation

Prior to the beginning of the day's testing routine and after each subject's test, every apparatus (anticipatory boxes, successive alleys, and open field chambers) was thoroughly sanitised with F10®SC veterinary disinfectant at 1:500, following the same procedure for each apparatus.

Chapter 3 – Effects of Psilocybin on Locomotor Activity

This experiment investigated the behavioural effects of a single dose of psilocybin (4 or 8 mg/kg) in three groups of adult female Sprague Dawley rats in an open-field locomotor chamber at three distinct time points: 3 hours, 24 hours, or 8 days after the injection.

The following sections will detail the supporting theory underlying the test chosen and the specific methodologies. Then, the main findings will be presented and, finally, the results will be discussed.

3.1- Theoretical Framework

3.1.1- Open Field Test

The open field test is a highly reliable and versatile behavioural animal test used in pre-clinical studies over the years. In its initial conception, the classic open field test (OFT) was developed to examine the emotional responses in rodents (Hall, 1934; Prut & Belzung, 2003). Presently, this classic test is utilised in diverse formats mainly to assess locomotor activity and exploratory behaviour, although it is also used to assess aspects of anxiety (Deacon, 2006).

It is hypothesised that evolutionary pressures have favoured a uniform response among animals, leading most species to exhibit anxiety-driven fear or flight reactions to certain stimuli. Rodents, for example, typically display strong aversions to large, brightly lit, open, and unfamiliar spaces. It is believed they have evolved to perceive these environments as threatening, as they make them vulnerable to predation. The test investigates an innate conflict between the exploration of a novel environment and the aversion to open and bright spaces. The unconditioned avoidance response can be quantified through the degree of exploration, locomotor activity, urination and defecation. (Seibenhener & Wooten, 2015).

The simplicity and versatility of the OFT allow the researcher to establish a dose-response curve before performing other tests. Moreover, installing this apparatus inside an enclosed chamber, along with various accessories (e.g. microphones and tracking systems), not only increases the amount of data that can be collected but also allows for more precise control over the experimental conditions. This, in turn, enhances the practical implications of the research, making it more relevant and impactful.

3.1.1.1- Ultrasonic Vocalizations

One additional measure that can be assessed in the OFT is ultrasonic vocalisations (USVs). Positioning a microphone above the chambers allows for recording vocalisations emitted during testing with minimal disruption to the experiment.

Rats emit USVs that communicate an animal's emotional state to other members of the social group. In assessing rodents' behaviour, the most frequently analysed aspects of USV data include the frequency and the number of vocalisations (calls) emitted. Research has shown that USVs above thirty kilohertz, often referred to as 'happy calls,' are linked to positive affective states in rats and appear to be triggered by predictive cues associated with rewarding stimuli, such as food (Ahrens et al., 2009). On the other hand, lower frequency calls or "sad calls", around 22 kilohertz, are related to states when the animals are stressed (Ahrens et al., 2009; Brudzynski et al., 2013), for example, when male animals face another male intruder.

Analysing the USVs emitted by animals during trials provides a non-invasive way to assess their emotional states, which can be categorised as positive or negative, in reaction to various environmental factors. (Ahrens et al., 2009; Brudzynski et al., 2013). Given psilocybin's complex effect on emotions (as discussed in Chapter 1), this offers a unique additional assessment.

3.1.2- Doses

Preclinical psychedelic research has been testing the effects of a very broad range of doses from 0.1 mg/kg up to 20 mg/kg (Jefsen et al., 2021). The decision to evaluate relatively high doses in this study draws upon compelling evidence from several studies. Firstly, research by Meinhardt et al. (2020) suggests that a minimum of 1 mg/kg of psilocybin is required to observe any behavioural effects in rats. Secondly, a study by Jefsen (2021) using doses ranging from 0.5 up to 20 mg/kg found that a single dose of psilocybin rapidly induces gene expression related to neuroplasticity (prefrontal cortex and hippocampus) in a dose-dependent manner. Lastly, comparative neurobiological analysis demonstrated that psilocin has a 15 times greater affinity for human 5-HT2 receptors compared to the same receptors in rodents. This finding suggests that to achieve similar neurochemical and behavioural outcomes in rodent models, a higher dosage of psilocybin may be required (Gallaher et al., 1993; Klein et al., 2018).

Additionally, as previously mentioned, most tests were conducted using male subjects. Consequently, it remains unclear whether the effects observed with specific doses in males will be replicated in female subjects.

3.1.3- Time of Tests

As discussed previously, clinical research consistently demonstrates some immediate effects of psilocybin shortly after administration and the positive therapeutic effects persist for months or even years after administration.

In the lab, the immediate post-administration effects in rodents include several behavioural and autonomic responses congruent with the serotonin syndrome, typically observed after administration of drugs that enhance 5-HT levels (Haberzettl et al., 2013). Among these responses are reduced locomotion and low body posture (Beer, 2021; Chafee, 2022), which makes immediate locomotion and exploration-based behavioural tests impractical.

Pharmacokinetic studies have found that the peak brain levels of psilocin occur between 25 to 90 minutes post-administration in mice (Horita & Weber, 1962) and at approximately 90 minutes following oral administration in rats (Musikaphongsakul et al., 2021).

This observation aligns with our laboratory's prior findings, indicating that animals generally return to regular locomotion within one hour following i.*p*. administration. Consequently, a three-hour interval between injection and testing appears suitable to circumvent the serotonin syndrome effects. Moreover, additional time points at 24 hours and 8 days were included to assess the drug's post-acute and long-term effects.

Previous studies suggest that the impact of Psilocybin on locomotion may be more pronounced in males than females (Tylš et. al, 2016), yet the evidence is limited, particularly for female subjects. Additionally, most of these studies assessed the behaviours within a short window following injection, typically between 10 and 30 minutes post-administration, despite peak brain activity occurring later, between 30 and 90 minutes.

This chapter aims to address these gaps by investigating the time-dependent effects of high doses of psilocybin in females on locomotion behaviour.

Therefore, we hypothesize that animals injected with psilocybin (4 and 8 mg/kg) will exhibit dose-dependent impaired locomotor activity and a reduced number of happy calls compared to saline-injected animals in all testing conditions (3h, 24h, and 8 days post-injection).

Prediction 1 - Compared to saline groups, animals injected with psilocybin will exhibit dose-dependent reduced levels of locomotion expressed by reduced central distance travelled (total and central), reduced counts of rears (total and central), reduced sad calls, increased happy calls, and increased thigmotaxis.

Prediction 2 - The differences observed in the behaviour of animals injected with psilocybin will be visible in all three test conditions (i.e. 3h, 24h, and 8 days post-injection).

3.2- Specific Methods

3.2.1- Animals

This experiment used 95 adult female Sprague Dawley rats born from 12 litters. They belonged to two cohorts born approximately 30 days apart with 55 and 40 animals, respectively. Within each cohort, animals were randomly assigned to one of three treatment groups: 4 mg/kg, 8 mg/kg, or saline. As demonstrated in Table 3.1, they were also assigned to 1 of the three testing conditions: 3 hours, 24 hours, and 8 days following a single i.p. injection with the designed treatment as illustrated in the table below. They were tested in a single 25-minute session in an open field arena to assess the effects of the treatment. Four animals were excluded from the data analysis due to inconsistent tracking during the test.

Table 3.1- Distribution of Animals in the Groups

Time of test /Treatment	Saline	4 mg/kg	8 mg/kg	Total
3 hours	11	11	10	32
24 hours	8	12	10	30
8 days	10	9	10	29
Total	29	32	30	91

3.2.2- Apparatus

The apparatus included four sound-insulated and ventilated locomotor activity chambers (MED Associates Inc.). An open field arena was placed within each chamber, measuring 45 cm x 45 cm x 40 cm, consisting of four transparent Plexiglas walls and a light grey PVC floor (Figure 3.1). Animals were tested in groups of 4, one in each chamber.

An ultrasound microphone (Avisoft Bioacoustics) was positioned directly above each arena and connected to a gate above the chambers. During the test, a transparent Plexiglas lid was placed over the arena to prevent jumping.

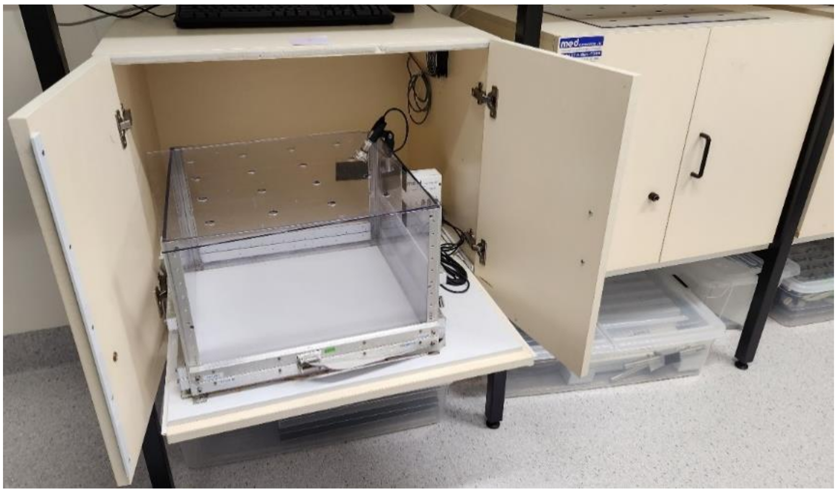

Figure 3.1- Open field arena inside the locomotor activity chamber.

Animal activities were tracked using the Activity Monitor Software (MED Associates, 2007), utilising the X and Y axes for positional tracking and the Z axis for rearing detection. To assess ambulation and rearing, two rows of infrared (I/R) sources and sensors were evenly spaced and juxtaposed around the walls on all four sides of the chamber. The sensors detect the subject's presence or absence at these coordinates.

An area in the centre of the arena, comprising 50% of the total (1012.5 cm^2), was defined on the map configuration (figure 3.2). This area was designated as a 'central area' of interest for later analysis. The remaining area, adjacent to the walls, was considered a 'residual area'. Together, both areas represented the 'total area' of the arena.

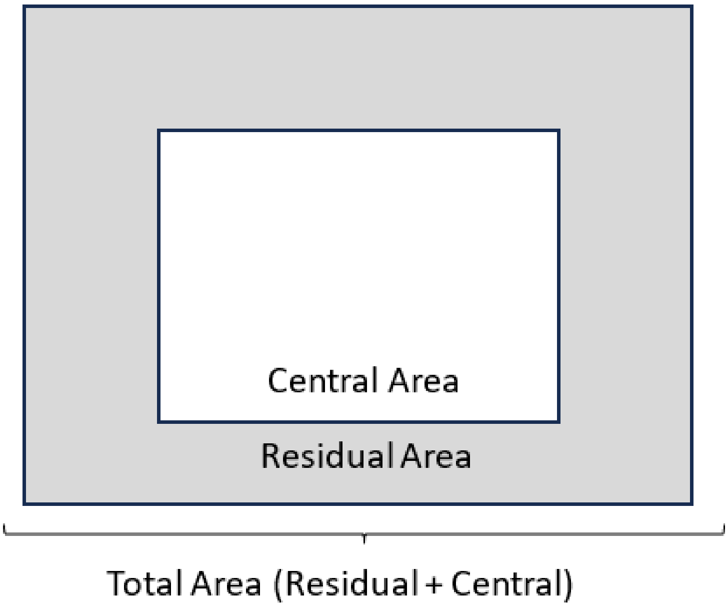

Figure 3.2- Arena's floor plan configuration.

3.2.3- Procedures

3.2.3.1 – Open Field Chambers

To habituate the animals to handling and to reduce stress, each rat underwent a 2-minute individual handling session for three consecutive days before the injection. These sessions occurred randomly between 9 a.m. and 4 *p*.m. and were evenly distributed among all groups to effectively manage the number of animals. Additionally, animals assigned to the 8-day testing condition received an extra 2-minute handling session the day before their test, within the exact 9 a.m. to 4 *p*.m. period. This extra session was designed to mitigate the possible effects of 8 days without experimenter contact.

The injections were administered 2 to 3 days after the cage-cleaning day to ensure consistent environmental conditions. All injections occurred between 9 a.m. and 1 p.m. and were performed by the same experimenter. The entire process, conducted in a testing room with dimmed lighting, lasted approximately 10 minutes per cage with 3 or 4 animals. During this time, animals were weighed, injected with the designated treatment, and returned to their home cage and dark housing room. Both cohorts follow precisely the sequence illustrated in Table 3.2 and described below.

Table 3.2 - Chronogram of Tests

Days/ Time of test	1	2	3	4	5	6	12	13
3 hours	Handling	Handling	Handling	**Injection** /Test				
24 hours		Handling	Handling	Handling	**Injection**	Test		
8 days		Handling	Handling	Handling		**Injection**	Handling	Test

Note. The numbers refer to the days when the specific activities, described below them, occurred.

Tests were conducted over three days for each cohort. At the beginning of each session, the home cage was moved from the housing room to a testing room illuminated with a red light. The animals were given 10 minutes to acclimate to the room in their home cages. After that each animal transferred to an individual testing chambers for a 25-minute trial. After the test, the animals were returned to the housing room in their cages. The open field were then cleaned with F10®SC veterinary disinfectant. This process was repeated for each subsequent cage according to a predefined schedule.

The measures analysed are described in the table below. Except for faecal boli count and USVs, all the remaining measures were acquired automatically using the activity monitor software.

Table 3.3 - Measures and Definitions

Measure	Definition
Central Area	
Central distance travelled	Centimetres travelled in the central area
Percentage of time in central area	The ratio of time (in seconds) spent in the central area/total time of the session
Central rears	Count of rears in the central area (unsupported)
Residual Area	
Thigmotaxis	The ratio of time spent in residual area/total time
Total Area	
Total distance Travelled	Total of centimetres travelled during the session.
Total rears	Total count of rears (supported and unsupported)
Faecal boli	Total count of faecal boli at the end of the session
USVs happy calls	The ratio of happy calls (above 25 kHz)/total calls
USVs sad calls	The ratio of sad calls (below 25 kHz)/total calls

Note: USVs = Ultrasonic Vocalizations. kHz = kilohertz.

3.2.3.2- Ultrasound Vocalizations

For USV recording, ultrasonic microphones were positioned directly above the open field arena to record USVs during the trial using Avisoft Recorder (version 4.3.5 Avisoft Bioacoustics Glienicke/Nordbahn Germany). The audio was later converted into a spectrogram using Avisoft SASLAB lite (Version 5.3.2.16 Glienicke/Nordbahn Germany) to classify the USVs based on frequency. Groups of USV calls with a variety of frequencies were excluded. (Figure 3.3).

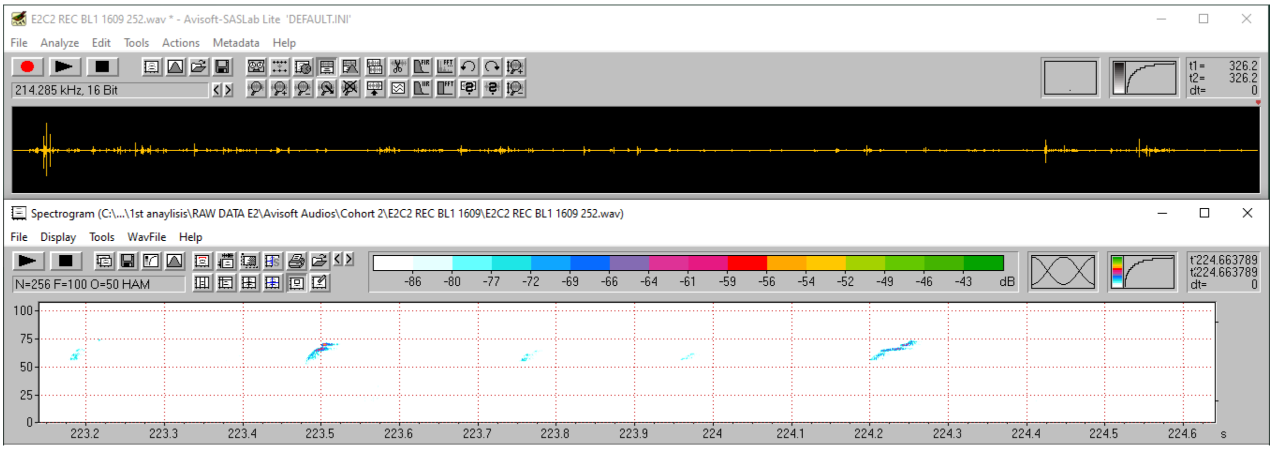

Figure 3.3 - Sonogram of calls recorded during a trial and the correspondent spectrogram presenting happy calls in a frequency between 50 and 75 kilohertz.

3.2.4- Statistical Analysis

The dataset evaluation for missingness showed a chi-square value of 0, with 30 degrees of freedom and a *p*-value of 1, suggesting that the data was missing completely at random (MCAR). This means that the pattern of missingness would not bias further analysis.

One-way ANOVA with the cohort as a grouping variable showed that five of the nine measures returned significant differences between cohorts: percentage of time in the central area, central rears, thigmotaxis, and total distance moved. However, the effect sizes (ω^2) associated with the cohort differences were small or medium, so the analysis proceeded with the two cohorts together to increase statistical power.

Normality assumption was analysed with Shapiro-Wilk tests for each measure and visual examination of both a Q-Q plot and histogram of residuals. Results returned violation of this assumption in five out of 9 measures. Homogeneity of variances, evaluated with Levene's test, was violated in one of the measures (thigmotaxis). However, the option to proceed with the analysis of variance was made, considering that the transformation of the measures could compromise the consistency of the data. This way, the analysis of variance (ANOVA) was the chosen test for evaluating the data set.

To evaluate the main effects and interactions between factors, nine two-way ANOVAs were performed, one for each measure, with treatment and time as fixed factors. Error bars in the bar charts always show ± standard errors of the mean (SEM). Holm Bonferroni correction was used in post-hoc analysis to address the heightened risk of Type I errors (i.e., incorrectly rejecting the null hypothesis) due to multiple pairwise comparisons.

3.3- Results

This experiment aimed to evaluate the effects of psilocybin on the locomotor activity of female rats using the open-field test. Adult female rats were injected with a single dose of psilocybin (4mg/kg or 8mg/kg) or saline and tested in a single session in the open field in one of the three different conditions: 3 hours, 24 hours, or 8 days after injection. Measures of ambulation, vertical behaviour, faecal boli count and USVs were acquired.

The hypothesis was that animals injected with psilocybin would exhibit impaired locomotor activity and a reduced number of happy calls compared to those injected with saline and that animals tested in the most extended interval after injection (8 days) would still exhibit the drug's effects.

The results will be presented in detail for each measure:

3.3.1- Central Distance Travelled

Anova tests showed that for distance travelled in the central area of the OFT, there were significant main effects of treatment $F(2, 82) = 7.81$, $p < 0.001$, and time $F(2, 82) = 8.57$, $p < 0.001$ which can be seen in table 3.4. Post hoc analysis of main effects demonstrated that animals treated with saline walked more than animals treated with 4 mg/kg of the drug ($p < 0.001$) and with 8 mg/kg ($p < 0.05$). In terms of time, treated animals tested 8 days after injection walked more than those tested 24h ($p < 0.001$) and those tested 3h after administration ($p < 0.001$), as seen in the graph.

Table 3.4- Summary of Analyses for Central Distance Travelled Split by Time and Treatment

	df	F	p
Time	(2, 82)	8.57	< 0.001
Treatment	(2, 82)	7.81	< 0.001
Time*treatment	(4, 82)	1.42	0.24

Additionally, post hoc evaluations demonstrated a significant treatment and time interaction. Animals treated with 4 mg/kg walked significantly less ($p < 0.05$), when tested 3 hours after injection compared to the saline group, as seen in Figure 3.4.

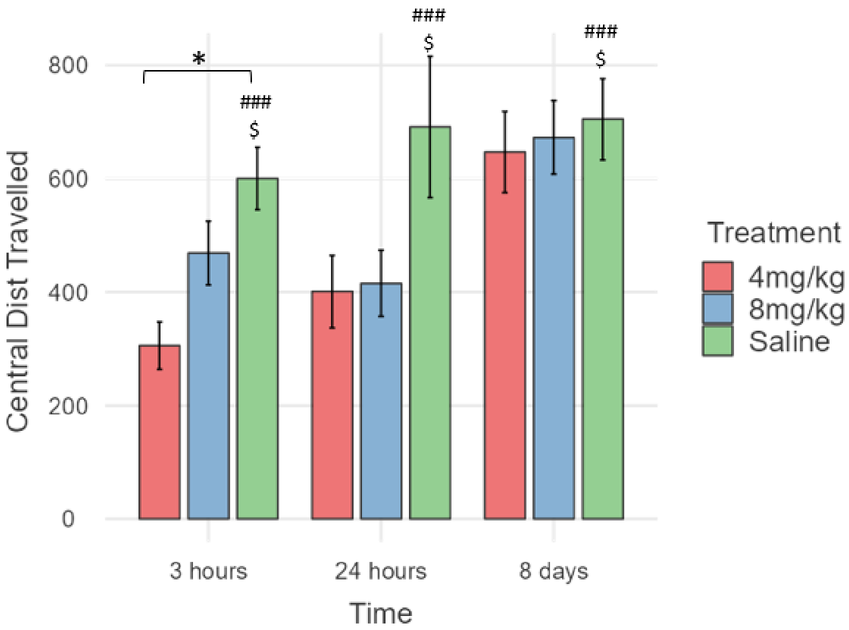

Figure 3.4- Means and ±SEM for central distance travelled split by time and treatment. A significant interaction (*) was observed in post hoc evaluations between the 4 mg/kg and saline groups at 3 hours. Main effects of treatment were noted when comparing saline to 4 mg/kg (#) and saline to 8 mg/kg ($).

3.3.2- Percentage of Time Spent in Central Area/Total Time

Time spent in the central area showed significant effects for treatment (table 3.5). Psilocybin-injected animals spent less time in the centre of the arena compared to saline ones.

Table 3.5- Summary of Analyses for Percentage of Time Spent in Central Area Split by Time and Treatment

	df	F	p
Time	(2, 82)	2.74	0.07
Treatment	(2, 82)	9.42	< 0.001
Time* treatment	(4 82)	1.17	0.33

Post hoc analysis revealed a significant difference in the effects observed in animals tested 3 hours after injection in both the 4 mg/kg group (p < 0.05) and the 8 mg/kg group (p < 0.05). Although not significant, figure 3.5 also demonstrates a clear difference when animals were tested 24h after injection, and an interesting reduction in 8 mg/kg treated animals tested 8 days after injection.

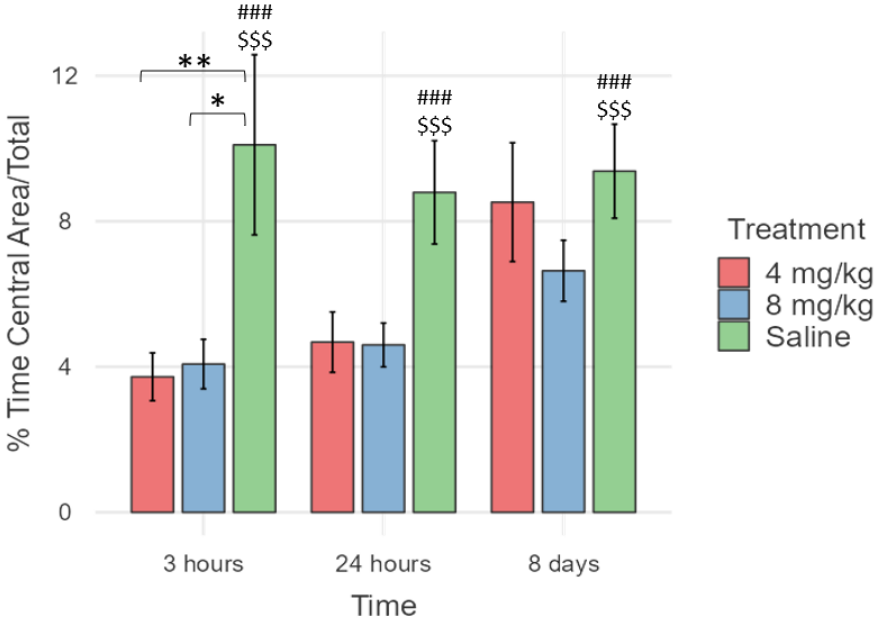

Figure 3.5 - Means and ±SEM for percentage of time spent in central area split by time and treatment. Significant interactions (*) were observed in post hoc evaluations between psilocybin doses and saline groups at 3 hours. Main effects of treatment were seen when comparing saline to 4 mg/kg (#) and saline to 8 mg/kg ($).

3.3.3- Central Rears

The analyses of the rears in the central area also demonstrated several significant effects comparing time, treatments, and the interaction between these two factors (Table 3.6). Saline bars are similar across conditions, but psilocybin animals reared differently depending on the time between injection and test. Animals tested 3 hours after injection reared less in the centre of the arena than in other time conditions.

Table 3.6- Summary of Analyses for Central Rears Split by Time and Treatment

	df	***F***	***p***
Time	(2, 82)	4.62	< 0.05
Treatment	(2, 82)	3.81	< 0.05
Time*treatment	(4, 82)	3.14	< 0.05

Post hoc evaluations demonstrated significant differences comparing animals tested 3h after injection. In this groups saline animals reared more than animals of the 4 mg/kg group ($p < 0.01$). Additionally, animals injected with 4 mg/kg of psilocybin reared more when tested eight days later than when tested 3 hours after the injection ($p < 0.05$), as demonstrated below.

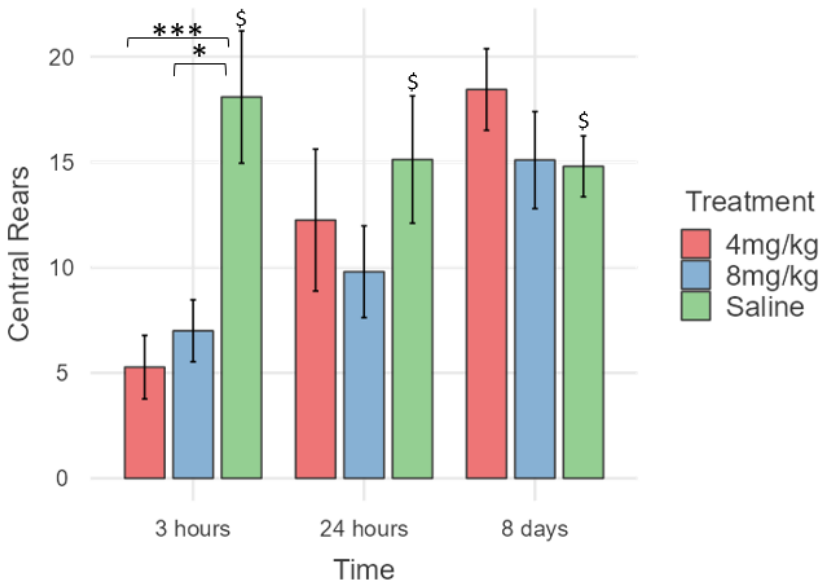

Figure 3.6- Means and ±SEM for central rears split by time and treatment. Significant interactions (*) were observed in post hoc evaluations between psilocybin doses and saline groups at 3 hours. A significant main effect of treatment was seen when comparing saline to 8 mg/kg ($).

3.3.4- Thigmotaxis

The percentage of time in the residual area considering the total session time revealed only a significant main effect of treatment $F(2, 82) = 9.42$, $p < 0.001$, as seen in the table and Figure 3.7. Post hoc evaluations demonstrated that animals injected with psilocybin and tested 3h later showed significantly more thigmotaxis than saline groups: 4 mg/kg $p < 0.01$ and 8 mg/kg $p < 0.05$.

Table 3.7 - Summary of Analysis for Thigmotaxis Split by Time and Treatment

	df	F	p
Time	(2, 82)	2.74	0.07
Treatment	(2, 82)	9.42	< 0.001
Time*treatment	(4, 82)	1.17	0.33

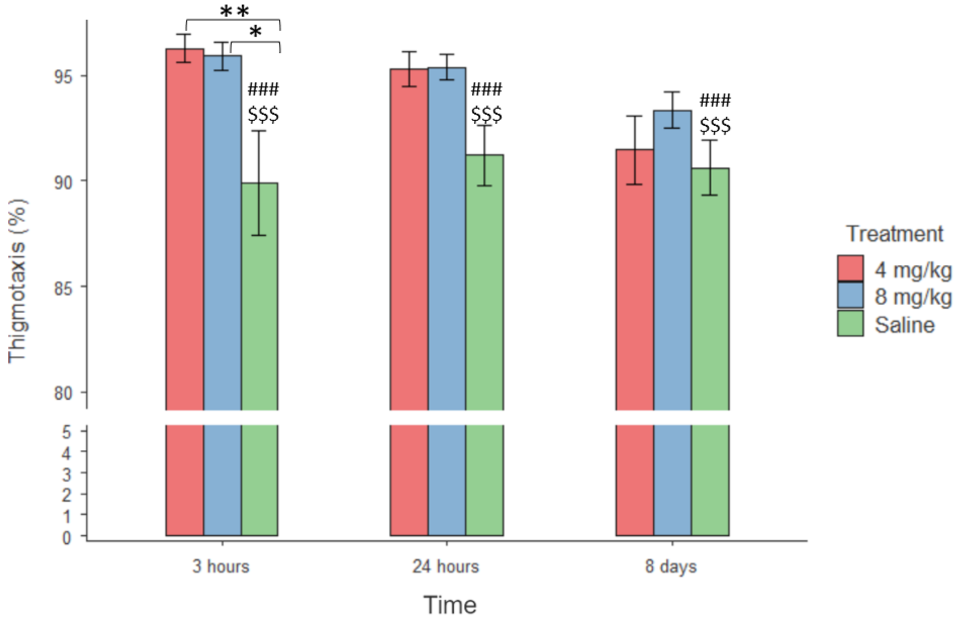

Figure 3.7- Means and ±SEM for thigmotaxis split by time and treatment. Significant interactions (*) were observed in post hoc evaluations between psilocybin doses and saline groups at 3 hours. Significant main effects of treatment were seen when comparing saline to 4mg/k (#) and 8 mg/kg ($) groups.

3.3.5- Total Distance Travelled

The distance travelled in total area (i.e., central + residual areas) presented significant main effects of treatment $F(2, 82) = 6.32$, $p < 0.01$, and time $F(2, 82) = 10.14$, $p < 0.001$. Saline groups walked more than both psilocybin groups (Table and Figure 3.8). Regarding treatment, animals that received saline walked more than those receiving 4 mg/kg ($p = 0.002$).

Moreover, animals tested 3 hours and 24 hours after injection walked significantly less than those injected 8 days prior. ($p < 0.05$) as seen in table and figure 3.8.

Table 3.8- Summary of Analysis for Total Distance Travelled Split by Time and Treatment

	df	F	p
Time	(2,82)	10.14	< 0.0001
Treatment	(2,82)	6.32	< 0.01
Time *treatment	(4,82)	1.05	0.39

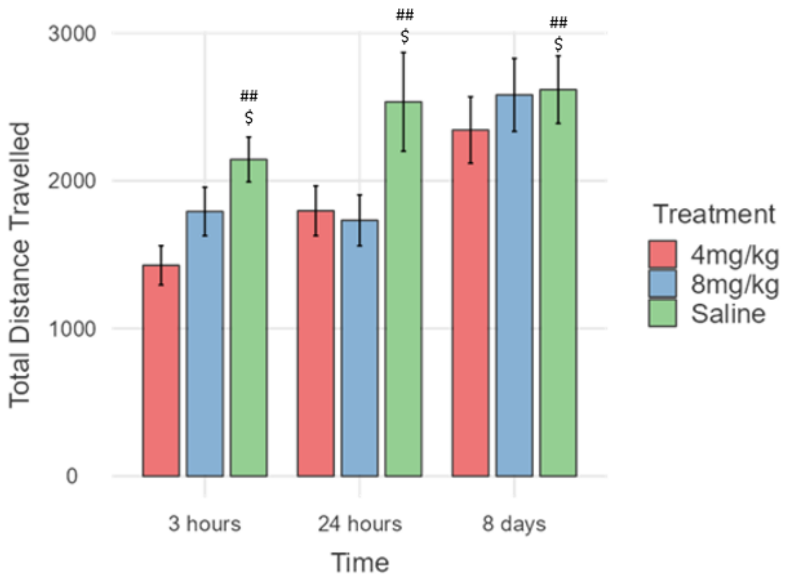

Figure 3.8- Means and ±SEM for total distance travelled split by time and treatment. In post hoc evaluations significant main effects of treatment were seen when comparing saline to 4mg/k (#) and 8 mg/kg ($) groups.

3.3.6- Total Rears

Another measure for the total area of the arena with significant differences in treatment, time and interaction between these two factors is the total number of rears in the arena, as seen in Table 3.9.

Table 3.9- Summary of Analysis for Total Rears Split by Time and Treatment

	df	F	p
Time	(2, 82)	16.31	< 0.001
Treatment	(2, 82)	10.46	< 0.001
Time*treatment	(4, 82)	4.74	< 0.01

Post hoc analysis demonstrated that in terms of time, animals tested 8 days after the injection reared significantly more than those tested 3 hours later $p < 0.001$, and then those tested 24 hours later $p < 0.01$. This effect was mainly driven by the effects of psilocybin on the earlier time point as seen in the figure below when treatment revealed significant differences between saline and 4 mg/kg group *($p <$* 0.001) and with 8mg/kg group *($p <$* 0.05).

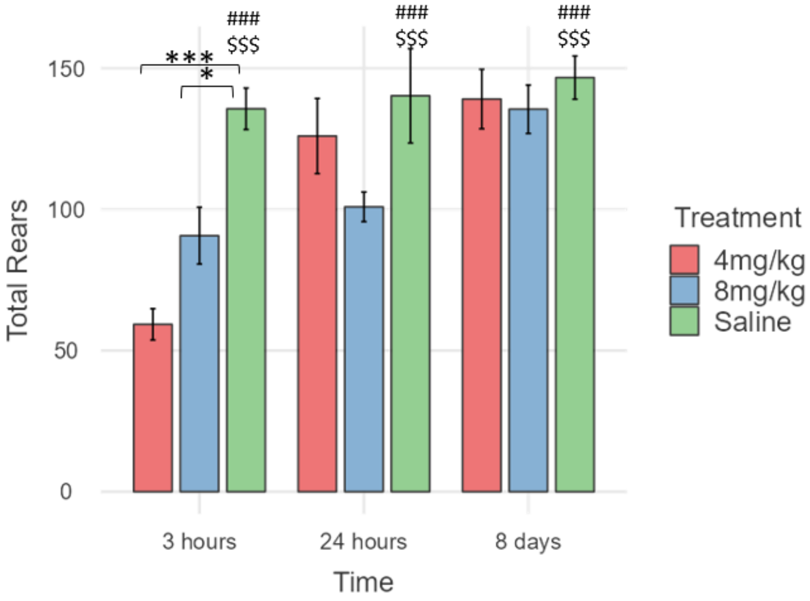

Figure 3.9- Means and ±SEM for total rears split by time and treatment. Significant interactions (*) were observed in post hoc evaluations between psilocybin doses and saline groups at 3 hours. Significant main effects of treatment were seen when comparing saline to 4mg/k (#) and 8 mg/kg ($) groups.

3.3.7- Faecal Boli Count

Table and Figure 3.10 shows the analysis of the faecal boli count, which revealed significant differences only in terms of time: $F(2, 82) = 5.16$, $p = 0.008$.

Table 3.10- Summary of Analysis for Faecal Boli Count Split by Time and Treatment

	df	**F**	**p**
Time	(2, 82)	5.16	< 0.01
Treatment	(2, 82)	0.15	0.86
Time *treatment	(4, 82)	1.52	0.21

Furthermore, post hoc analysis demonstrated that the difference is significant when comparing the 3h with the 8-day conditions p < 0.01, with larger counts occurring in more extended intervals between treatment and test.

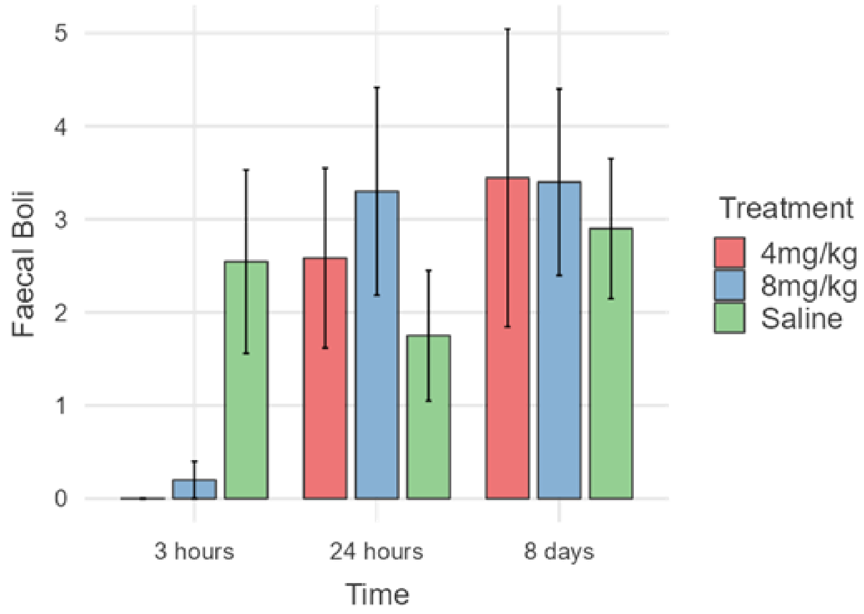

Figure 3.10- Means and ±SEM for faecal boli Count split by time and treatment. No main effects of treatment or interactions with treatment were seen.

3.3.8- Ultrasonic Vocalizations (Happy and Sad calls)

No significant main effects or interactions were seen in these measures (table3.11); however, the graphs (Figures 3.11 and 3.12) show slightly opposite trends between sad and happy calls. The 4 mg/kg-treated animals also differ somewhat from the other two groups in happy calls in the 3-hour condition. In comparison, the 8 mg/kg group differentiates in sad calls for 24h and 8 days conditions.

Table 3.11- Summary of Analysis for USVs – Happy and Sad Calls Split by Time and Treatment

		df	F	p
Happy calls	Time	(2, 81)	1.54	0.22
	Treatment	(2, 81)	0.25	0.78
	Time *treatment	(4, 81)	1.13	0.35
Sad calls	Time	(2, 81)	0.23	0.80
	Treatment	(2, 81)	2.08	0.13
	Time *treatment	(4, 81)	0.9	0.47

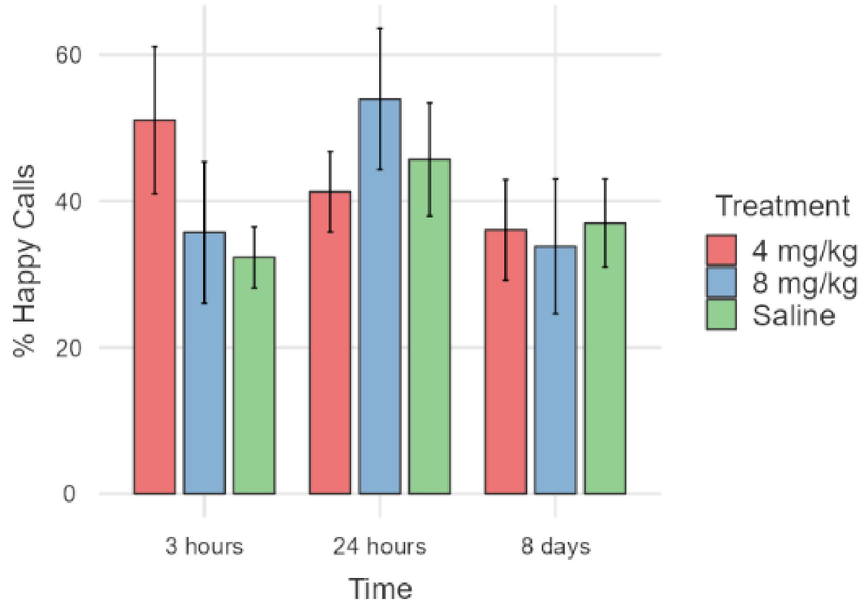

Figure 3.11- Means and ±SEM for USVs happy calls split by time and treatment. No main effects of treatment or interactions with treatment were seen.

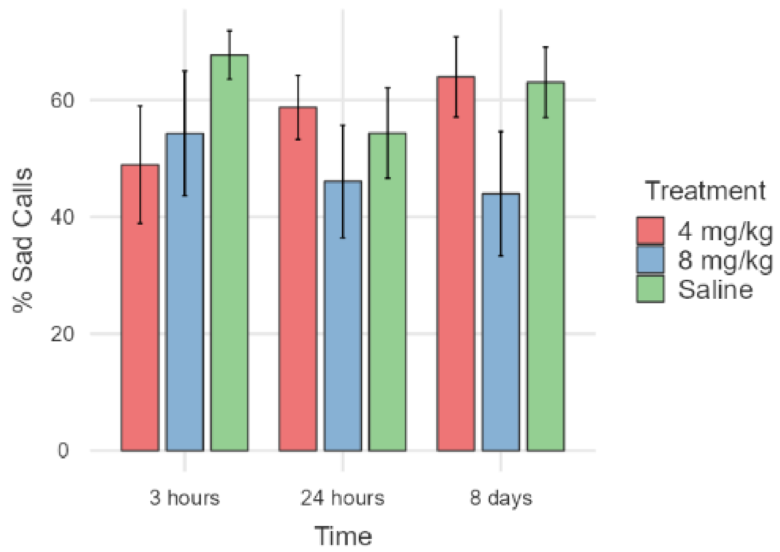

Figure 3.12- Means and ±SEM for USVs sad calls split by time and treatment. No main effects of treatment or interactions with treatment were seen.

3.4- Discussion

Two significant gaps in current research on psilocybin are the effects of high doses of psilocybin and its impact on female subjects. This experiment chose OFT to begin addressing these gaps.

The OFT is a well-established behavioural assay that evaluates an animal's ambulation (movement), exploration, and thigmotaxis (wall-hugging behaviour) within an arena. These behaviours serve as indicators of the animal's motor capabilities and emotional state.

The effects of psilocybin treatment on the animal's behaviour were evident in the present study. Almost all the measures analysed, except for USVs, exhibited significant differences between psilocybin-treated animals and saline-treated ones. Psilocybin groups generally walked less, spent proportionally less time in the central area, more time in the peripheral area of the arena, and reared less compared to saline groups. Additionally, five measures showed significant time effects: total distance moved, percentage of central distance moved, rears (total and central), and faecal boli count. Notably, interactions between time and treatment for rearing behaviour showed that a 4 mg/kg dose had a more pronounced effect on animals tested on the same day of injection. Although other time and treatment interactions were not significant, the figures show this to be a consistent pattern: Psilocybin's effects were strongest at 3 h, intermediate at 24 hr and virtually absent at 8 days.

Regarding psilocybin treatment, most studies suggest an impairment in locomotor activity measures following injections of doses higher than 1 mg/kg (Wojtas & Gołembiowska, 2024) due to increased GABA levels (Wojtas et.al., 2022). This effect seems more pronounced in males than in females (Tylš et al., 2016). However, most tests showing this result occurred on the same day of the injection, between 10 and 30 minutes after administration. Considering that the peak action in the brain occurs between 30 and 90 minutes (Horita & Weber, 1962; Musikaphongsakul et al., 2021), the results of the present study are compelling . The present findings align with the literature showing a reduction in locomotion on the same day of administration even though tests happened 3 hours after injection but also demonstrated a significant reduction in activity when tests were conducted the following day, contrary to the literature suggesting no effects in this condition with male subjects (Wojtas, 2022). This provides robust evidence of a prolonged impact of psilocybin on animal behaviour compared to previous preclinical trials.

Although the primary objective of this experiment was to examine the effect of high doses on locomotion, the percentage of time spent in the central area and the percentage of distance covered in the central area are often used as anxiety measures. In this sense, the findings from this study align with several findings in the literature showing that serotonin-like drugs, such as psilocybin (Nutt, 2020), may exhibit anxiogenic properties upon acute administration and anxiolytic effects following chronic administration or prolonged testing. These varying responses could be influenced by factors associated with the pharmacological profiles of serotonergic agents (e.g., the time-dependent nature of the effects and specific receptor interactions), for instance, Gepirone, a 5-HT1A receptor partial agonist, and fluoxetine, a selective serotonin reuptake inhibitor, show divergent effects in the EPM, suggesting that specific receptor interactions significantly influence the drugs' anxiolytic effects. Additionally, the time-dependent nature of these effects underscores that responses to these drugs evolve with the duration of treatment. (Silva & Brandão, 2000). These divergences highlight the need for varied approaches, including repeated testing and diverse treatment protocols, to address them effectively.

In this same direction, a review by Prut and Belzung (2003) demonstrates that SSRIs and tricyclics often fail to produce anxiolytic effects in the open field test and occasionally induce anxiogenic effects. They note that 5-HT2 receptor antagonists, such as ritanserin, show anxiolytic effects in 75% of studies, while other antagonists like Ketanserin do not, highlighting the variability and specificity of receptor-mediated responses. Moreover, non-specific 5-HT receptor agonists consistently induce anxiogenic responses. In contrast, 5-HT1A receptor antagonists are anxiolytic in 56% of cases, and administration of 5-HT1A agonists in limbic structures consistently induces anxiolysis in this test.

It is important to emphasise that methodological variations among the studies significantly contribute to the lack of consensus in the test results. Differences in animal strains, apparatuses, and experimental procedures complicate the analysis and comparison across different studies.

Moreover, the predominance of male subjects in the studies (Altemus et al., 2014)) overlooks the potential influence of sex differences, including the effect of female hormones on the outcomes of the studies, thereby introducing a bias in data interpretation. This scenario makes it challenging to set consistent parameters for evaluating results from new research but also advocates for the importance of studies with innovative design.

The second hypothesis of this study was related to time. Five measures achieved statistical significance in terms of the effect of the time between treatment and test on the behaviour of the animals. These measures are central distance travelled, total distance travelled, faecal boli count, and rears (both in central and total areas). A trend observed in these measures is that the longer the time elapsed between injection and testing, the higher the scores were for these measures, suggesting that the effects of the treatment were more pronounced 3 or 24 hours following injection. Conversely, results from tests conducted 8 days later showed more similar outcomes between psilocybin and saline groups.

The data suggest that contrary to observations from clinical trials, the effect of psilocybin in this study does not persist over an extended period. One possible explanation for this lack of chronic effect can be related to the dose. It is possible that the high doses tested were high enough for acute effects but not for sustained changes, which might require a higher dose. Alternatively, the test chosen may not be adequate to detect subtle or prolonged effects of the treatment.

The difference in the counts of faecal boli between the 3 hours conditions and the other two can be explained by the fact that serotonergic drugs can induce autonomic responses, including diarrhoea. This is likely due to increased serotonin (5-HT) levels in intestinal tissue (Haberzettl, R., 2013; Squires, L.N., 2007). This finding aligns with observations from other experiments of this work and previous studies conducted in our lab (Chafee, H. 2022) when animals treated with the drug and observed for 30 min invariably presented diarrhoea.

The only variables showing significant interactions between treatment and time were the two related to rearing behaviour in the arena's central and total areas. The groups treated with psilocybin consistently showed fewer rear counts than those treated with saline. Regarding time, animals tested 8 days after injection exhibited more rearing activity than those tested 3 hours or 24 hours post-injection. These results demonstrate the impact of psilocybin on the animals' exploration behaviour.

Significant interactions between treatment and testing conditions were observed, primarily driven by the 4 mg/kg dose, which exhibited more pronounced effects 3 hours post-injection compared to later time points, particularly in total rears. This measure includes all rearing behaviours (supported and unsupported) in any part of the arena. Generally, the impact of psilocybin diminished with the lengthening of the interval between injection and testing. This trend was evident as the rearing counts of animals tested 8 days post-injection closely approached those of the saline-treated groups, suggesting a reduction in the drug's effects over time.

Interestingly, rearing behaviour was the only measure showing a difference between the two doses of psilocybin. All the other measures did not show differences in the effects of the doses, with both 4mg/kg and 8mg/kg inducing similar effects in the animals. This suggests that rearing may be a particularly sensitive measure to detect the effects of psilocybin and may explain why a significant interaction between treatment and time was only observed with this measure. However, other measures showed similar (non-significant) trends.

Finally, USVs were the only measures showing no significant main effects or interactions among factors. However, 4mg/kg animals emitted visibly more happy calls in the 3h condition, and 8mg/kg treated animals exhibited fewer sad calls with tests happening later (e.g. 24h and 8 days). The decision to use a percentage of happy calls rather than the absolute number was made, considering that the number of USVs emitted varies greatly among individual subjects. This large degree of individual variety may explain the absence of results, especially considering a test design with a single trial.

As previously discussed, moderate to high doses of psilocybin (at least 1 mg/kg) are required to elicit behavioural effects (Meinhardt et al., 2020). Additionally, the 15-fold greater affinity of psilocin for human 5-HT2 receptors compared to rodent receptors (Gallaher et al., 1993; Klein et al., 2018) suggests that higher doses are needed in animal studies to produce effects comparable to those observed in humans with smaller doses. Consequently, slightly higher doses may induce more pronounced behavioural effects, as demonstrated by Jefsen et al. (2021), who observed dose-dependent increases in plasticity-related genes with treatments ranging from 0.5 to 20 mg/kg.

Moreover, it is possible that motor behaviours are more sensitive to psilocybin's effects than emotional responses. Thus, to effectively observe changes related to depression- and anxiety-like behaviours, higher doses of psilocybin might be necessary.

Overall, the findings highlight the effects of psilocybin on locomotor activity, particularly at 3 and 24 hours post-injection. The percentage of time and distance spent in the centre of the arena indicate an anxiogenic effect of both doses when animals were tested on the same day and the day following injection. While most measures revealed no significant differences between the doses, rearing behaviour showed a stronger effect with a 4 mg/kg dose compared to 8 mg/kg when assessed on the day of injection. However, the lack of significant results across other measures limits the ability to draw firm conclusions about dose-dependent differences. This emphasizes the importance of employing diverse experimental approaches, including model variations, dosing strategies, and consideration of sex-specific responses, to better understand psilocybin's behavioural effects.

Chapter 4 – Effects of Doses of 8 and 16 mg/kg of Psilocybin on Anxiety and Depression-like Symptoms

This chapter describes the behavioural effects of higher doses of psilocybin (8mg/kg and 16mg/kg) using the affective disorders test (ADT) with rats submitted to Early Maternal Separation (EMS) to elicit anxiety and depression-like symptoms.

The coming sections will detail the supporting theory underlying the models chosen, their specific methodologies, and the doses utilised in the experiment. After that, the main findings and discussion will be presented.

4.1- Theoretical Framework

4.1.1- Animal Models

Animal models are essential in replicating human biological processes, diseases, or behaviours, providing insights into the pathology and potential treatments. These models range from those inducing symptoms corresponding to human conditions to those mimicking the complex biological processes underlying these conditions. This diversity in model design enriches research, allowing for a more comprehensive exploration of diverse aspects of disorders. The present experiment will employ the EMS model, the ADT, to evaluate the effects of EMS and psilocybin.

4.1.1.1- Early Maternal Separation

The early life of most mammals is spent in close contact with their mother, and this period is crucial not only for nurturing and protection but also for the development of the offspring's behavioural and neurologic phenotype. (Vetulani, 2013). Disruption in this maternal bond produces critical behavioural, endocrine and neurochemistry changes that can persist into adulthood (Lehmann & Feldon, 2000; Huot et al., 2001)

In humans, early childhood stress, such as neglect or loss of caregivers, is linked to increased susceptibility to mental health conditions like depression and anxiety later in life (Vetulani, 2013). Similarly, in rodents, separating pups from their mother during a critical developmental phase leads to changes in the functioning of the hypothalamic-pituitary-adrenal (HPA) axis and disturbances in various behavioural aspects, including increased fear response, anxiety and depression-like behaviours (Nishi M., 2020). Because of these similarities between humans and rodents in terms of the effects of stress in early life, EMS is a widely recognised animal model that mimics early-life stress in humans.

From postnatal day (PND) 4 to PND 14, this pivotal neurodevelopmental stage in rodents is known as the stress Hyporesponsive Period (SHRP). This period is marked by a reduced response to stress, diminished Adrenocorticotropic Hormone (ACTH) and corticosterone levels, and limited activation of the HPA. This phase is essential for synaptic pruning and the formation of neuronal networks, which later play a crucial role in managing stress responsiveness (Vásquez, 1998). Maternal care during this period is essential for reducing HPA activation (Kambali et al., 2019). EMS is applied in the pup's first weeks after birth to disrupt this period of reduced sensitivity (Vetulani, 2013; Sapolsky & Meaney, 1986).

A vast literature has linked EMS to the development of depression and anxiety-like symptoms, as evidenced in behavioural and neurobiological studies in rats. For instance, Wang, D. et al. (2020) found in their systematic review and meta-analysis that rodents subjected to EMS exhibited increased defensive behaviours in the open field test (OFT) and elevated *p*lus maze (EPM). The increased defensive behaviours were interpreted as indicative of the anxiogenic effects resulting from early-life stress. Moreover, animals submitted to EMS showed a higher propensity for depression-like behaviours in adulthood, as measured, for example, in the forced swim test (FST) (Marais, 2008; Wei, 2018). The normalisation of anxiety and depression-like behaviours in EMS rats with antidepressant treatments (Hout et al., 2001; A. El Khoury et al., 2006) further validates the model's effectiveness in simulating early-life stress and the resultant anxiety and depression-like symptoms in adulthood.

However, despite the substantial evidence supporting this model's utility, the debate regarding its validity persists, especially considering the varying results across different sexes and animal strains (Romeo et al., 2003; Schmidt et al., 2011). In addition, the variability in research outcomes can also be attributed to the differences in EMS protocols employed across studies. One example of this variation is concerning the separation's duration (ranging from 1 to 6 hours daily) and the duration of the separation period, typically starting between PND 1 and 4 and ending on PND 14.

This context reinforces the importance of employing EMS, particularly in female rats, to further assess its reliability in inducing anxiety and depression-like behaviours since there is evidence that the phase of the cycle influences the outcomes, as seen in Romeo (2003), for example, which found reduced anxiety and fear behaviour in the OFT, but only when the females were in the dioestrus phase of their oestrous cycle.

4.1.1.2- Affective Disorders Test

In the present study, the affective disorders test (ADT) (Kidwell, 2019; Blackburne, 2021) was used to assess depression and anxiety-like symptoms in the subjects. This model was developed at the Victoria University of Wellington Behavioural Neurogenetics Laboratory based on two preexisting paradigms: anticipatory pleasure (Makowska & Weary, 2016) and the Successive alleys test (SAT) (Deacon, 2013). This combination allows for the repeated assessment of measures for both conditions, depression and anxiety.

The diagnosis of anxiety and depression depends on determining how symptoms such as anhedonia and excessive fear affect daily life functioning. To evaluate symptom-like effects, pre-clinical models employing rodents utilise correlates of these symptoms that can be manifested in various behavioural changes, such as decreased or increased locomotor activity, decreased motivation, decreased social contact, impulsivity, and decreased sensitivity to rewards (Willner & Muscat,1991). These can be measured in a variety of experimental apparatus and designs. Understanding these behavioural indicators in animals provides a valuable framework for studying similar manifestations of depression in humans, particularly the concept of anhedonia, a fundamental aspect of depression as recognised in human diagnoses (Association, 2013).

In human patients, anhedonia manifests as a diminished capacity to experience pleasure, characterised by two main aspects: anticipatory pleasure, which is the enjoyment derived from the expectation of receiving a reward, and consummatory pleasure, the joy felt upon actually receiving the reward (Winer et al., 2019). Here, a "reward" can be any outcome or experience generally regarded as positive, enjoyable, or fulfilling, eliciting a sense of pleasure or satisfaction. Furthermore, growing evidence suggests that the anhedonia observed in individuals with depression is primarily due to a decreased ability to experience anticipatory rather than consummatory pleasure (Hallford, 2020).

In the laboratory context, anhedonia-like behaviours can be observed in rodents through a variety of behavioural and physiological changes accessed in different animal models, for example, a decrease in sucrose consumption in the sucrose preference test (Strekalova T. et al., 2004) or reduced anticipatory behaviour in the open field test (Brenes & Schwarting, 2015).

Compared to other behavioural assays, the ADT allows the distinction between anticipatory and consummatory pleasure. The measures that can be acquired from this first component of the ADT are measures of ambulatory activity, vertical behaviour (rears), and USVs. The rationale behind the use of these measures is that under normal circumstances, a steady increase in these measures is expected in response to an approaching reward suggestion (van der Harst, Fermont, Bilstra, & Spruijt, 2003). With no or a more blunted increase being indicative of reduced anticipatory pleasure, one of the core components of anhedonia.

Regarding anxiety, various physical signs of stress or anxiety often appear in anticipation of potential future threats. Although animal models of anxiety cannot replicate every aspect and symptom of specific anxiety disorders, they can induce a state that closely resembles human anxiety disorders.

One of the most commonly used animal models to assess anxiety-like symptoms is the EPM (Rodgers et al. et al., 1997). The apparatus consists of a plus-shaped platform elevated above the ground, with two open arms and two closed arms that extend from a central square. The open arms have unprotected edges, which can induce anxiety in the animals due to the exposure to heights and open spaces. In contrast, the closed arms have walls on three sides, offering a more secure environment. Inspired by the EPM, the SAT follows a similar concept but features a different design. It consists of four alleys arranged in a linear sequence, with each alley becoming successively narrower and having lower walls (as shown in Figures 4.1 and 4.2). This gradual increase in openness along the alleys is designed to produce a progressively greater anxiogenic effect.

Both models are based on the approach-avoidance conflict since they expose the animal to the conflict between its natural exploratory impulse and its aversion to open and brighter spaces (Cryan & Sweeney, 2011). This conflict results in observable anxiety-related behaviours, such as the extent to which animals avoid more anxiogenic areas.

Moreover, both models encounter a common issue when it comes to accurately assessing anxiety, which is the one-attempt tolerance (OTT) phenomenon. The OTT, first described by File, Mabbutt, and Hitchcott (1990), suggests that the type of anxiety experienced in subsequent trials is qualitatively different from the initial anxiety seen in the first trial due to the absence of a reward for exploring anxiety-inducing areas. As a result, after a second exposure to the test, animals are more likely to spend increased time in the closed arms. This behaviour complicates the analysis of results, potentially leading to inaccuracies in measuring anxiety levels, particularly as we are interested in the long-term effects of psilocybin on anxiety-like behaviour.

An adaptation was introduced to address this issue in the SAT (Kidwell, 2019). Before testing, a food reward (half a piece of Froot Loops®) is placed in each alley, and the animals are maintained on a food deprivation regime throughout the experiment. This modification ensures that the animals remain motivated to explore the increasingly anxiety-inducing alleys, even after several rounds of testing. Consequently, this assay has been renamed the modified successive alleys test (mSAT).

For the mSAT, the measures taken are distance and velocity, latencies to eat, number and time of visits in alleys, and score for the location of reward consumption.

Although the animals' behavior was monitored for 30 minutes in the anticipatory box after treatment administration, this data was not included in the present study. The results section will focus on comparing baseline data (i.e., five testing days before injection), acute effects (i.e., three days immediately following injection), and chronic effects (i.e., the last six days of testing), as illustrated in Table 4.2. This approach minimizes any potential influence of perceptual impairments on animal behavior, as the first acute effects test occurred at least 18 hours after peak brain levels were reached in the injection day. A detailed investigation into perceptual confounds is beyond the scope of this study, and they are not expected to have played a substantial role in the design of the two experiments using the ADT.

4.1.2- Doses

Using the same principles mentioned previously concerning the use of high doses in psilocybin research (Meinhardt et al.,2020; Jefsen, 2020; Gallaher et al.,1993 and Klein et al., 2018) and observing the results of the previous experiments where we found alterations in locomotor activity but not in USVs, higher doses (i.e., 8mg/kg and 16mg/kg) were evaluated in the present experiment.

Given the scarce literature on the effects of high doses of psilocybin on anxiety and depression-like measures in female subjects this experiment intends to evaluate the effects of high doses of psilocybin in measures of anxiety and depression taken from the affective disorders test and using animals submitted to early maternal separation.

The hypothesis formulated is that at baseline, subjects in the EMS groups will exhibit significantly more anxiety and depression-like behaviours compared to the ones in control groups. Additionally, both doses of psilocybin (8 and 16mg/kg) will significantly reduce these symptoms compared to those administered saline, as evidenced by the ADT measures.

Prediction 1 – At baseline, compared to controls, EMS animals will exhibit reduced anticipatory pleasure in a dose-dependent manner, as indicated by decreased distance moved, increased velocity, and reduced total, free, and wall rears counts in the anticipatory box. They will also show heightened anxiety-like behaviours, reflected in the modified successive alleys test by reduced distance moved, increased velocity, lower frequency and cumulative duration in alley 4, increased latency to eat the first reward and reward number 4, and lower mSAT scores.

Prediction 2 – Both doses of psilocybin (8 and 16 mg/kg) will increase, in a dose-dependent manner, anticipatory behaviours, as indicated by increased distance moved, decreased velocity, and increased total, free, and wall rear counts in the anticipatory-box test. Psilocybin will also reduce anxiety-like behaviours in EMS animals, reflected by increased distance moved, lower velocity, increased frequency and cumulative duration in alley 4, decreased latency to consume the first reward and reward number 4, and higher mSAT scores at both acute and chronic testing time points.

4.2- Specific Methods

Building on the general methods described previously, this section will present the specific procedures tailored for this experiment.

4.2.1- Animals

Adult Sprague Dawley rats were paired in four sequential cohorts roughly every 30 days. Three to four mating couples were paired, producing 14-30 animals per cohort. Different breeding pairs were selected based on non-relation from the colony maintained at the Victoria University of Wellington animal facility and paired for 5-6 days before separation. The distribution of animals per group of conditions and treatments is presented in the table below.

Table 4.1 – Distribution of Animals in Groups.

Condition/Treatment	Saline	8 mg/kg	16 mg/kg	Total
EMS	11	9	9	29
Control	13	9	9	31
Total	24	18	18	60

Note. EMS= Early maternal separation.

4.2.2- Animal Models

4.2.2.1- Early Maternal Separation Procedures

In each cohort, on PND - 0, the litters were randomly allocated to either the EMS or control conditions. Using the model described in Vetulani (2013), EMS was performed from PND 4 until PND 14. Thus, dams whose pups were designed for this condition were separated from their litters for 3 hours a day for 11 consecutive days. The dams were housed individually in a new box (same throughout all 11 days) and placed in another housing room during the 3-hour separation period. On the other hand, dams of litters designed for the control condition were handled for 1 minute for the same 11 days (i.e., once a day from PND 4 to PND 14) to control the effect of animal transfer. All four cohorts had animals designed for both conditions.

All litters were weaned at PND 21 when females from both conditions (i.e., EMS and control) were chosen to form the four cohorts of the experiment, which varied in numbers from 11 to 17 subjects with a total of 60 animals. As mentioned above, the chosen animals were housed in standard polycarbonate cages in groups of two to three animals per cage (of the same condition) with water and food ad libitum until the preparation for the test started and the animals started the food deprivation regimen.

4.2.2.2- Affective Disorders Test

4.2.2.2.1- ADT - Apparatus

As discussed above, the ADT uses two apparatuses: the anticipatory box (locomotor activity box) and the mSAT. Only the locomotor activity box was used for the habituation phase in the first three days of tests. Both apparatuses were used for the baseline, acute, and chronic effects phases.

The locomotor activity box consists of a square chamber made of black polycarbonate measuring 40 cm x 40 cm x 40 cm. An UltraMic microphone (250K 16 bi; Java Sound) was placed over the anticipatory box to record the USVs.

The mSAT is made of 4 glossy black polycarbonate alley slots, assembled to form a complete unit that stands 1 meter tall from the floor. Each alley is 45 cm long, totalling 180 cm, with a gradual reduction in runway width and wall height. Alley one has dimensions of 29 cm in height and 9 cm in width; alley two is 6 cm by 8 cm; alley three measures 2.5 cm by 6.5 cm; and alley four is 1 cm by 3.5 cm (see Figures 4.1 and 4.2).

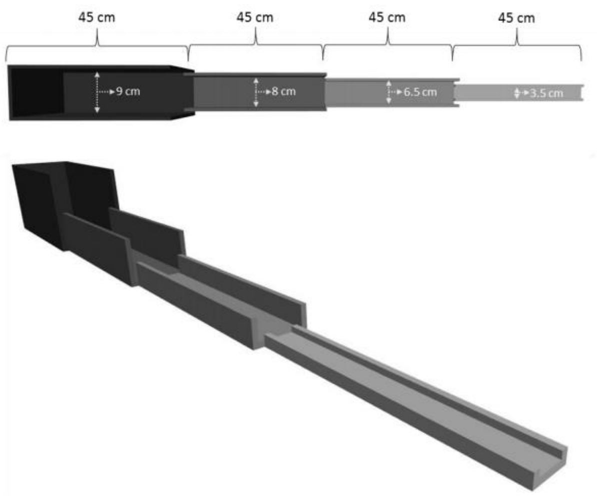

Figure 4.1- Successive alleys.

Figure 4.2- Successive alleys and anticipatory box.

The tests took place in the testing room, where the apparatuses were set up. The room had dimmed lighting, progressing in brightness towards alley 4 of the mSAT (approximately 15 lux measured in alley 1 of the mSAT and about 40 lux in alley 4). A webcam (640 x 480 resolution, 30 fps; C170Logitech Lausanne, Switzerland) was placed with a birds-eye view over both apparatuses. Both apparatuses were placed on a black cover.

Additionally, on the day of injection, a set of 4 locomotor activity boxes identical to the ones used on testing days was used to accelerate data collection on this day (Figure 4.3).

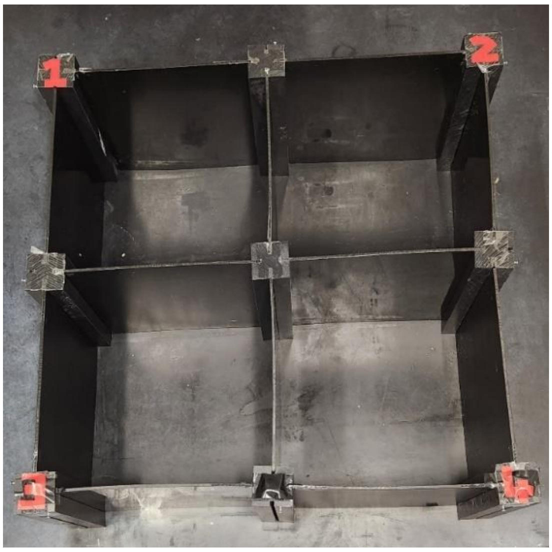

Figure 4.3- Anticipatory boxes for injection day.

4.2.2.2.2- ADT - Procedures

The entire timeline of the experiment can be visualised in the table below.

Table 4.2– Experiment Chronogram

Preparation			**Pre-test**			**Test**					
			1	2-5	6-7	8-10	11-15	16	17-19	20-27	28-33
Day of birth	EMS or Handling	Weaning	Food Deprivation								
			Weighing	Interval	Handling	Habituation ant. box	ADT baseline	Injection	ADT acute effects	Interval	ADT chronic effects

Note. The numbers refer to the days when the specific activities, described below them, occurred. EMS = early maternal separation. Ant. = Anticipatory. ADT = affective disorders test.

The experiment commenced after the EMS procedure and a subsequent interval of approximately 60 days that allowed the animals to reach maturity. From day 1 until the end of the experiment, the animals were subjected to a food deprivation regimen. They were weighed every other day to ensure their weight was maintained at 85 to 90% of their initial weight recorded on Day 1. The animals were fed daily, exclusively by the experimenter, between 4 *p*.m. and 5:30 *p*.m., always at least 30 minutes after the last test session and after the weighing. They received 10 to 13 grams of standard chow each.

For two days (6 and 7), the animals received a 2-minute handling session in the housing room to make them familiar with the experimenter.

The next phase was habituation to the anticipatory box. This habituation happened once a day for three consecutive days (days 8 to 10). Every box with 2 or 3 animals was placed in the testing room and left untouched to habituate to the room for 10 minutes. After that, each animal was individually placed inside the anticipatory box for 10 minutes. During this period, rearing behaviour was manually scored, ultrasound vocalisations were recorded, and locomotor activity was tracked using EthoVision XT v15 software (Noldus, Netherlands). The objective of the habituation phase is to diminish the animals' novelty exploration of the anticipatory box. Additionally, over these three days, the animals were given four halves of Froot Loops (each) in their home cages at some point each day. This was done after the test session and before their regular feeding time to familiarise them with the sugary reward.

After the habituation phase, a five-day baseline testing period (11 to 15) was conducted using the anticipatory box and successive alleys. Following a 10-minute habituation period in the room, an animal was placed in the anticipatory box for 5 minutes. Immediately afterwards, it was transferred to the first alley of the mSAT, facing the back wall. USVs were recorded, locomotor activity was monitored using EthoVision and rears, and place and time of consumption of froot loops were scored.

On the sixteenth day, the animals received the treatment (either Psilocybin or saline). They were divided into groups of four animals per trial. Both EMS and control animals were randomly assigned to one of four groups, receiving either psilocybin at doses of 8 or 16 mg/kg or a solution of NaCl (9%) at doses of 1 or 2 ml/kg. Immediately following the injections, the animals were placed in one of the four locomotor activity boxes (figure 4.3), each identical to the anticipatory box used during the habituation and baseline phases. Over the next 30 minutes, their ultrasonic vocalisations (USVs) were recorded, and their locomotion was tracked with EthoVision.

On the following three days (17-20), the same procedure (5 minutes in the anticipatory box and 5 minutes in the successive alleys) was used for the baseline recording. Subsequently, the animals had an 8-day break before the final 6 days of chronic effects testing (28 – 33), following the same procedures.

Throughout all the experiment phases, a counterbalanced order of cages and animals within each cage ensured that each animal was tested at different times of the day on each experimental day.

Conducting two sets of tests separated by an interval compares the drug's effects at two distinct times: immediately after psilocybin is eliminated from the body and after 12 days, mimicking the long-lasting effects observed in clinical trials.

The table below lists all the measures taken from the two apparatuses:

Table 4.3- Measures of the ADT

Measure	Definition
Depression-like measures (anticipatory box)	
Box distance moved	Total of centimetres travelled during the session
Box velocity	Average of centimetres/second during the session
Free rears	Standing on the hind legs without touching the wall. One rear is finalised when four paws touch the floor
Wall rears	Standing on the hind legs with the support of a wall. One rear is finalised when four paws touch the floor
Total rears	The sum of wall and free rears
Anxiety-like measures (mSAT)	
Alleys distance moved	Total of centimetres travelled during the session
Alleys velocity	Average of centimetres/second during the session
Frequency in alley 4	Number of visits in alley 4
Cumulative duration in alley 4	Total time spent in alley 4
Latency to eat 1st FL (any alley)	Time until eat the first reward (in any alley)
Latency to eat FL #4	Time until eating the reward located in alley 4
mSAT score	The sum of the product of each reward's value (ranging from 1 to 4) and the value assigned to its consumption location (also from 1 to 4. This calculation is represented as mSAT = (Reward 1 Value × Location 1 Value) + (Reward 2 Value × Location 2 Value) + (Reward 3 Value × Location 3 Value) + (Reward 4 Value × Location 4 Value) Ex. If the subject ate rewards placed in alleys #1, #2 and #3 in alley 1 and reward #4 in alley 2, its score would be: =SUM (1*1) +(2*1) +(3*1) +(4*2) = 14

Note. mSAT = modified successive alleys test. FL = Froot loops (half of a piece)

Unfortunately, a technical issue resulted in the loss of all USV data from subjects in cohort 1. This fact could potentially affect the assessment of the entire dataset, and consequently, USVs were not analysed for this experiment.

4.2.3- Statistical Analysis

Initially, an analysis was conducted to address missing data. Little's Mcar test was performed with the complete data frame returning a result of $X2\ (182) = .000$ Sig = 1.000, demonstrating that the missing values have a random distribution and, consequently, the further analysis will not be biased by the pattern of missingness.

After the preliminary analyses, to reduce the volume of data, the values of each phase (i.e., baseline, acute, and chronic) were averaged, resulting in a single value for each time point. Values for the first two days of baseline were not included to reflect more stable and consistent behavioural patterns after acclimatisation to the two apparatuses.

Secondly, repeated-measures ANOVAs with the cohort as a between-subject factor were conducted to assess cohort effects. Except for the measures of box velocity, total rears and cumulative duration in alley 4, all other measures showed significant differences among the cohorts. However, it is noteworthy that these differences were consistent across all cohort comparisons, suggesting that no specific cohort stood out from the others. Moreover, due to the small number of subjects within each group in each cohort (ranging from 9 to 13), it was decided to combine the cohorts to enhance statistical power.

After that, the normality assumption was analysed with Shapiro-Wilk tests and visual examination of Q-Q plots and histograms of residuals for each measure and time point. In normality tests, except for total rears, wall rears, frequency in alley 4 and cumulative duration in alley 4, all the other measures returned $p < 0.05$ for at least one of the time points, although most graphs did not present massive distortions. Homogeneity of variances, evaluated with Levene's test, returned violation in at least one time point for 4 out of 12 measures (Box distance moved, box velocity, latency to eat first and mSAT score). However, it was decided not to perform selective transformation in the data as it can lead to inconsistencies between transformed and non-transformed data in terms of interpretation of the results, especially because different methods of transformation would be necessary. This way, we chose not to transform idiosyncratically to preserve consistency in measurement metrics. Instead, we analysed all data untransformed data.

Then, One-Way ANOVAs were performed for each measure to evaluate differences between saline dosages (i.e., high and low) and condition (i.e., EMS and control) during the baseline phase for each measure. The results will be discussed in the next section.

The primary analysis was conducted using repeated measures ANOVA, which examined the interactions among two factors and three time points in a matrix 2 x 3 (table 4.4). Factor condition encompasses the EMS and Control groups; the factor treatment, includes the two groups treated with psilocybin (i.e., 8 mg/kg and 16 mg/kg) and the saline group. The timepoints are the test phases: baseline (before injection) and acute and chronic phases.

Table 4.4- Anova Components

ANOVA	**Factor**	**Level**
Within subjects	Time	Baseline, acute and chronic
Between subjects	Condition	EMS and control
Between subjects	Treatment	8 mg/kg, 16 mg/kg and saline

Note. ANOVA – Analysis of variance. EMS = early maternal separation.

When necessary, post hoc evaluations for main effects and interaction were made using Holm-Bonferroni correction to address the heightened risk of Type I errors (i.e., incorrectly rejecting the null hypothesis) due to multiple pairwise comparisons. Complete tables with ANOVA results for main effects and interactions are in Appendix 2B.

4.3- Results

The main objective of this experiment was to enhance understanding of the behavioural effects of psilocybin, specifically on anxiety and depression-like symptoms.

The first hypothesis of this study was that at baseline, subjects in the EMS groups would exhibit significantly more anxiety and depression-like behaviours compared to the control group. The second hypothesis was that psilocybin treatment would lead to a notable reduction in symptoms compared to those administered saline, as evidenced by the Anxiety and Depression-like measures.

The results will be presented in detail, starting with saline volume comparisons, condition analysis (i.e. EMS and Controls), effects of psilocybin on depression-like measures and anxiety-like measures, and finally, results comparing time points.

4.3.1- Saline Dosage

The first analysis aimed to evaluate whether animals that received a low volume of saline differed significantly from those that received a high volume. The relatively high dose of psilocybin, combined with its limited solubility, required us to use two different volumes for the low and high-dose groups. Aiming to match the volume of psilocybin doses, the saline doses were split into low and high-volume groups (i.e., 1 or 2 ml/kg).

The one-way ANOVA showed no significant differences between the two groups across all measures of interest for both acute and chronic phases (table 4.5). As a result, both groups, the high and low volume of saline, were combined into a single group referred to as the saline group for all subsequent analyses.

Table 4.5 – Comparative Analysis between Saline Dosages

	Acute Phase			Chronic Phase		
	F	**df**	**p**	**F**	**df**	**p**
Box distance moved	0.04	(1, 21.84)	0.85	0.05	(1, 20.31)	0.83
Box velocity	0.01	(1, 21.97)	0.91	0.1	(1, 19.99)	0.76
Total rears	0.25	(1, 17.67)	0.62	4.57	(1, 20.55)	0.04
Wall rears	0.27	(1, 21.59)	0.61	3.02	(1, 21.58)	0.1
Free rears	0.02	(1, 20.9)	0.89	1.11	(1, 21.71)	0.3
Alleys distance moved	0.62	(1, 20.76)	0.44	0.34	(1, 22)	0.57
Alleys velocity	0.75	(1, 21.71)	0.4	0.3	(1, 21.68)	0.59
Frequency alley 4	1.57	(1, 21.97)	0.22	1.52	(1, 21.96)	0.23
Cum duration alley 4	0.44	(1, 21.75)	0.51	1.61	(1, 21.8)	0.22
Latency to eat first FL	0	(1, 21.97)	0.95	1.62	(1, 11.2)	0.23
Latency eat FL#4	0.38	(1, 18.83)	0.55	0.38	(1, 15.09)	0.55
mSAT score	0.58	(1, 21.69)	0.46	0.02	(1, 21.43)	0.89

Note: FL = Froot Loops (half of a piece), mSAT = modified Successive Alleys Test.

4.3.2- Condition: EMS and Control

The analysis of the effectivity of EMS indicated that the only measure reaching significance when comparing both groups is the cumulative duration in alley 4, all the other 12 measures did not present significant differences between EMS and control groups (table 4.6).

Table 4.6- Comparative Analysis between Conditions (EMS and Control) for all the Measures and Time Points

Measure in baseline	df	F	p
Box distance moved	(1, 57.95)	1.59	0.213
Box velocity	(1, 57.21)	1.37	0.246
Total rears	(1, 57.78)	1.27	0.265
Wall rears	(1, 57.21)	0	0.968
Free rears	(1, 57.97)	2.31	0.134
Alleys distance moved	(1, 56.55)	0	0.999
Alleys velocity	(1, 56.97)	0	0.947
Frequency alley 4	(1, 55.99)	1.81	0.184
Cum duration alley 4	(1, 57.67)	4.49	0.038
Latency to eat first FL	(1, 54.66)	0.39	0.537
Latency eat FL#4	(1, 52.77)	0.27	0.605
mSAT score	(1, 45.14)	1.13	0.294

Note: FL = Froot Loops (half of a piece), mSAT = modified Successive Alleys Test.

Repeated measures ANOVA demonstrated a significant effect of condition on the cumulative duration in alley 4 $F(1,54) = 7.16$, $p = 0.00986$, with EMS animals spending more time in alley 4 than controls.

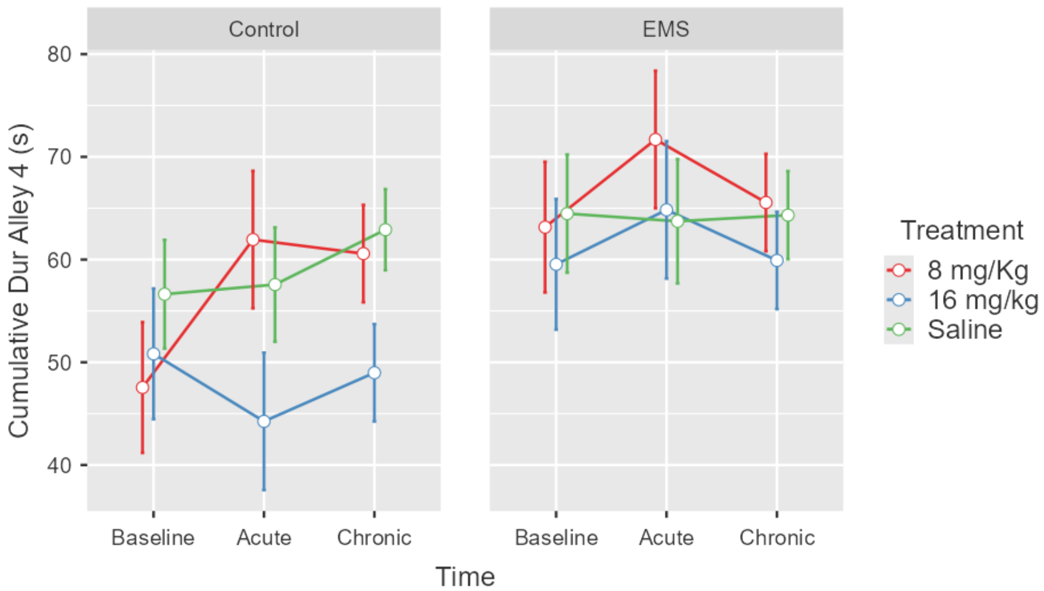

Figure 4.4- Means and ±SEM for cumulative duration in alley 4 across all three time points, split by condition and lines representing treatments.

4.3.3- Psilocybin

4.3.3.1- Psilocybin - Depressive-like measures

Analysis of the main effects and interactions of time, condition, and treatment of the measures acquired in the anticipatory box (e.g., box distance moved, box velocity, total rears, wall rears and free rears) showed a significant interaction between factors of time and condition for two measures: box distance moved and velocity. The post hoc tests only showed the effect of time. EMS and control animals presented higher scores in the chronic phase than in others 2 phases.

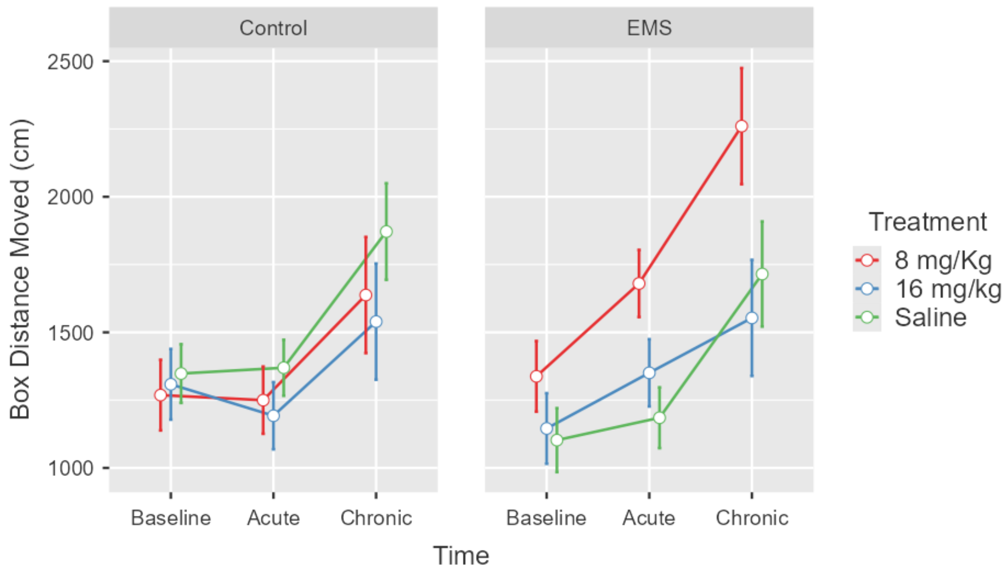

Figure 4.7- Means and ±SEM for box distance moved across all three time points, split by condition and lines representing treatments.

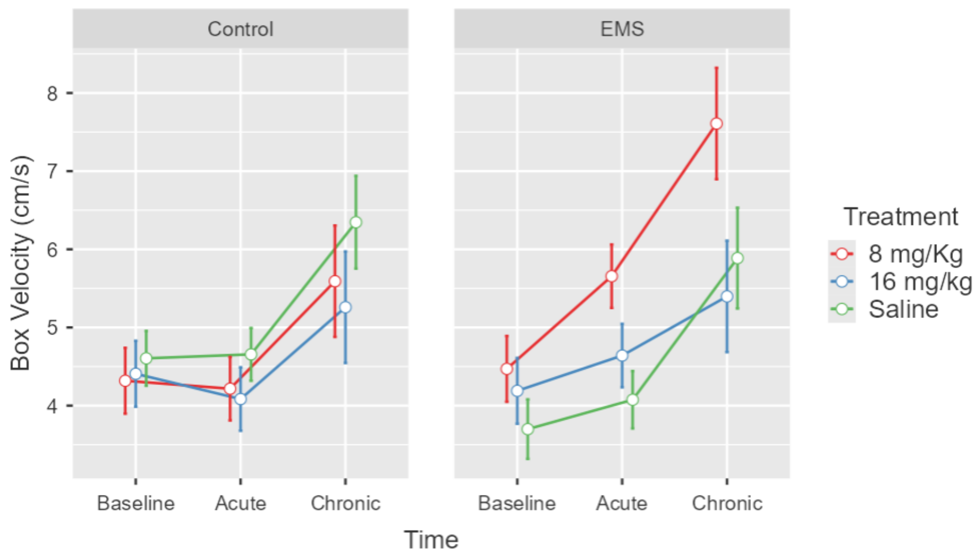

Figure 4.8- Means and ±SEM for box velocity across all three time points, split by condition and lines representing treatments.

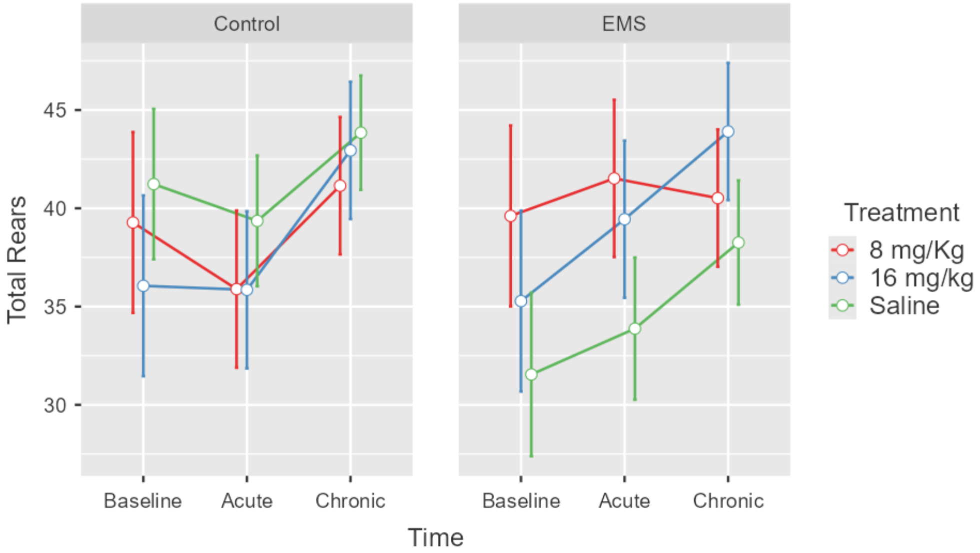

Figure 4.9- Means and ±SEM for total rears across all three time points, split by condition and lines representing treatments.

Although not significant, there is a discernible surge in the levels of ambulatory behaviour measures for EMS animals treated with psilocybin, especially with 8mg/kg. Moreover, significant main effects of time were also seen in all measures.

Table 4.7 – Main Effects of Time in Depression-like Measures

Measure	*df*	*F*	*p*
Box distance moved	(2, 108)	41.95	< 0.001
Box velocity	(2, 108)	44.47	< 0.001
Total rears	(2, 108)	6.01	0.003
Wall rears	(2, 108)	3.34	0.039
Free rears	(2, 108)	6.31	0.003

4.3.3.2- Psilocybin - Anxiety-like measures

Repeated measures ANOVA for anxiety-like measures obtained from the mSAT showed no significant main effects or interactions among factors for this group of measures, except for the factor time, with 6 out of 7 measures reaching a significant level (Table 4.2).

Table 4.1- Main Effects of Time in Anxiety-like Measures

Measure	*df*	*F*	*p*
Alleys distance moved	(2, 108)	6.92	< 0.01
Alleys velocity	(2, 108)	11.39	< 0.01
Frequency alley 4	(2, 108)	12.00	< 0.01
Cumulative duration alley 4	(2, 108)	1.19	0.31
Latency to eat first FL	(2, 108)	8.11	< 0.01
Latency eat FL#4	(2, 108)	16.69	< 0.01
mSAT score	(2, 108)	4.60	< 0.01

Note: FL = froot loops (half of a piece), mSAT = modified successive alleys test.

Post hoc analysis once again demonstrated that the chronic phase differed significantly from the other phases of the test. In this phase, animals walked more and faster, visited alley four more, exhibited reduced latency to eat 1st and 4th FL 4, and finally had a higher mSAT score.

Despite not demonstrating significant results, some patterns among the figures in this set of measures are still evident. For instance, comparing the baseline phase and acute effects, an increase in the distance moved, velocity, frequency of entries in alley 4, cumulative duration in alley 4 and mSAT score is consistently observed in subjects injected with 8 mg/kg. In the same period, saline-injected subjects exhibit a similar trend, albeit less prominently, while those administered with 16 mg/kg present near opposing tendencies. Conversely, comparisons between acute and chronic effects presented a more uniform pattern for all the groups, as seen in Figures 4.10 and 4.11.

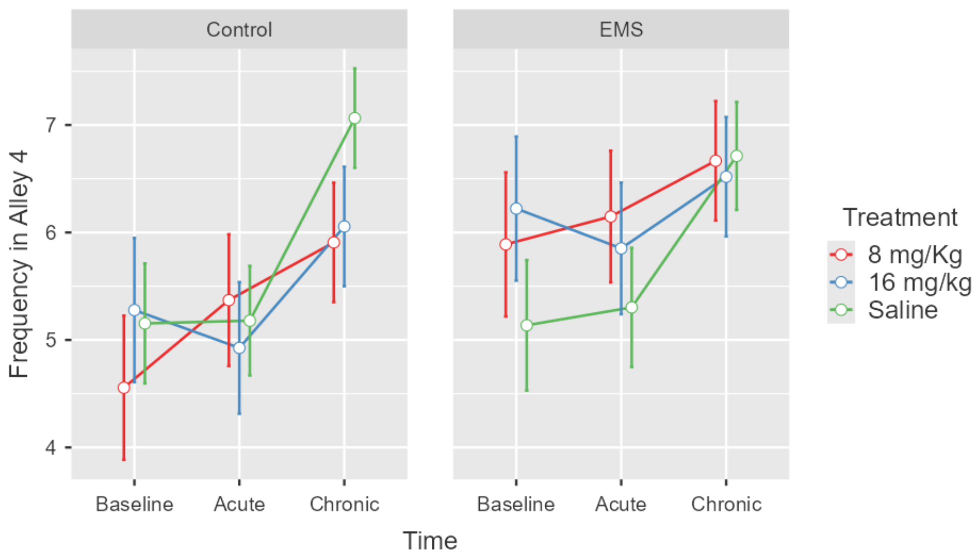

Figure 4.10- Means and ±SEM for frequency in alley 4 across all three time points, split by condition and lines representing treatments.

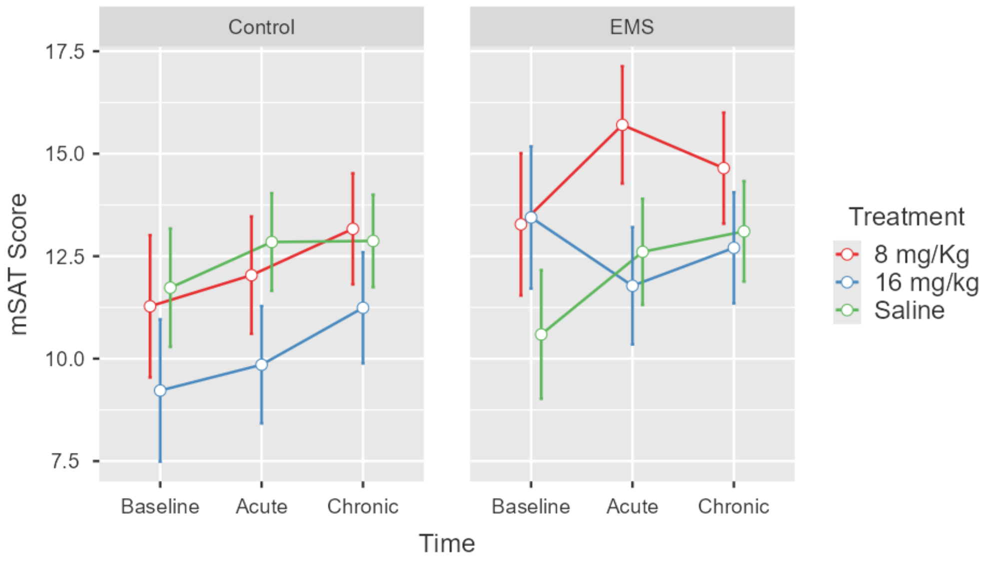

Figure 4.11- Means and ±SEM for mSAT Score across all three time points, split by condition and lines representing treatments.

4.4- Discussion

As demonstrated in the previous experiment, psilocybin affected the animals' locomotor activity, but it was unclear whether the doses used were enough to elicit the changes in emotional behaviour. In this experiment, more complex animal models were chosen to evaluate the drug's impacts: the EMS to elicit anxiety and depression-like symptoms and the ADT to evaluate the effects of 8 and 16mg/kg of the drug.

4.4.1- EMS and Control Animals

The first hypothesis was that EMS (compared to control animals) would induce reduced ambulation activity and exploration in both apparatuses, signalling the presence of anxiety and depression-like symptoms.

Contrary to expectations, the comparisons between EMS and control groups before the treatment revealed a significant difference in only one measure: the cumulative duration in alley 4, with EMS animals spending more time in alley 4 compared to control ones. The singular nature of this finding, which is not corroborated by other measured parameters, suggests a need for cautious interpretation, possibly diminishing its conclusiveness due to the absence of supportive evidence from the remaining 11 measures.

In anxiety-like measures, although not significant, some differences were observed between the groups. EMS animals presented increased latencies before consuming rewards but, conversely, augmented mSAT score, frequency and cumulative duration in alley 4. These results, suggest that while EMS subjects had greater latency to eat the first reward and reward number 4, they visited less frequently and spent less time in the more anxiety-inducing alley of the mSAT.

The absence of conclusive findings may be attributed to several factors, such as the nature of the data, the design chosen and sex differences.

The notable variability within the data (particularly the cohort effects), identified early in the statistical analysis and the reduced sample sizes in the groups may have interfered with the analysis. High variability makes it challenging to detect actual effects, especially with smaller sample sizes, contributing to an increased risk of type II errors where genuine effects may go unnoticed due to limited statistical power.

Another point that could have contributed to the lack of statistical results is that the ADT model may not have been sensitive enough to observe the differences between the conditions. In this respect, Chafee (2022) obtained similar results with male subjects exposed to the same EMS and ADT protocols. Indeed, the ADT is still a new model that needs to be tested in several different contexts.

Moreover, it is crucial to consider the sex dimension in this analysis. Many studies validating the effectiveness of EMS involve only male animals. This potential oversight may not adequately account for sex-specific differences in the neurodevelopmental responses of the animals submitted to early stress and its posterior manifestations. It underscores the importance of acknowledging and exploring potential variations that could contribute to the nuanced outcomes encountered in the present study.

In fact, evidence suggests that the effects of EMS on the brain and behaviour may be sex dependent. Arnold & Siviy (2002) found that separation increased activity in a novel open field in female rats, while Rees (2006) reported a lower corticosterone stress response in juvenile female rats. Nonetheless, neither study found a significant impact on behaviour, pointing to the complex interplay of factors influencing the outcomes of early maternal separation.

The outcomes observed in this experiment using the Early Maternal Separation (EMS) model may be influenced by methodological factors specific to this study's design. Unlike standard protocols where separation typically begins on postnatal day (PND) 2, separation in this study commenced on PND 4. This change was made to coincide with the sensitive hypothalamic-pituitary-adrenal (HPA) response period (SHRP), which starts only in PND 4 and extends until PND 14, as discussed earlier in this chapter. This adjustment adheres to the principles of animal welfare outlined by Russell & Burch (1959), aiming to minimise significant and unnecessary harm to developing animals. Additionally, no evidence affirms that a delay in separation reduces the likelihood of inducing symptoms in the subjects.

Additionally, experimental manipulations may have been insufficient to produce a strong phenotype. Evidence suggests that the impact of EMS on rodent behaviour in adulthood largely depends on maternal behaviour immediately before and after the separation period (Schmidt et al., 2011), however, there is no consensus in how and how much these differences can impact the litter behaviour later in life.

Furthermore, patterns of maternal care exhibit significant variability among strains and even among individual dams (Schmidt et al., 2011). This variability introduces an additional factor influencing the effects of EMS in adulthood. Standard protocols, including the one used in this experiment, do not account for this variability. It is important to note that the effects of EMS in the present experiment were assessed in adulthood, weeks after the EMS procedure ended. This means that events occurring during the intervening period inevitably influence the long-term outcomes of the paradigm.

While the lack of statistical significance requires caution in drawing firm conclusions, the consistent trend observed in some of the measures, particularly depression-like ones, hints at the potential effects of EMS exposure on animal behaviours.

4.4.2- Psilocybin

The second premise of this experiment suggested that animals treated with psilocybin doses would show reduced anxiety and depression-like behaviours compared to those in the saline groups, with a more pronounced difference expected in EMS animals. Indeed, on the injection days, animals that received 8 mg or 16 mg/kg doses of psilocybin consistently exhibited reduced locomotor activity and flat body posture (FBP) (data not shown). In contrast, animals in the control groups exhibited normal behaviour (Beer, 2021).

However, the statistical analysis of tests in the ADT did not find significant effects of psilocybin compared to the saline-treated groups in both the anticipatory box and the mSAT assays. Nonetheless, it is essential to note that trends among variables were observed. Specifically, a pattern emerged in 4 out of 5 depression-like measures analysed. EMS groups presented a more apparent separation between treated and non-treated animals. Groups receiving psilocybin (8 or 16 mg/kg) exhibited more ambulatory behaviour (distance moved and velocity) and rears (free and total but not wall) in almost all phases of the test compared to saline groups. Additionally, EMS 8 mg/kg animals walked more and reared more (free and total but not wall rears) compared to animals injected with saline or 16 mg/kg and compared to all the groups in the control condition in the acute phase of the test, even when they presented closer scores in the baseline phase.

The fact that EMS animals administered with 8 mg/kg showed higher ambulatory and rearing behaviour in the acute phase compared to other groups, despite having similar baseline scores, suggests a differential response to the drug dosage, with the 8 mg/kg group exhibiting heightened behavioural effects compared to the 16 mg/kg, saline and all the three control groups. Ultimately, this can suggest a ceiling effect, with 8 mg/kg being an optimal dosage to elicit the expected results. This assumption, although important, must be taken carefully since it represents only a trend in the results without robust statistical support.

Drawing conclusions about anxiety-like measures from the mSAT is a little more challenging due to the diverse patterns observed. For instance, while critical measures in the study, namely mSAT score and frequency in alley 4, revealed different tendencies between baseline and acute effects, there is a general increase towards the chronic phase. In contrast, latencies to eat the rewards generally demonstrated a steady decrease from baseline to acute and from acute to chronic phases.

Additionally, an interesting trend reversal occurred between treatments in the acute phase. Subjects treated with 8 mg/kg exhibited a visible increase in ambulatory measures, visits to alley 4, cumulative duration in alley 4 and mSAT score for both conditions. In contrast, for 16 mg/kg subjects, generally, a decrease was seen in the same period for the same measures, except for the cumulative duration EMS group and the mSAT score control group. Saline groups present a more blunted increase or stability.

This finding, although not statistically significant, again points to a difference in the effects of the different doses in the acute phase.

The complex pattern in the results may indicate that multiple factors could have influenced the animals' anxiety-related behaviour, which evolved throughout the experiment, not only the treatment. Once more, it is essential to note that characteristics inherent in the variability of the data set could have played a role in influencing the attainment of statistically significant results.

Possible explanations for these results include the fact that the EMS protocol did not create the expected phenotype to clearly show the effects of the doses utilised, as discussed in the previous section. Second, the chosen doses and timing could not have been adequate to provoke the expected effects, and third, the use of female subjects, impacted in a lesser degree by psychedelics.

Different results were reported in terms of dosage. Wojtas et al. (2022) observed no behavioural effects at doses of 2 and 10mg/kg in various tests, including the OFT, light-dark box, and FST, even though there was an increase in dopamine, serotonin, glutamate and GABA extracellular levels in the frontal cortex. These mixed results again highlight the complexity and often contradictory nature of research in this area. As already discussed, it remains unclear how different dosages of psilocybin interact with several factors.

Finally, concerning sex, the underrepresentation of female subjects in research studies introduces a notable gap in our understanding, particularly concerning sex-specific responses that could elucidate distinct mechanisms of action or behavioural effects of drugs. If the general research scenario is mainly built with male subjects, the lack of studies with females makes the scarce literature even more inconsistent. Such oversight is critical in studying substances like psilocybin, where the complexity of its effects and the nuanced nature of behavioural responses across different settings demand a comprehensive approach that includes both sexes.

Sex disparity potentially complicates the assessment of drug effects in traditional animal models based on exploratory behaviour. It also challenges the assumption that both sexes will respond similarly to experimental variables, highlighting the variability in how such variables may influence different sexes. In line with these findings, sex differences were seen in response to psilocybin treatment in mice Alper et al. (2023) and in rats Tylš et al. (2016), with females being less affected by the effects of the drugs than male subjects. However, female hormones may play a role in the differences found (Tylš et al., 2016; Bubeníková et al., 2005; Páleníček et al., 2010).

The existing yet limited and contradictory literature on high doses of psilocybin's effects in preclinical trials complicates the contextualisation of its impact on anxiety and depression-like symptoms observed in this study. While specific studies have reported positive effects in animal models using tests like the OFT, FST and EPM (Jones et al., 2020; Hibicke et al., 2020), others have documented contrary findings in similar assessments (Risca, 2021; Horsley, 2018; Wojtas et al., 2022). Even studies using the same animal model, such as FST, have reported varied outcomes. This scenario reinforces the necessity for more studies with different designs.

Concerning the factor time, the statistical evaluations consistently demonstrated that it was the only one exerting a significant effect across nearly all measures, with clear and significant differences comparing the phases of the study. Post hoc analysis indicates that these differences were more marked when comparing chronic and the other two phases.

Except for the cumulative duration in alley 4, all 11 measures presented significant results for this factor. By examining this factor independently and comparing the baseline and/or acute phase with the chronic phase, an increase in ambulatory parameters (i.e., distance moved and velocity) in both apparatuses, an enhancement in vertical behaviour (total, wall and free rears), an increase in the number of visits in the more anxiogenic area of the mSAT (alley 4), a higher mSAT score and reduced latencies to eat rewards could generally be seen.

Specifically talking about anticipatory behaviour, the results of the factor time suggest that ADT is effective in measuring anticipatory pleasure, as seen in previous studies (Chafee, 2022; Blackburn, 2021). Following five consecutive days of baseline when the animals associated being placed in the box with the posterior reward offered in the mSAT, their anticipatory behaviours increased consistently throughout the study. In other words, through the temporal association of imminent access to a reward (i.e., Froot Loops) with the contextual stimuli of the box, the ADT effectively triggered anticipatory behaviour in the context of the present study.

Overall, the results of this experiment demonstrate no significant effects of EMS procedures or treatment with doses of 8 and 16 mg/kg of psilocybin, as measured by the ADT. However, trends across all measures in the anticipatory box suggest some effects from early life stress and from both doses of the drug, with more pronounced effects of 8 mg/kg in depression-like measures.

Although these findings do not support our initial hypothesis, they are consistent with some aspects of the existing literature on psilocybin research. This study sheds light on critical variables such as dosage, animal models, and the drug's effects on female subjects. However, other factors, such as the potential influence of oestrous cycles on the results, remain unclear. The insights gained from this research help to fill a significant gap in the current literature and advocate for the use a broader range of research methods that capture diverse aspects of the phenomenon. A comprehensive approach is essential for understanding the complex dynamics of psychedelic studies and ensuring that the findings are relevant and applicable. Ultimately, this research enhances our understanding of psilocybin's effects on behaviour indicating new directions for investigation.

Chapter 5 – Effects of Doses of 4 and 8 mg/kg of Psilocybin on Anxiety and Depression-like Symptoms and Impacts of the Oestrus Cycle

This chapter describes the behavioural effects of psilocybin (4 mg/kg and 8 mg/kg) using the ADT model and heterozygous female SERT knockout rats. Initially, the analysis considered only the factors of genetic condition and treatment. However, for a secondary analysis, the rats were divided based on their oestrous cycle phase at the injection time. As this leads to a relatively small group, the analysis should be regarded as a pilot study rather than a full investigation of the influence of the oestrous cycle on the behavioural effects of psilocybin.

The following sections will outline the theoretical basis for the models used and describe the methodologies used. The main findings of the principal analysis will then be presented, followed by a discussion of the results. Finally, a subchapter will be presented with results and discussion, including the oestrus factor.

5.1- Theoretical Framework

5.1.1- Animal Models

The present study utilised SERT knockout rats to induce anxiety and depression-like symptoms and the affective disorders test (ADT) to assess these symptoms. Additionally, microscopic cytology was utilised to determine the oestrous cycle of the subjects.

5.1.1.1- SERT knockout rats

A common genetic variation linked to the symptomatology of mental disorders is the 5-HTTLPR (5-HT transporter-linked polymorphic region), a polymorphism in the serotonin transporter gene *SLC6A4*. This mutation involves a 44-base pair deletion/insertion in the promoter region of the serotonin transporter gene. The result can be a short (S) or long (L) form of the gene, respectively. The 50% reduction in SERT activity in the homozygous S genotype compared to the homozygous L genotype increases extracellular serotonin levels in the synaptic cleft during neurodevelopment (Nakamura et al., 2000).

Several studies have linked the S variant of the 5-HTTLPR to increased stress sensitivity in the development phase, leading to a propensity for depression and anxiety not only in humans but also in non-human primates, as demonstrated by Nakamura et al. (2010) in their review. Although this mutation does not occur naturally in rodents, genetic manipulation techniques make it possible to reproduce this variation, which was created using ENU mutagenesis (Smits et al., 2006).

Comparing homozygous ($^{-/-}$) and heterozygous ($^{+/-}$) SERT knockout rats with wild types ($^{+/+}$), Homberg et al. (2007) found a gene dose-dependent reduction in SERT levels, with homozygous knockouts showing a 100% reduction and heterozygous showing a 50% reduction when compared to wild-type rats (thus roughly equivalent to humans with the s allele of the 5-HTTLPR). They also found a nine-fold increase in extracellular serotonin concentrations in the hippocampal region of SERT$^{-/-}$ rats. However, there were no compensatory changes in tryptophan hydroxylase (the enzyme responsible for serotonin synthesis) or monoamine oxidase (the enzyme responsible for serotonin metabolism). Furthermore, there were no significant changes in other monoamines, particularly dopamine and noradrenaline.

Additionally, they found that the SERT$^{+/-}$ rat exhibits intermediate phenotypes concerning SERT mRNA levels, SERT protein density, and SERT function. However, despite these findings, SERT $^{+/-}$ rats do not differ from wild-type animals regarding 5-HT tissue levels and the reactivity of serotonergic neurons. This fact aligns with observations in SERT$^{+/-}$ mice, where no changes were found in intracellular 5-HT levels. (Homberg et. al 2007; Kim et al., 2005).

Although the exact mechanisms involved in this process are not fully understood, the observed phenotypes could be linked to the abnormally absent serotonin transporter (i.e., SERT) and, consequently, high serotonin levels in the synapses during development. This may lead to a reduction in both the number of serotonergic neurons and their firing rates in the dorsal raphe nucleus (Lira et al., 2003; Golebiowska et al., 2019; Krishnan & Nestler, 2011).

In humans, several studies connected the S variant of 5-HTTLPR to stress sensitivity and, either by itself or in combination with early stressful life events, to a propensity for depression and anxiety, as demonstrated by Caspi et al. (2010). Parallelly, in pre-clinical trials, SERT Knockout rats developed through an ENU mutagenesis technique show increased depression and anxiety-like (Olivier et al., 2008). However, most of these studies used homozygous SERT rather than heterozygous knockout rats. All these findings strongly support the validity of this strain as a model to investigate anxiety and depression-like symptomatology.

It is essential to say that in the laboratory, homozygous SERT knockout rats with a complete deletion of the SERT protein are expected to manifest severe disruption of serotonergic neurotransmission, which is not seen in humans. This will result in a phenotype markedly different from the symptoms observed in humans with the naturally occurring SERT gene polymorphism.

Additionally, the evidence of the validity of SERT knock rats as a valid model for depression and anxiety in behavioural analysis is predominantly based on studies in male (homozygous) SERT knockout rats. This approach overlooks the potential interplay between female hormones and alterations in the serotonergic system and neglects the effects of partial deletion of the SERT protein.

This is certainly a gap that needs to be better explored. Joeyen-Waldorf (2009) found no significant behavioural differences between female heterozygous and wild-type mice in the open field test and elevated plus maze, while Roversi (2020) tested males and found significant differences in the EPM and in the sucrose preference test compared to $SERT^{+/-}$ and wild types. Olivier (2007), comparing $SERT^{-/-}$ and wild types in depression and anxiety models, found significant genotype differences in the behaviour of males and females in all the paradigms tested except for the EPM, with no significant interactions between genotype and sex.

These examples illustrate the need for more comprehensive research, particularly using females with a partial reduction of SERT protein, to represent better human conditions related to anxiety and depression prevalence. Considering this scenario, the present experiment testing female subjects included SERT-/+ and wild-type rats (Macchi et al., 2013).

5.1.1.2- Affective Disorders Test

In the present experiment, the ADT, as described in Chapter 5, was again used to assess anxiety and depression-like symptoms (Kidwell, 2019; Blackburne, 2021). However, one additional measure, ultrasonic vocalisations (USVs), will be explored in the anticipatory box. Additionally, the procedure underwent a few adjustments (described in the specific methods).

5.1.1.2.1- Ultrasonic Vocalization

As mentioned before, analysing the ultrasonic vocalisations (USVs) emitted by animals during trials offers a non-invasive method for investigating rodents' emotional states, which can be classified as positive or negative, in response to different environments (Buck et al., 2014; Brudzynski, 2013). This measure can be easily incorporated into a wide range of tests.

Just as locomotion and rearing behaviour tend to increase in response to the anticipation of a reward (Barbano & Cador, 2007), it is expected that, in typical circumstances, 'happy calls' counts will consistently increase due to the prediction of an incoming reward. Conversely, a more blunted or absent increase in these calls could indicate reduced anticipatory pleasure (Boulanger-Bertolus & Mouly, 2021). A noticeable decline in anticipatory pleasure is a crucial aspect of anhedonia, which is a core symptom of depression and anxiety in humans. USVs can, therefore, be a potent instrument in pre-clinical studies targeting anxiety and depression-like behaviours.

5.1.2- Oestrous Cycle

The determination of the oestrous cycle can find valuable application in pre-clinical research to evaluate, for example, the reproductive availability of females or to have an index of the functional status of the neurological regions related to the reproductive system (Cora et al., 2015). Another important application is to evaluate the potential interactions between female hormones and the drugs being tested once there is considerable variation of hormone levels in the phases of the cycle.

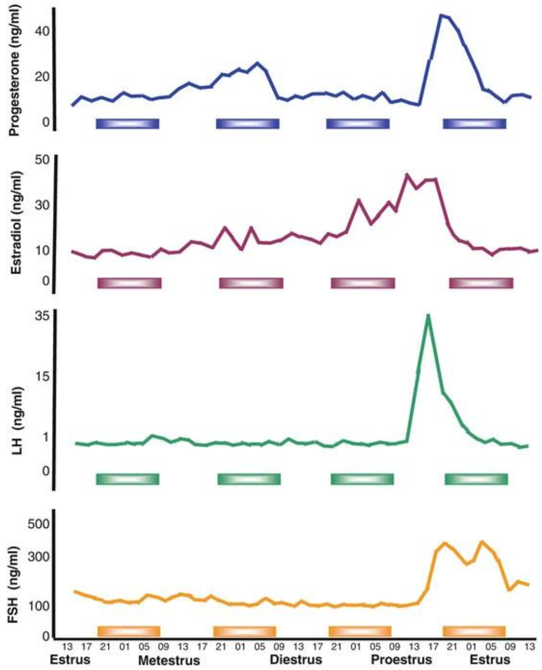

Figure 5. 1- Hormone concentrations (progesterone, oestradiol, luteinising hormone—LH, and follicle-stimulating hormone—FSH) across the estrous cycle of rats (Antunes et al., 2016).

There is a considerable variety of methods and techniques to determine the stage of the oestrous cycle in rodents, including the examination of the cytological morphology of vaginal smears under a microscope. This analysis primarily focuses on the proportions of various cell types within these smears to accurately identify each cycle phase (Byers, 2012). Building on this foundational understanding, Palenicek et al. (2010) suggest a classification with four stages based on the level of hormones and proportion of cells found in the smears. The phases are proestrus and oestrus, with more elevated levels of female hormones and dioestrus and metoestrus phases, with lower levels, as seen in the figures below. To simplify this classification, two categories can be created with high levels of hormones (i.e., PE) and with lower levers (i.e., DM)

The types of cells and their proportion in each phase of the cycle, represented in the figures below, generally is: 1- In the proestrus stage, mostly nucleated and some cornified epithelial cells are present in vaginal smears. Few leukocytes may be present at the beginning of this stage. 2- In Oestrus - Cornified epithelial cells are predominant in the sample. 3- Metestrus is when the corpora lutea forms but does not fully luteinise due to a lack of progesterone. Cornified epithelial cells and polymorphonuclear leukocytes are seen in the samples. Some nucleated epithelial cells will also be present in late metoestrus. 4- Finally, dioestrus predominates in the polymorphonuclear leukocytes and a few epithelial cells during late dioestrus. Leukocytes remain the predominant cell type after removing cellular debris, as seen in Figure 5.2. (Byers et al., 2012).

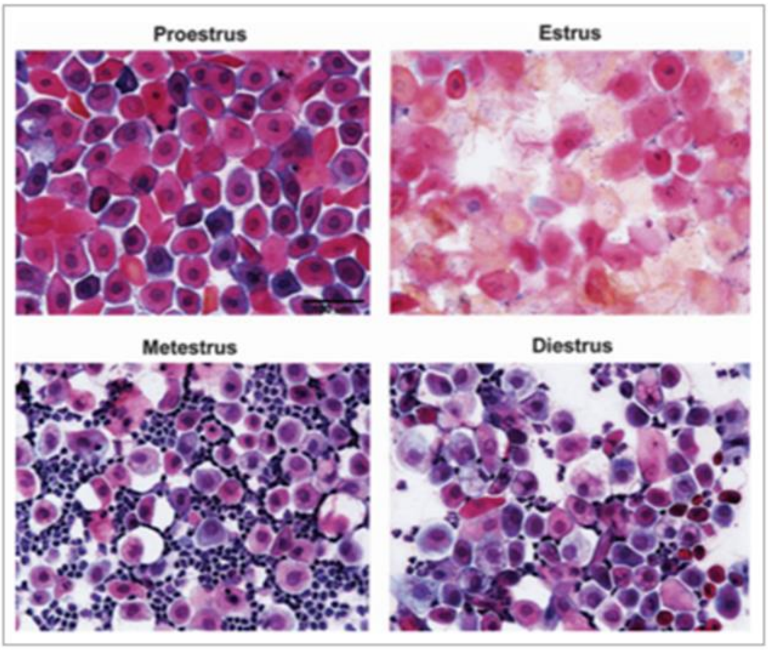

Figure 5.2- Kind of cells in the four stages of cycle. Adapted from (Hubscher et. al., 2005).

The goal of using oestrous determination in this study is to differentiate between the cycle's phases marked by high and low hormone levels. The classification method described earlier simplifies this process, focusing on a critical feature when analysing the samples under the microscope: a high proportion of leukocytes.

Figure 5.3 represents the distribution of distinct kinds of cells in the distinct stages of the cycle. One evident feature is the presence of leukocytes in the stages of lower hormone levels, metoestrus and dioestrus.

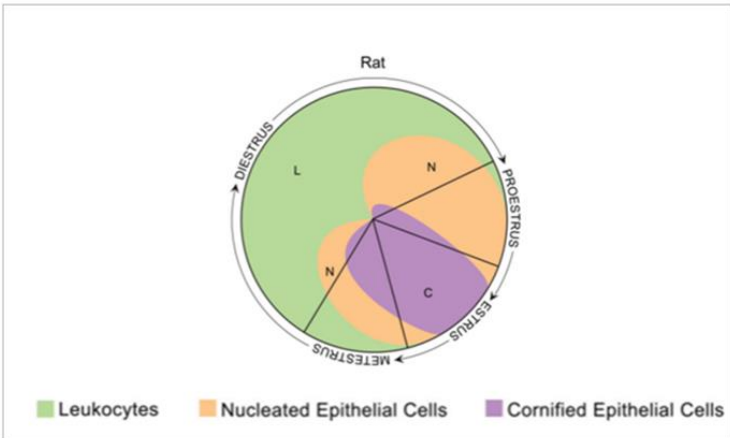

Figure 5. 3- Cell type distribution across oestrous cycle phases. Adapted from Cora et al. (2015).

5.1.3- Doses

Still considering the rationale from our previous experiment and informed by findings from Meinhardt et al. (2020), Jefsen (2021), Gallaher et al. (1993), and Klein et al. (2018), doses of 4 and 8 mg/kg were selected for this experiment. This decision was based on prior results suggesting that 8 mg/kg was more effective, whereas 16 mg/kg did not yield discernible benefits.

Continuing investigating the effects of high doses of psilocybin in anxiety and depression-like symptoms, this experiment used the ADT and heterozygous SERT knockout animals. Additionally, to evaluate if the phase of the oestrus cycle would interfere in the results, microscopy was used to determine in which phase the animals were injected.

It was hypothesised that at baseline, SERT Knockout (KO) animals will exhibit significantly more anxiety and depression-like behaviours compared to wild-type groups. Both doses of psilocybin (4 and 8 mg/kg) will significantly reduce these symptoms compared to saline administration in a dose-dependent manner, as measured by ADT outcomes. Finally, higher hormone levels will attenuate the effect of psilocybin.

Prediction 1 – At baseline, compared to controls, SERT KO animals will exhibit, in a dose-dependent manner, reduced anticipatory pleasure, as indicated by decreased distance moved, increased velocity, reduced total, free, and wall rear counts, and a reduced number of happy calls in the anticipatory box. They will also exhibit heightened anxiety-like behaviours, reflected in the modified successive alleys test by reduced distance moved, increased velocity, lower frequency and cumulative duration in alley 4, increased latency to consume the first reward and reward number 4, and lower mSAT scores.

Prediction 2 – Both doses of psilocybin (4 and 8 mg/kg) will increase, in a dose-dependent manner, anticipatory behaviours, as indicated by the increased distance moved, decreased velocity, and increased total, free, and wall rear counts and happy calls in the anticipatory box. Additionally, psilocybin will reduce anxiety-like behaviours in SERT KO animals, as reflected in increased distance moved, lower velocity, increased frequency and cumulative duration in alley 4, decreased latency to consume the first reward and reward number 4, and higher mSAT scores at both acute and chronic testing time points.

Prediction 3 – Animals tested during the lower hormone phases of the cycle (Diestrus and metestrus phases) will exhibit a greater reduction in depression- and anxiety-like symptoms following psilocybin administration when compared to animals injected in the high hormone level phases (proestrus and oestrous phases).

5.2- Specific Methods

Building on the general methods described previously and in specific methods of experiment 2 (ADT), this section presents the specific procedures used in this experiment.

5.2.1- Animals

Wistar SERT knockout heterozygous males were paired with wild-type females in four consecutive cohorts, each separated by approximately 50 days. This choice of pairing SERT$^{+/-}$ males with wild-type (WT) females stemmed from the consideration that the absence or reduction of the serotonin transporter might potentially impact maternal behaviour, thereby introducing an uncontrolled and novel variable into the study. Moreover, this specific combination would produce only heterozygous and wild-type offspring, so all (female) animals could be used for the experiments.

Four or five couples were paired for each cohort, producing litters ranging from 5 to 18 animals with two possible genotypes: heterozygous or wild-type animals.

After weaning the animals at PND21, ear tissue samples were collected for in-house genotyping. For each cohort, groups with heterozygous and wild-type females were formed, ranging in size from 13 to 21 subjects. This resulted in 60 animals for the complete experiment (31 heterozygous and 29 wild types), as seen in Table 5.1.

Table 5. 1- Distribution of Animals in the Group

Condition/ Treatment	Saline	4 mg/kg	8 mg/kg	Total
SERT $^{+/-}$	9	11	11	31
Wild type	7	11	11	29
Total	16	22	22	60

Note. SERT = Serotonin Transporter (Protein). +/- = Heterozygous.

The chosen animals were housed in standard polycarbonate cages in groups of three or four animals per cage (of the same genotype) with water and food ad libitum until PND 48-52, when preparation for the test began. Then, the animals started the food deprivation regimen as described previously.

5.2.2- Procedures

5.2.2.1- Affective Disorders Test

The same apparatus and procedures described in Chapter 4 were used in this experiment, with only minor differences described below.

Before the beginning of the tests, a protocol for vaginal sample collection started, happening every two days alternately with the weighting days. The time for the habituation to anticipatory box trial was also reduced from 10 to 5 minutes. Finally, the testing days were altered to four days for baseline, four days for acute effects, seven days for interval and four days for chronic effects, accompanying the regular duration of the oestrous cycle.

Additionally, since no significant differences were found between the two volumes of saline injected in the previous experiment, only the low volume was injected in the present study.

The timeline of the experiment is shown in the table below.

Table 5.2- Experiment Chronogram

Preparation			Pre-test				Test					
			1	2-3	4-5	6-7	8-10	11-14	15	16-19	20-27	28-31
Day of birth	Weaning	Geno typing			Food deprivation							
						Vaginal smears collecting (alternate days)						
			Weighing	Interval	Handling	Interval	Habituation antic. box	ADT baseline	Injection	ADT acute effects	Interval	ADT chronic effects

Note. The numbers refer to the days when the specific activities, described below them, occurred. Weaning happened in PND 21. Genotyping happened around PND 25. Ant. = Anticipatory. ADT = affective disorders test.

The measures acquired from the model are described in Table 5.3

Table 5.3- Measures of the ADT

Measures	Definition
Depression-like measures (Anticipatory Box)	
Box distance moved	Total of centimetres travelled during the session
Box velocity	Average of centimetres/second during the session
Free rears	Standing on the hind legs without touching the wall. One rear is finalised when four paws touch the floor
Wall rears	Standing on the hind legs using the wall as support. One rear is finalised when four paws touch the floor
Total rears	The sum of wall and free rears
USVs [a]	The ratio of happy calls (above 30 kHz) happy-sad/happy + sad
Anxiety Like measures (mSAT)	
Alleys distance moved	Total of centimetres travelled during the session.
Alleys velocity	Average of centimetres/second during the session
Frequency in alley 4	Number of visits in alley 4
Cumulative duration in alley 4	Total time spent in alley 4
Latency to eat 1st FL (any alley)	Time until eating the first reward (in any alley)
Latency to eat FL #4	Time until eating the reward located in alley 4
mSAT	The sum of the product of each reward's value (ranging from 1 to 4) and the value assigned to its consumption location (also from 1 to 4). This calculation is represented as mSAT = (reward 1 value × location 1 value) + (reward 2 value × location 2 value) + (reward 3 value × location 3 value) + (reward 4 value × location 4 value). Ex. If the subject ate rewards placed in alleys #1, #2 and #3 in alley 1 and reward #4 in alley 2, its score would be: =SUM (1*1) +(2*1) +(3*1) +(4*2) = 14

Note. USVs = Ultrasonic vocalizations. mSAT = modified successive alleys test. FL = Froot loops (half of a piece). a - Considering the massive volume of data collected in the 900 trials in the anticipatory box (4500 minutes of recordings), only USVs collected in the trials of four time points, considered crucial, were included in the analysis: the first and last day of baseline, the first day of the acute phase, and the first day of chronic phase.

5.2.2.2- Vaginal Samples Collection

The determination of the phase of the cycle was made through microscopic analysis of vaginal smears (Cora et al., 2015; Mohammed & Sundaram, 2018).

Five collections were conducted at the end of the day, before feeding time, starting from day 6 of the interval period until the last day of the baseline. On the morning of the test day, a sixth collection was performed at least two hours before drug administration. This series of collections allowed a prediction about the animals' cycle phase on the day of injection and facilitated habituation to the procedure, thereby reducing stress on the injection day.

The vaginal samples were collected using a swab wetted with ambient temperature physiological saline onto the vagina of the animal. The collected material was transferred to a dry glass slide air-dried, stained with crystal violet, and analysed under the microscope (Olympus BX-51 20x magnification) (Ajayi A.*F*., 2020) to determine the oestrus phase.

As stated before, considering the prevalence of leukocytes in the smears, after analysis, the samples will be classified in either PE (when proestrus or oestrus were identified) or DM (when Diestrus or metoestrus were identified).

5.2.3- Statistical Analysis

Initially, an analysis was conducted to address missing data. Little's Mcar test was performed with the complete data frame returning a result of X2 (182) = .000 Sig = 1.000, demonstrating that the missing values have a random distribution and, consequently, the further analysis will not be biased by the pattern of missingness.

After the preliminary analyses and to reduce the volume of data, the values of each phase (i.e., baseline, acute, and chronic) were averaged, resulting in a single value for each time point. Values for the first two days of baseline were not included to reflect more stable and consistent behavioural patterns after acclimatisation to the two apparatuses.

Secondly, repeated-measures ANOVAs with the cohort as a between-subject factor were conducted to assess cohort effects. The results indicated that all other measures showed significant differences among the cohorts except for the measures of total, wall and free rears, latency to eat FL #4 and mSAT Score and USVs. However, it is noteworthy that these differences were consistent across all cohort comparisons, suggesting that no specific cohort stood out from the others. Moreover, due to the relatively small number of subjects within each group in each cohort, it was, nonetheless, decided to combine the cohorts to enhance statistical power.

After that, Shapiro-Wilk tests and visual examination of Q-Q plots and histograms of residuals for each measure and time point returned violation of normality assumption for several measures. Except for cumulative duration in alley 4, all measures returned $p < 0.05$ for at least one of the time points, although most graphs did not present massive distortions. Homogeneity of variances, evaluated with Levene's test, returned violation in at least one time point of 8 out of 13 measures (except for total rears, wall rears, free rears, alleys distance moved and frequency in alley 4). However, the same decision taken for experiment 2 was made for this set of data, which is not performing selective transformation in the data as it can lead to inconsistencies between transformed and non-transformed data in terms of interpretation of the results, especially because different methods of transformation would be necessary. This way, we chose not to transform idiosyncratically to preserve consistency in measurement metrics. Instead, we analysed all data untransformed data.

Then, One-Way ANOVA was performed for each measure to evaluate differences between conditions (SERT$^{+/-}$ and Wild Types) during the baseline phase for each measure. The results will be discussed in the next section.

The primary analysis was conducted using repeated measures ANOVA, which examined the interactions among two factors and three time points in a matrix 2 x 3 (table below). Factor condition encompasses the SERT$^{+/-}$ and wild type groups; the factor treatment, with the two groups treated with psilocybin (i.e., 4 mg/kg and 8 mg/kg) and the saline group. The timepoints are the test phases: baseline (before injection) and acute and chronic phases. The USVs measure exceptionally has four time points.

Table 5. 2- Anova Components

	Factor	Level
Within subjects	Time	Baseline, acute and chronic
Between subjects	Condition	SERT$^{+/-}$ and wild types
Between subjects	Treatment	4 mg/kg, 8 mg/kg and saline
Between subjects	Oestrous phase (second analysis)	PE and DM

Note. SERT = Serotonin Transporter (Protein). +/- = Heterozygous. DM = Group Oestrous phase dioestrus and metestrus phases. EP = Group Oestrous phase proestrus and oestrous phases.

When necessary, post hoc evaluations for main effects and interaction were made using Holm-Bonferroni correction to address the heightened risk of Type I errors (i.e., incorrectly rejecting the null hypothesis) due to multiple pairwise comparisons. Complete tables with ANOVA results for main effects and interactions are in Appendix 3B.

Finally, for the subchapter including the oestrus factor, the repeated-measures ANOVA was made with a 3 x 3 matrix having the same three time points and condition, treatment and oestrus classification as factors. Complete tables with ANOVA results for main effects and interactions are in Appendix 3C.

5.3- Results

This experiment aims to understand the effects of 4 and 8 mg/kg doses of psilocybin on anxiety and depression-like symptoms in heterozygous SERT knockout and wild-type rats. The symptoms were assessed using the ADT.

The first hypothesis of this study was that the animals in the condition SERT $^{+/-}$ would exhibit significantly more anxiety and depression-like behaviours compared to the wild-type group. The second hypothesis posited that psilocybin treatment would result in a significant reduction in anxiety and depression-like symptoms compared to saline administration. The third hypothesis, presented in a separate analysis, is that phases of the oestrus cycle would interfere with the effects of psilocybin.

The results will be presented in detail, starting with condition analysis, the effects of psilocybin on depression-like measures and anxiety-like measures, and then the results comparing time points. After that, the results and discussion of the second analysis with the oestrus cycle will be presented.

5.3.1- Condition: SERT $^{+/-}$ and Wild Type

The comparison between 2 groups of the factor condition (i.e., SERT$^{+/-}$ and Wild Type) shows no differences between groups in the baseline phase (Table 5.3).

Table 5.3- Comparative Analysis between Conditions (SERT $^{+/-}$ and Wild Type) for all the Measures

	F	df	p
Box distance moved - baseline	0.9	(1, 52.67)	0.346
Box velocity - baseline	1.03	(1, 52.83)	0.315
Total rears - baseline	0.03	(1, 57.33)	0.87
Wall rears - baseline	0.01	(1, 57.13)	0.937
Free rears - baseline	0.25	(1, 56.78)	0.621
Alleys distance moved - baseline	1.61	(1, 55.89)	0.21
Alleys velocity - baseline	0.04	(1, 57.98)	0.833
Frequency in alley 4 - baseline	0.1	(1, 57.94)	0.751
Cumulative duration alley 4 - baseline	0.01	(1, 57.58)	0.939
Latency eat 1st - baseline	0.39	(1, 44.65)	0.535
Latency eat #4 - baseline	0.01	(1, 56.7)	0.932
mSAT score - baseline	0.55	(1, 52.77)	0.462
USVs- baseline	0.08	(1, 44.45)	0.783

Note. Note: FL = Froot Loops (half of a piece), mSAT = modified successive alleys test. USVs = ultrasonic vocalizations.

However, although not statistically significant, it is possible to notice in the graphs (figures 5.4 and 5.5) a pattern where SERT$^{+/-}$ animals generally walked faster and had more total and wall rears than those in the wild-type groups, especially in acute and chronic phases.

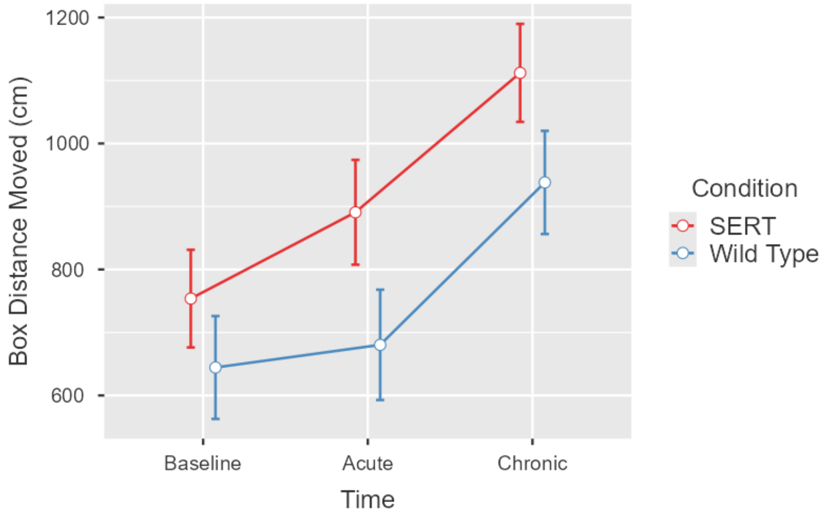

Figure 5. 4- Means and ±SEM for box distance moved across all three time points with lines representing conditions

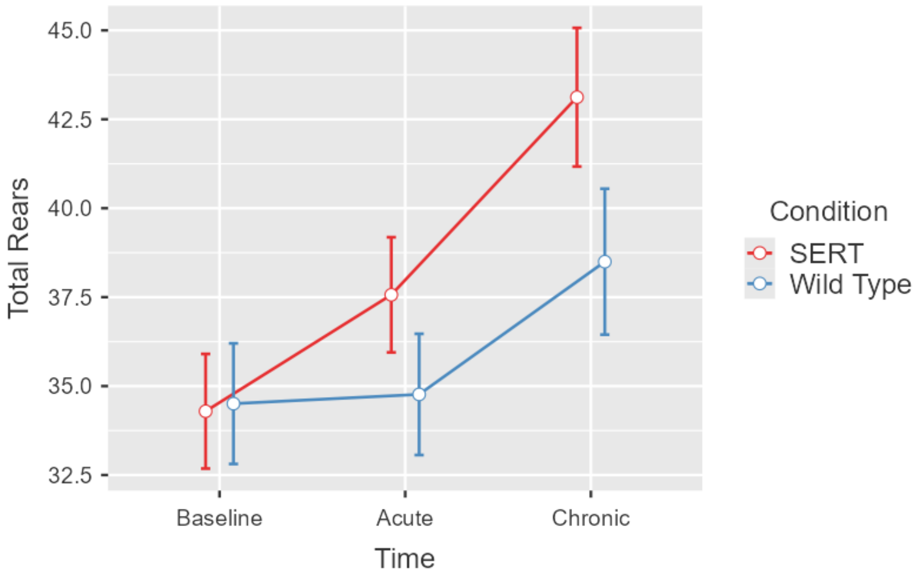

Figure 5. 5- Means and ±SEM for total rears across all three time points, with lines representing conditions

5.3.2- Psilocybin

5.3.2.1- Psilocybin - Depression-like Measures

Repeated measures ANOVA for the main effects and interactions of time, condition, and treatment of the measures acquired in the anticipatory box revealed significant main effects of time in all the measures except for free rears. Post hoc analysis generally showed higher scores for chronic effects compared to baseline and acute phases and for acute compared to baseline. No significant post hoc results were found for USVs.

Table 5. 4 Main Effects of Time in Depression-like Measures

Measure	df	F	p
Box distance moved	(2, 108)	19.193	0.00
Box velocity	(2, 108)	23.45	< .001
Total rears	(2, 108)	19.02	< .001
Wall rears	(2, 108)	34.65	< .001
Free rears	(2, 108)	0.09	0.92
USVs	(3, 69)	4.09	0.01

Interactions between time and condition were observed in wall rears $F(2, 108) = 3.61$, $p = 0.03$ as demonstrated in figure 5.6. Post hoc evaluation pointed out the effect of the time in this measure.

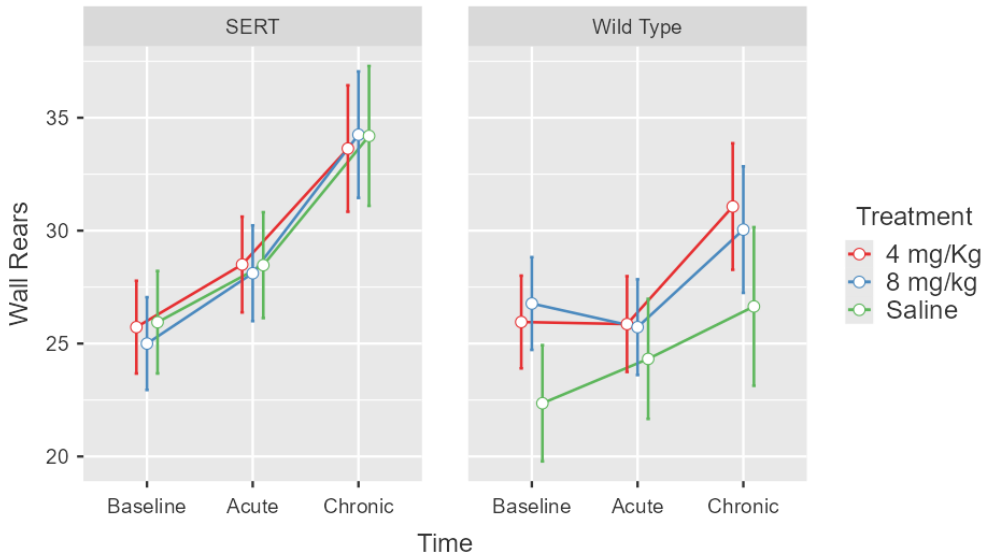

Figure 5. 6- Means and ±SEM for wall rears across all three time points, split by condition and lines representing treatments.

Finally, interaction between condition and treatment was observed in USVs $F(2,23) = 3.37$, $p = 0.05$ (Figure 5.7). However, post hoc evaluations did not return significant differences.

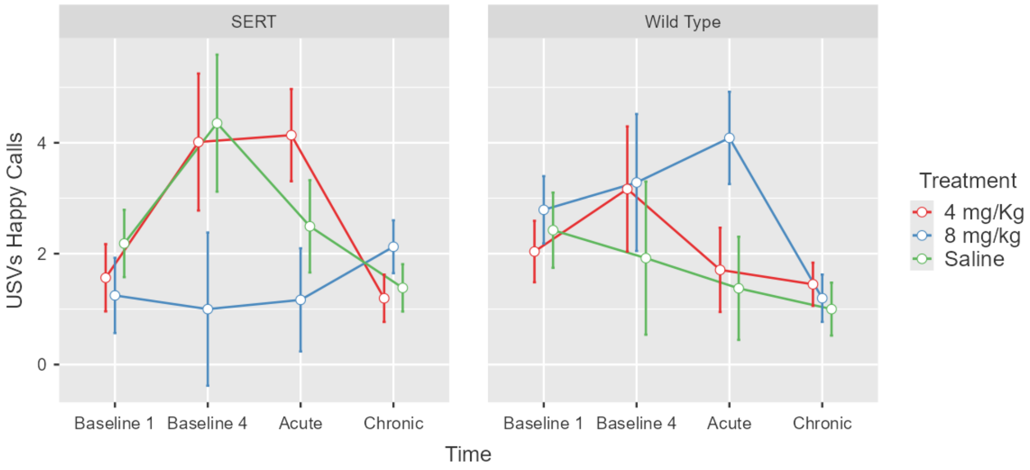

Figure 5. 7- Means and ±SEM for USVs happy calls across all three time points, split by condition and lines representing treatments

5.3.2.2- Psilocybin - Anxiety-like measures

Four out of seven measures taken from the mSAT showed significant main effects of time (Table 5.5). Post hoc analysis shows that time generally has higher scores for the chronic phase. Alleys velocity and mSAT did not reach the significance level.

Table 5. 5- Main Effects of Time in Anxiety-like Measures

Measure	*df*	*F*	*p*
Alleys distance moved	(2, 108)	0.82	0.443
Alleys velocity	(2, 108)	1.34	0.266
Frequency in alley 4	(2, 108)	8.79	< .001
Cumulative duration in alley 4	(2, 108)	15.23	< .001
Latency to eat 1st FL	(2, 108)	16.97	< .001
Latency to eat FL #4	(2, 108)	50.96	< .001
mSAT	(2, 108)	0.82	0.443

Additionally, an interesting pattern was observed in the acute phase of the study. Wild-type animals treated with 8 mg/kg presented a noticeable increase compared to other groups in three measures: alleys distance moved, velocity and mSAT Score as seen in figures 5.8 and 5.9.

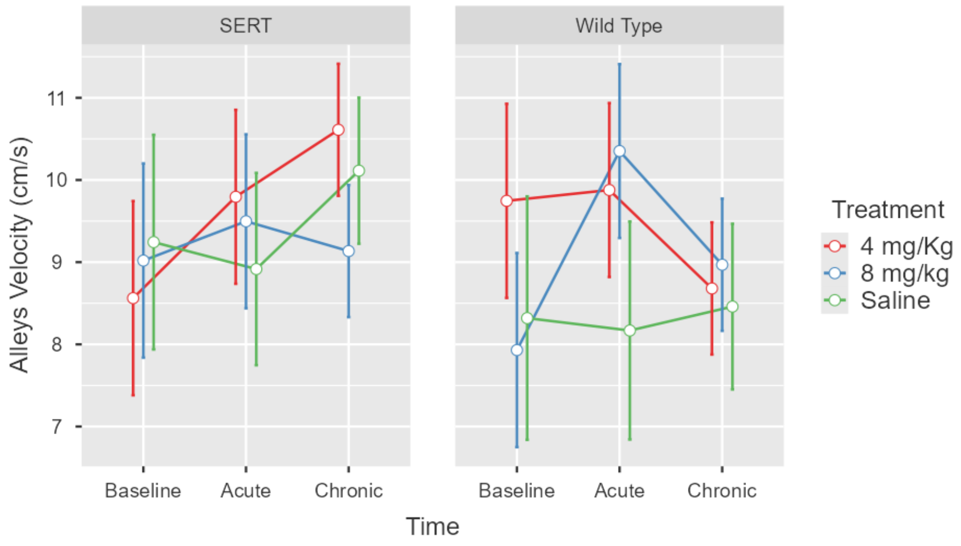

Figure 5. 8- Means and ±SEM for alleys velocity across all three time points, split by condition and lines representing treatments.

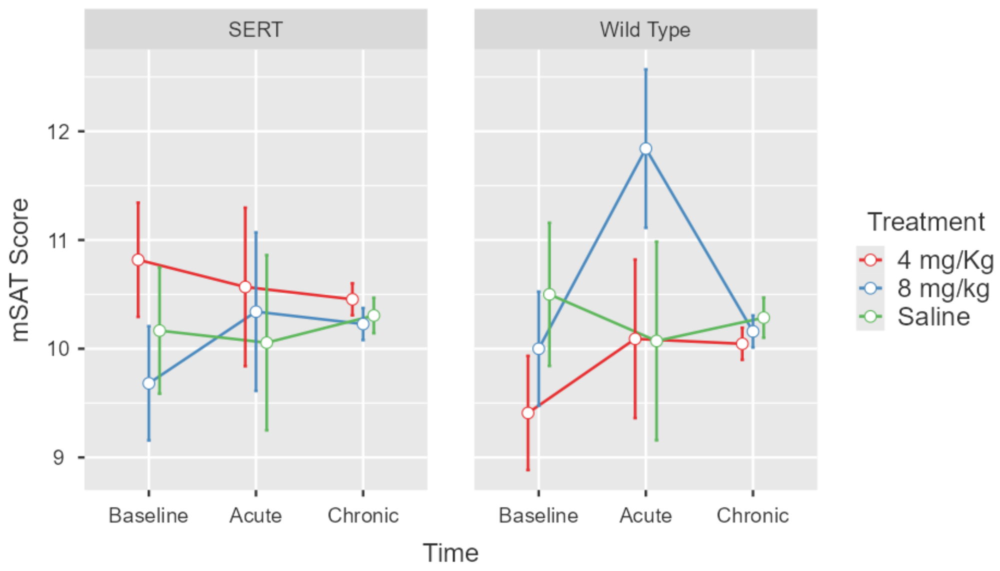

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Figure 5. 8- Means and ±SEM for mSAT score across all three time points, split by condition and lines representing treatments

Moreover, the frequency in alley 4 demonstrated a surge in the acute phase for both doses (4 and 8 mg/kg) in both conditions compared to saline groups (figure 5.10).

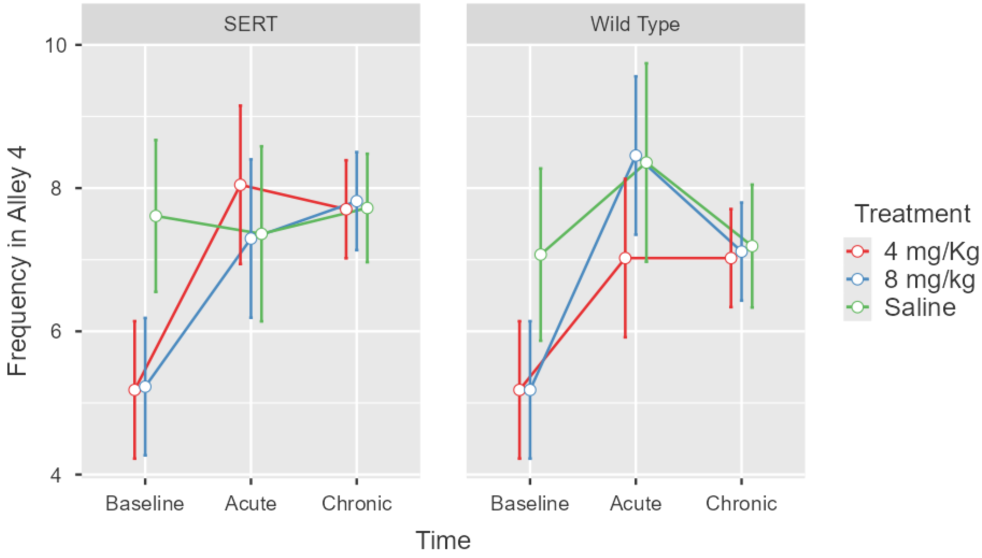

Figure 5.9- Means and ±SEM for frequency in alley 4 across all three time points, split by condition and lines representing treatments

5.4 Discussion

The findings from this study provide significant insights into understanding the behavioural effects of psilocybin. While not all anticipated outcomes were observed, the results underscore the complexity of psilocybin treatment and its varied effects on behaviour. This section will discuss the implications of these findings and juxtapose them with the current body of literature.

5.4.1- SERT Knockout and Wild Types

In baseline it was anticipated that SERT $^{+/-}$ animals, when compared to wild-type counterparts, would exhibit reduced ambulatory activity vertical behaviour, USVs, mSAT score, cumulative duration, and frequency in alley 4. It was also anticipated that there would be increased latencies to consume the first reward and reward #4, indicating higher levels of anxiety and depression-like symptoms compared to wild-type animals. However, as demonstrated in the results section, no measure in this experiment during the baseline phase (i.e., before treatment) showed a significant difference between the two conditions.

These findings could be due to several factors. While the SERT knockout model is a well-established technique for eliciting anxiety and depression-like behaviours, as highlighted by Golebiowska et al. (2019) and Krishnan & Nestler (2011), some evidence suggests that heterozygous animals do not exhibit as pronounced a phenotype as homozygous animal do, resulting in subtler differences that are less detectable in the model employed to assess the symptoms (i.e. ADT). This possibility aligned with findings from previous studies when no significant differences between SERT $^{+/-}$ and WT were found, for example, in critical measures of the ADT, such as cumulative duration in alley 4 and latencies to eat FL in the mSAT of the ADT (Kidwell, 2019), while significant differences were found with homozygous SERT knockouts (Blackburne, 2021). However, as homozygous SERT animals are not representative of the human situation, we choose to use heterozygous animals for this study.

Additionally, results from other tests suggest that heterozygous animals may not always show apparent behavioural differences from their wild-type counterparts in specific assessments. Notably, Homberg's study in 2007 (2) demonstrated that behavioural phenotypes of heterozygous SERT rats were mainly similar to those of wild-type controls across various tests, including the resident–intruder test, the five-choice serial reaction time task, and serial reversal learning, indicating no significant differences between the conditions.

Likewise, investigations by Ellenbroek, August, and Youn (2016) revealed no discernible genotype differences between wild types and SERT$^{+/-}$ rats in tests such as the EPM and the novelty-suppressed feeding task.

Furthermore, most studies validating SERT knockout rats for behavioural analysis used male subjects, overlooking the role of female hormones in serotonergic system alterations. Among the few studies with females, Joeyen-Waldorf (2009) found no significant behavioural differences between female heterozygous and wild-type mice in the OFT and EPM. Similarly, Olivier (2007) observed significant genotypic differences in depression and anxiety models for both sexes, except in the EPM, but no significant interactions between genotype and sex.

In this study, the fact that the condition analysis did not evidence the propensity for anxiety and depression-like behaviours in SERT$^{+/-}$ animals challenges prior assumptions about the SERT knockout model's effectiveness in replicating this specific symptomatology. This discrepancy demonstrates the complexity of genetic models in behavioural neuroscience, which highlights the necessity for continued exploration and validation of these models for a better understanding of the intricate mechanisms underlying anxiety and depression.

It is essential to clarify that the decision to use heterozygous animals in this study rather than homozygous ones was based on three considerations. First, a complete 100% deletion in the SERT protein is associated with a phenotype markedly different from what is observed in humans with the naturally occurring mutation of the SERT gene (Macchi et Al., 2013). Thus, heterozygous animals more accurately reflect the conditions of patients with depression and anxiety. Secondly, previous studies have demonstrated visible and significant differences between SERT$^{+/-}$ and wild-type animals in behavioural models (Bartolomucci, A., 2010; Macchi et al., 2013). However, only a few studies have been conducted with the ADT and SERT knockout rats, and, as mentioned before, few studies have included female subjects. Thirdly, the principle of Reduction in animal welfare (Russell & Burch, 1959) was considered, given that female homozygous pups are very scarce in litters, which would require breeding a significantly larger number of pairs to obtain the necessary number of subjects for the study.

One alternative explanation for the lack of statistical difference between the two conditions is the sensitivity of the ADT measures, which may not be sufficient to detect more nuanced differences. This suggestion is supported by parallel findings in earlier experiments of the present work, where rats subjected to EMS were assessed using the ADT. It is also consistent with Chaffee's (2022) findings, highlighting limitations in the ADT's sensitivity for detecting nuanced behavioural differences. These insights suggest that while the ADT provides valuable data, its utility may be limited by its ability to capture the full spectrum of anxiety and depression-like behaviours. This necessitates developing or incorporating more sensitive and nuanced behavioural assessment tools or combinations in future research.

5.4.2- Psilocybin

The second premise of this study posited that animals treated with psilocybin would exhibit reduced anxiety and depression-like behaviours compared to those in saline groups, as measured by the ADT across various phases of the test. This effect was anticipated to be more pronounced in SERT $^{+/-}$ groups. However, the repeated measures ANOVA did not reveal significant main effects or treatment interactions for any study measure, even though trends in the graphs suggested changes in anxiety-like measures.

Although not significant, an interesting trend was seen in three anxiety-like measures: alleys distance moved, alleys velocity and mSAT score, which showed a notable increase in the wild-type groups following psilocybin administration. This increase was observed in animals treated with 8 mg/kg . Additionally, EMS and $SERT^{+/-}$ groups receiving psilocybin presented visible increase in frequency in alley 4, a critical measure of the ADT.

This finding suggests an anxiolytic effect of the drug, especially for Wild-Type animals, and the consistency across both automated and manually taken measures reinforces this potential effect of the drug in this context.

The lack of statistical significance in the results could stem from two factors. Firstly, as previously discussed, the combined use of ADT and $SERT^{+/-}$ knockout models may not be sensitive enough to detect subtle behavioural changes. Secondly, the administered doses might not have been adequate to produce the expected effects.

Moreover, the observation of a trend in behavioural effects (even if not significant) at both 4 mg/kg and 8 mg/kg underscores the importance of dose consideration. Previous research indicates that higher doses of psilocybin are required to achieve significant behavioural changes. For instance, studies administering doses up to 20 mg/kg have reported positive results in various behavioural assessments (Meinhardt et al., 2020; Gallaher et al., 1993; Klein et al., 2018).

Conversely, some studies report no significant outcomes with high doses ranging from 1-10 mg/kg (Wojtas, 2022), while others have found positive effects with low or microdoses between 0.05-0.1 mg/kg (Higgins et al., 2021). These inconsistent findings, especially in studies involving female subjects, highlight the complexities of evaluating findings from new research. They also emphasise the need for further studies with diverse designs to investigate the effects of psilocybin thoroughly.

One additional factor, which is not critical but remains significant in this analysis, is time. Except for free rears, alleys distance moved and velocity, and mSAT score, all measures displayed a significant main effect of time that, in most cases, was observed when comparing the chronic phase with the other two phases, exhibiting general trends similar to those seen in previous experiments with this model. As discussed in Chapter 4, these results emphasise the effectiveness of ADT in measuring anticipatory behaviour.

In summary, the data found no significant differences between the $SERT^{+/-}$ and Control groups, nor from treatments with 4 and 8 mg/kg doses of psilocybin. However, trends were observed in four anxiety-like measures—alleys distance moved, alleys velocity, mSAT Score, and frequency in alley 4. These trends show a notable increase in wild-type groups post-psilocybin administration, particularly with the 8 mg/kg dose affecting the first three measures and both doses influencing frequency in alley 4, a key ADT metric, in the 2 conditions (i.e. $SERT^{+/-}$ and WT). However, these are only trends and not statistically significant findings.

Collectively, these results demonstrate once again the complex interplay of factors involved in the effects of psilocybin. While part of the literature is coherent with the present findings, some points of attention are raised and need to be addressed in the future in terms of doses and in the combination of animal models sensitive enough to demonstrate potential changes caused by the drug.

5.5- Sub-chapter - Impact of the Oestrus Cycle on the Effect of Psilocybin

As stated before, in this experiment, the oestrous cycle phase was determined on the day of the injection, and two subgroups within each treatment and condition group were formed. Due to the small size of these subgroups, only an exploratory analysis including this third factor (i.e., oestrous cycle) and its interplay with treatment could be performed, and no firm conclusions can be drawn from this.

Out of the 60 subjects initially included in the experiment, one was excluded from the oestrus analysis due to inconclusive results from a vaginal smear collected on the injection day (SERT 4 mg/kg). The distribution of the remaining 59 subjects across groups is shown in the table below.

Table 5.5.1- Distribution of Animals in the Groups

Condition/Treatment	Saline		4 mg/kg		8 mg/kg		
Cycle	DM	EP	DM	EP	DM	EP	Total
SERT $^{+/-}$	5	4	5	5	4	7	30
Wild type	4	3	5	6	6	5	29
Total	9	7	10	11	10	12	59
Total	16		21		22		

Note. DM = Group Oestrous phase dioestrus and metestrus phases. EP = Group Oestrous phase proestrus and oestrous phases. SERT = Serotonin Transporter (Protein). $^{+/-}$ = Heterozygous.

5.5.1- Results

The premise for this part of the experiment was that the phase of the cycle when the subjects were injected would influence the treatment effects. It was hypothesised that animals injected in the proestrus and oestrous phases (PE), characterised by higher hormone levels, would exhibit a less pronounced effect of psilocybin compared to animals injected during the dioestrus and metestrus phases (DM), which presents lower hormone levels. Results presented several measures for both anxiety and depression-like symptoms that reached the level of significance for this factor.

5.5.1.1- Psilocybin - Depression-like measures and Oestrus

The main effects of the oestrus cycle reached a significance level for total rears $F(1,47) = 7.29$, $p < 0.05$, wall rears $F(1,47) = 4.89$, $p < 0.05$, and free rears $F(1,47) = 3.91$, $p < 0.05$ as demonstrated in figures 5.5.1 and 5.5.2.

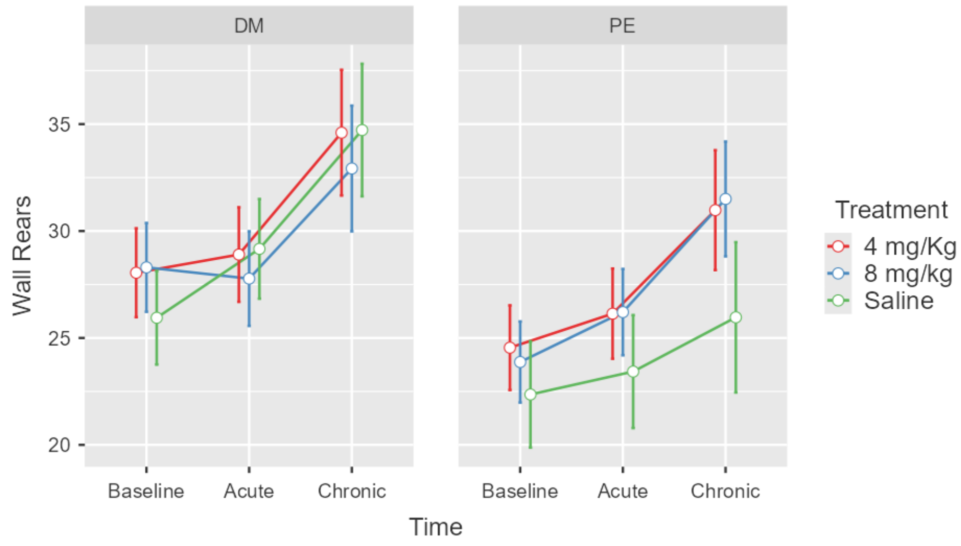

Figure 5.5.1- Means and ±SEM for wall rears across all three time points, split by group phase of the cycle and lines representing treatments.

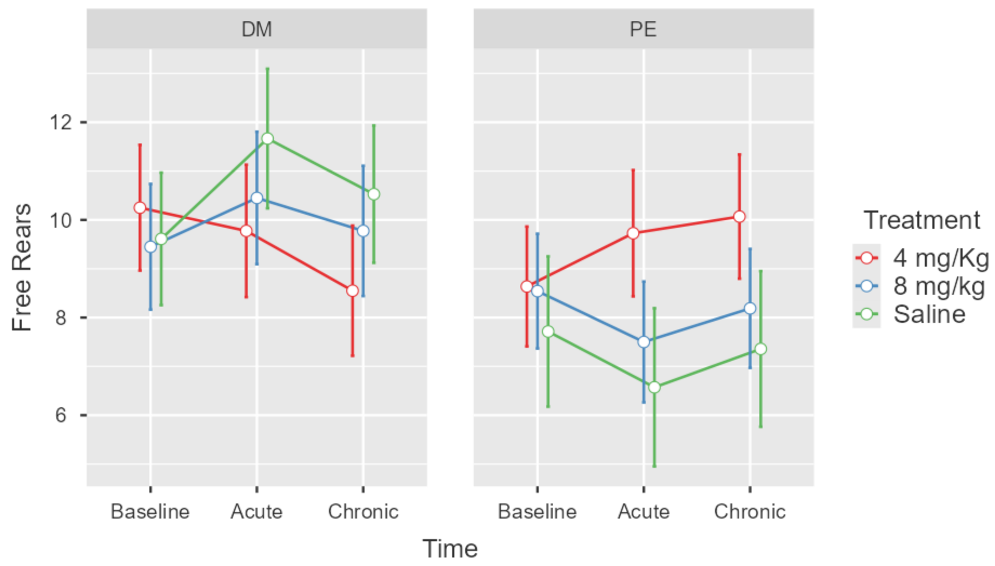

Figure 5.5.2- Means and ±SEM for free rears across all three time points, split by group phase of the cycle and lines representing treatments.

A one-way ANOVA for this factor demonstrated that total rears differed between oestrus groups in the baseline (p = 0.02) and acute phases (p = 0.01); free rears differed in the acute phase (p = 0.02), and wall rears differed only in the baseline (p = 0.02), with animals in the DM group showing higher rearing counts.

Ambulatory measures presented significant interactions between treatment and oestrous groups for Box distance moved $F(2,47) = 4.16$, $p = 0.02$ and velocity $F(2,47) = 4.11$, $p = 0.02$.

Although post hoc evaluations did not identify the source of the significance of these interactions, it is interesting to see markedly different patterns between oestrus groups. In DM groups, 8 mg/kg-treated animals presented a clear increase in acute and then chronic effects. In contrast, 4 mg/kg ones only exhibited the same dramatic increase after the acute phase. On the other hand, PE groups presented a less potent effect of psilocybin with a slight increase for 4 mg/kg towards the chronic phase and for 8 mg/kg treated, a decrease followed by an increase in the measures (Figures 5.5.3 and 5.5.4).

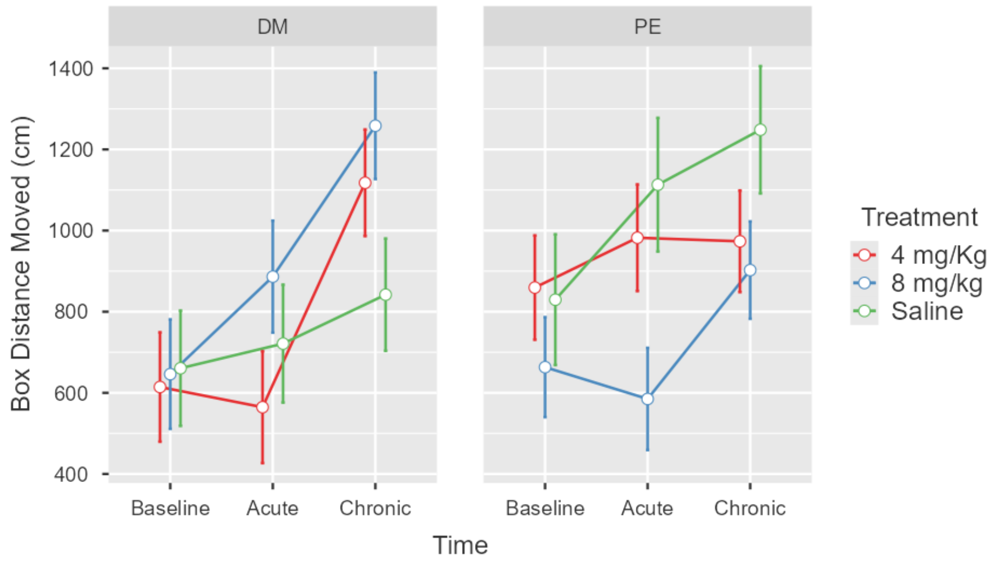

Figure 5.5.3 Means and ±SEM for box distance moved across all three time points, split by group phase of the cycle and lines representing treatments.

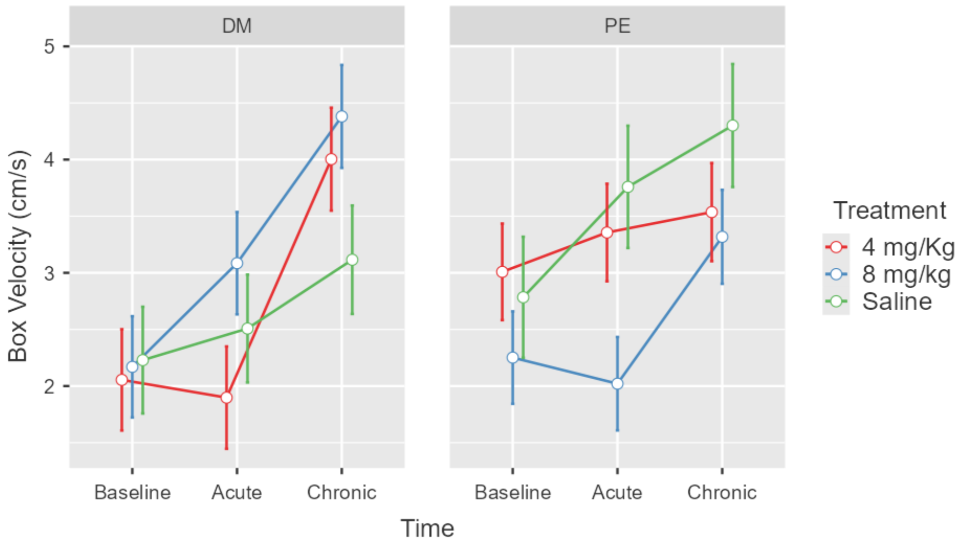

Figure 5.5 4 - Means and ±SEM for box velocity across all three time points, split by group phase of the cycle and lines representing treatments.

Additionally, interactions between condition and oestrus groups were also seen in ambulatory measures from the anticipatory box. box distance moved $F(1{,}47)7.19$, $p < 0.056$ and box velocity $F(1.47) = 6.84$, $p < 0.05$.

Graphs demonstrate opposite trends among groups and post hoc analysis confirmed that animals in the DM group walked more and faster when they were in the SERT$^{+/-}$ condition.

5.5.1.2- Psilocybin - Anxiety-like measures and Oestrus

No main effects or interactions involving treatment were significant; however, as seen in Figures 5.5.5 and 5.5.6, two anxiety-like measures presented significant results for interactions between the condition and oestrus. Alleys distance moved and velocity returned significant values for the interaction between condition and oestrus with $F(1{,}47) = 8.38$, $p < 0.05$ and $F(1{,}47) = 6$, $p < 0.05$, respectively. Post hoc evaluations confirmed that SERT$^{+/-}$ animals in the DM group walked more ($p = 0.02$) than WT, however the velocity measure did not reach a significance level.

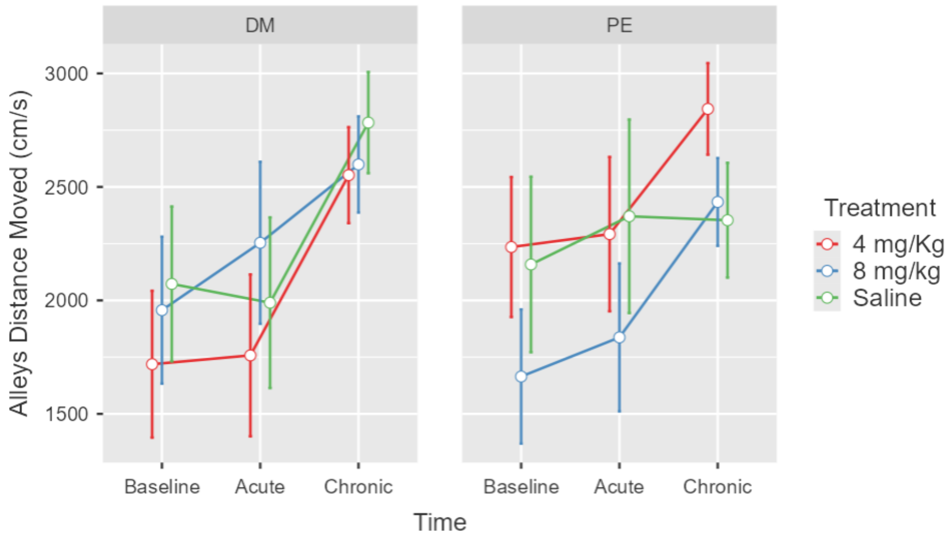

Figure 5.5 5- Means and ±SEM for alleys distance moved across all three time points, split by group phase of the cycle and lines representing treatments.

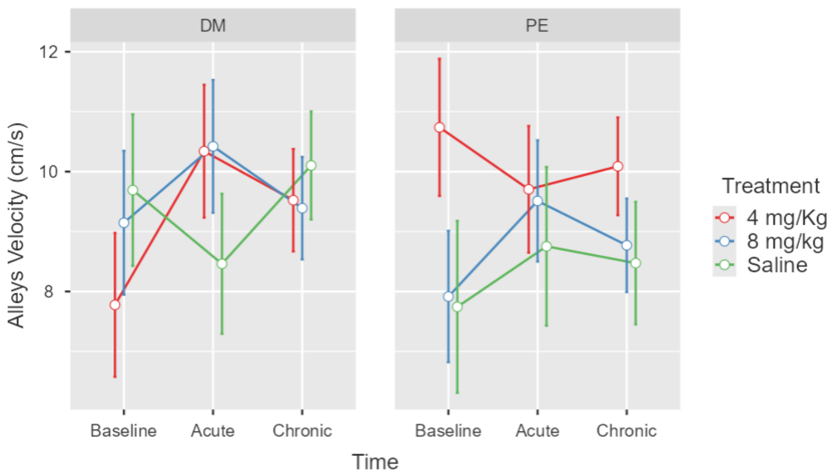

Figure 5.5 6- Means and ±SEM for alleys velocity across all three time points, split by group phase of the cycle and lines representing treatments.

5.5.2- Discussion

Including the oestrus cycle in the study revealed several significant results in this exploratory analysis, with main effects and interactions with treatment for depression-like measures and interactions with condition for depression and anxiety-like measures. This suggests an interplay between these factors and the effects of psilocybin, which is aligned with our hypothesis and certainly warrants further, more extensive analysis.

Significant interactions between treatment and oestrus were seen in ambulatory measures from the anticipatory box (i.e. box distance moved and velocity). Animals in the DM group who received psilocybin (8mg/kg) walked significantly more and faster than those in the PE group after injection. A sharp increase was also seen in the DM group injected with 4mg/kg, but only after the acute phase. Less pronounced effects were seen in the other groups. This enhanced response to the treatment in the DM group suggests a hormonal modulation of the drug's efficacy.

The anticipatory box showed significant main effects of the oestrus group in rearing behaviour (i.e., total, wall, and free rears). Animals in the DM group had higher rearing counts than those in the PE group. Notably, differences in total and free rears were significant in the acute phase of the cycle following the injection. Once again, animals injected in dioestrus and metestrus seem to exhibit a more substantial drug effect.

These results demonstrate that the increased anticipatory behaviour after psilocybin was more potent in the DM group, which suggests that, in this scenario, high levels of hormones potentially reduced the effect of psilocybin.

The literature does not present a consensus on how oestrous cycle phases influence female behaviour in behavioural tasks. Inconsistent results have been observed across tasks such as EPM, OFT, and light/dark boxes. For instance, some studies report reduced anxiety levels during the proestrus phase (when hormone levels are high) compared to the Diestrus phase (Rodríguez-Landa et al., 2021; Lovick et al., 2021), while others find no significant variations across different cycle phases in similar models (Chari, T., 2020; Lovick et al., 2021). It is important to note that methodological differences, including how cycle phases are classified, complicate these comparisons.

Exploratory behaviour in rats, such as rearing, is known to be influenced by several factors, including environmental novelty, anxiety levels, and internal factors, for instance, the phase of the oestrous cycle (Sturman, O., 2018). Regarding anticipatory behaviour, previous studies have not evaluated the impact of the oestrous cycle. In the ADT, typically, an increase in rearing behaviour is expected uniformly across all groups throughout the testing days. In this study, however, the observed differences in rearing behaviour between the oestrous groups during the acute phase strongly demonstrate the influence of the oestrous cycle, suggesting that it significantly modulated the exploratory behaviour in the anticipatory box.

Consistent with these results, previous findings in psychedelic research, though not extensive, indicate distinct drug sensitivities linked to oestrous phases. For instance, research by Bubeníková (2005) demonstrated increased sensitivity to MDMA in MD females during PPI tests. Similarly, Páleníček (2010) observed a reduced impact on locomotor activity and exploratory behaviour in MD females treated with LSD. Tylš et al. (2016) also noted diminished effects of moderate doses of psilocin (0.25 mg/kg, 1 mg/kg) on subjects in the PE phase. All these findings suggest a protective effect of oestrus in PE phases from the effects of psychedelics, including psilocybin.

One hypothesis for the varying responses to psilocybin across different phases of the menstrual cycle is that oestradiol, which varies throughout the cycle, selectively stimulates the expression and increases the density of 5-HT2A receptors, as Summer & Fink (1995) suggested.

Furthermore, oestrogen impacts other receptors involved in the responses of psilocybin to a lesser extent. For instance, the response to a 5-HT1A agonist during the PE phases tends to be reduced. Also, it is well-documented that oestrogens decrease both the expression of 5-HT1A mRNA and the number of these receptors, including auto receptors in the dorsal raphe nucleus. Conversely, an increase in dopamine system activity in females, influenced by oestrogens and progesterone, is also reported (Tylš et al., 2016).

Moreover, in the present study, locomotor activity measures, namely, distance moved and velocity in the anticipatory box and in the mSAT, also showed significant interactions between the oestrous phase and condition. Specifically, SERT animals demonstrated heightened anticipatory behaviour than WT in DM groups while simultaneously demonstrating reduced anxiety walking more and faster in the mSAT.

The interaction between the oestrous cycle phase and the genetic modification in SERT animals suggests that the impact of serotonin reduction on behaviour might also be influenced by hormonal status. This interaction may provide insights into how hormonal fluctuations affect behaviour in animals with reduced serotonin levels, which mimic the human condition.

To summarise, the study's findings reveal the impact of the oestrus cycle in several measures. Concerning depression-like measures, DM animals treated with 8 mg/kg of psilocybin had higher ambulation in the anticipatory box than PE groups in the acute and chronic phases. DM animals treated with 4 mg/kg also presented increased ambulation, but only after the acute phase. Additionally, significant effects of oestrous status were noted in rearing, with DM group animals displaying higher rearing counts, (total and free) during the acute phase. Moreover, interactions between condition and oestrus were seen in ambulatory measures in both the anticipatory box and in the mSAT, with SERT animals in the DM groups exhibiting more anticipatory behaviour and reduced anxiety.

Taken together, these findings suggest that the oestrous phase significantly modulates behaviour, aligning with existing literature that the oestrus phase influences female animals on depression and anxiety-like measures. While a robust statistical significance was not possible due to small sample sizes and high heterogeneity among groups, the effects clearly suggested that the doses used had a visible impact on behaviour on anxiety-like responses and even more on anticipatory behaviour. This confirms the initial hypothesis and highlights the influence of the oestrous phase. However, the limited group sizes necessitate caution in interpreting these results. Future studies should involve larger sample sizes and more sensitive measures to determine the effects of the treatment on these behaviours.

Chapter 6- General Discussion

6.1- Summary of Main Findings

The first experiment demonstrated that Psilocybin affected the locomotion of the animals in the open field test (OFT). Animals injected with either test dose (4mg/kg or 8mg/kg) showed reduced locomotion in the arena, especially when the test happened closer to the injection (i.e., 3 hours and 24 hours after the injection). Notably, animals treated with 4mg/kg appeared more affected (e.g. in rearing behaviour) when tested 3 hours post-injection, suggesting a dose-time interaction.

These findings confirm the hypothesis that psilocybin inhibits locomotor activity, although the effects did not, extend until the longest testing period. These results are important because they align with previous research indicating that serotonin-acting drugs in acute effects often reduce locomotor activity in classic models, often exhibiting the opposite effect (Prut & Belzung, 2003).

In the second experiment, animals subjected to Early Maternal Separation (EMS) and tested in the affective disorders test did not exhibit a robust depression or anxiety-like phenotype, nor did they show significant effects from psilocybin treatment as measured by the ADT.

Although the results were not statistically significant, there were some indications that EMS animals had reduced ambulation at baseline, and animals treated with both doses of psilocybin (8mg/kg and 16mg/kg) showed increased ambulation in the anticipatory box during acute and chronic phases. Notably, animals treated with 8mg/kg displayed increased rearing behaviour in the anticipatory box and slightly less anxiety-like behaviour the mSAT. Despite the lack of statistical significance, these observations align with the hypothesis that EMS would elicit the expected phenotype in baseline and that psilocybin would increase anticipatory behaviour in the locomotor box and produce anxiolytic effects in the mSAT. Additionally, the more noticeable effects of the lower dose of psilocybin (8mg/kg) indicate that this dose might be more adequate.

The final experiment, which also utilised the ADT and tested doses of 4mg/kg and 8mg/kg using SERT$^{+/-}$ knockout rats to elicit symptoms, yielded results similar to the second experiment regarding the absence of significant main effects and interactions between factors. However, an interesting pattern in the interaction between dose and condition was observed in anxiety-like measures, while wild-type animals treated with

8mg/kg psilocybin showed less anxiety in the mSAT expressed with increased distance moved and greater mSAT scores. Additionally, both groups (i.e. $SERT^{+/-}$ knockout and WT) presented increased scores of frequency in alley 4 for both doses administered, again suggesting an anxiolytic effect in this critical measure for the model. Although not statistically significant, these results align with the expected outcomes of the test.

A secondary analysis was made including the oestrous cycle as a factor. This analysis revealed that animals injected during the Diestrus and Metestrus (DM) phases characterised by lower hormone levels showed significant and positive effects on anticipatory behaviour for both doses when compared to those injected din the PE phases. Notably, the 4mg/kg dose demonstrated this effect only 8 days after injection. Additionally, $SERT^{+/-}$ DM animals exhibited more anticipatory behaviour and reduced anxiety. Although small group sizes compromise the impact of the results, they clearly indicate an interaction between treatment and oestrus phase as anticipated in the hypothesis and already seen in previous works.

The results of this work make evident the intricate interplay in psilocybin's effects, revealing complex interactions that extend beyond initial hypotheses proposed and offering a deeper understanding of its impact on mental health treatment. Although not all findings reached statistical significance, the data nevertheless highlight critical factors such as dosage, duration of effect, and hormonal status in female subjects, (frequently underrepresented in the literature). These insights are especially relevant considering the differential responses in locomotion observed at high doses and the modulatory role of female hormones, pointing to a gender-specific aspect of psilocybin's effects.

The significance of these findings lies not only in identifying potential therapeutic doses but also in suggesting that psilocybin's effects may be more variable than previously understood and therefore, requiring more investigation both in clinical and preclinical environments. This variability could influence treatment protocols, potentially requiring adjustments based on individual factors like hormonal status, thereby supporting a more personalized approach to psychedelic therapy.

Furthermore, the observed trends in the ADT once again evidence the necessity of more studies with different combinations of animal models for a deeper comprehension of the variables involved in the results which will result in a more consistent literature base on the behavioral effects of this drug.

Ultimately, these findings contribute to the groundwork for future research aimed at optimizing the therapeutic application of psilocybin. Further investigation with larger sample sizes and different experimental designs could elucidate the exact mechanisms by which dose, duration, and hormonal interactions influence treatment outcomes. Together, these insights advance the growing body of literature on psychedelics, emphasizing the importance of a cautious, research-driven approach to developing psilocybin as a safe, effective alternative for managing anxiety and depression.

6.2- Limitations

While the results of this study are promising, some limitations must be acknowledged. Specifically, about animal models, the doses utilised, the exclusive use of female subjects and the characteristics of the data set.

It is clear the models chosen in a design play a crucial role in the posterior evaluation of the effects of a given drug. In the experiments 2 and 3 of the present work, given that neither EMS nor the genetic reduction induced a visible depression or anxious phenotype in the ADT, it might make it more difficult to see any behavioural normalisation with psilocybin.

As mentioned before, three factors may have contributed to the lack of significant differences between EMS animals and Controls. The first is sex differences since the model has been validated mainly with male subjects, although sex differences have been found (Arnold, 2002; Rees, 2006). It is possible that females do not exhibit as clear a difference between conditions as males would. Additionally, methodology issues can interfere with the outcomes considering factors occurring during the process of EMS.

Given that neonatal rats rely on the mother for thermoregulation, nutrition, urination stimulation, and protection during the first two weeks of life, disruptions during this period may interfere with neurodevelopment. However, the effects of EMS on the offspring's adult behaviour are highly dependent on maternal behaviour before and after separation. The evidence says that maternal separation and handling increase maternal care upon reunion. (Schmidt et al., 2011). This compensatory behaviour may serve to overcompensate for the time off-nest with increased compensatory increases in licking, grooming, and arched-back nursing (Orso et al. 2019).

Nonetheless, as Orso and colleagues (2019) state in their review, several patterns of maternal care can be observed among different strains and even among individual dams. Therefore, there is no consensus on how these different patterns impact long-term outcomes in the pups, particularly regarding behavioural symptoms in adulthood. The inconsistent results in the investigations of these impacts may stem from the significant variability in maternal responsiveness after each period of separation. (Schmidt et al., 2011).

Several works have attested SERT knockout strain's effectivity in various behavioural essays in the pharmacology field (Golebiowska, 2019; Krishnan & Nestler, 2011). However, in the majority of the works, this validation as a model to elicit anxiety and depression-like symptoms was made with male subjects. Results including females are scarcer and tend not to present as robust results as those with males (Joeyen-Waldorf, 2009; Olivier, 2007). In this way, it is possible to presume that female subjects may not exhibit the expected phenotype so clearly or that the model utilised was not sensitive enough to demonstrate it, as already discussed in the EMS section.

Building on the discussion of EMS variability, it is important to note that observations from our laboratory (unpublished data) indicate poor and inconsistent maternal care in SERT KO dams. This variability in maternal care may have influenced neurodevelopmental outcomes, contributing to increased variability among offspring from different litters in the study. Consequently, this variability might explain the absence of significant differences in comparisons between SERT KO and wild-type subjects.

Another factor is, as discussed previously, the differences observed in animal models, including in the ADT, are more robust when using homozygous animals (Kidwell, 2019; Blackburne, 2021). Indeed, studies have demonstrated a very similar pattern of behaviour between SERT$^{+/-}$ and wildtypes in behavioural assays (Ellenbroek, 2016; Joeyen-Waldorf, 2009). However, using SERT $^{-/-}$ animals in this experiment was not considered since the heterozygous with a 50% reduction in the serotonin protein are more similar to human conditions related to anxiety and depression (Macchi et al., 2013).

Another possible explanation for the lack of difference between EMS and controls and SERT$^{+/-}$ and wild types is that was not sensitive enough to detect differences between conditions in both models. Similarly, Chaffee (2022), who combined EMS and ADT to test high doses of psilocybin in male rats, also did not observe significant differences between EMS and control animals.

A second important aspect to highlight in terms of limitations is the administration of a single dose of psilocybin, up to 16 mg/kg. Dose-dependent effects have been observed, with increased plasticity noted at doses as high as 20 mg/kg (Jefsen et al., 2021). Psilocybin promotes dendritogenesis and spinogenesis in cortical neurons, leading to enhanced synaptic connections. In humans, this structural remodelling may facilitate the formation of new neural pathways, potentially improving cognitive flexibility and emotional regulation, which can underlie the alleviation of anxiety and depression symptoms (Calder & Hasler, 2023). Remarkably, a single dose can produce symptom improvements in humans lasting up to six months, and no studies to date have shown that repeated doses or chronic administration of high doses yield more effective or prolonged effects. However, as previously discussed, differences in serotonin receptor affinity (Gallaher et al., 1993; Klein et al., 2018) indicate that rodents may require higher doses than humans to elicit pronounced effects in animal models. In the present work, while the results were not statistically significant, there were observable trends suggesting a reduction in anxiety and depression-like symptoms, which suggests that tests with even higher doses could produce better results.

A third important aspect to be mentioned is that all three studies in this work exclusively involved female subjects, preventing any comparison between sexes under identical procedures by the same researcher. Furthermore, assessing the effects of the oestrous cycle in experiments one and two would enable an evaluation of how this factor interacts with a different test and the behavioural paradigm used to induce symptomatology. While a more comprehensive approach could enhance understanding of the drug's effects, it would also introduce methodological challenges due to an increased volume of tests potentially requiring more than one researcher.

Lastly, another significant limitation of this study involves data variability between cohorts and the small sample sizes in the second and third experiments. As already discussed in the correspondent discussion sections, these characteristics compromised the statistical evaluation, potentially reducing the study's statistical power and increasing the risk of failing to detect actual effects. Additionally, the nature of the ADT test, which includes repeated testing and a lengthy daily protocol, complicates the expansion of group sizes, thereby making it challenging to improve the dataset's characteristics, although in terms of cohorts' effects, previous studies do not report huge variability. Future studies should consider involving a team of researchers to facilitate testing larger sample groups without significantly extending the study's overall duration.

6.3- Future directions

The use of psychedelics for mental health disorders, particularly in preclinical research, still leaves several unanswered questions, including sex differences and hormonal interactions. These are vital areas requiring further clarification.

In light of the findings from this study, future research should address the identified limitations for a better understanding of the effects of psilocybin. Firstly, in terms of animal models, combining the two paradigms used to elicit symptomatology (SERT knockout and EMS) could potentially produce different outcomes. However, it will require a more significant commitment to animal welfare. A combination of a genetic paradigm with a behavioural one would likely result in a more evident depressive and anxious-like phenotype. Alternative possibilities include using other models such as Flinders Sensitive Line or Wistar Kyoto strains and chronic unpredictable stress.

Additionally, while the ADT is a robust assay that explores both anxiety and depression-like symptoms in acute and chronic phases, it is a relatively new test and has demonstrated validity mainly with homozygous SERT knockout rats. Consequently, this test may not be sensitive enough to identify subtle variations in animal behaviour resulting from pharmacological treatments and using SERT heterozygous animals. Its combination with other models to elicit symptoms may return more visible effects.

As emphasised previously, neuroscience studies often neglect female subjects, assuming that findings from male subjects are fully applicable to females. A more realistic approach would involve comparing the effects of psilocybin in both male and female animals subjected to the same paradigms and doses. Considering that anxiety and depression are twice as prevalent in women as in men (Altemus, 2014), sex comparisons would help elucidate sex-specific responses and enhance the generalizability of the findings.

Furthermore, understanding the interaction between psilocybin treatment and female hormones is still in its infancy and, as with most topics in preclinical research about this drug, has produced conflicting results. Evidence suggests that the oestrous cycle stages significantly influence HPA axis responses (Rhodes, 2002), which are involved in the symptoms of depression and anxiety and intensely involved in the effects of psilocybin. Although no clinical trials with psilocybin have found this evidence, preclinical research is still too limited to fully address this issue.

Finally, one crucial point that still needs further investigation is whether the psychedelic experience is necessary for the effects of psilocybin and other psychedelics on mental health disorders. The psychedelic experience, particularly the sense of oceanic boundlessness, ego dissolution, and mystical-type experiences, is considered by many researchers to be crucial for the therapeutic effects of these drugs in humans. These subjective experiences are challenging, if not impossible, to directly assess in animal models. This presents a significant limitation in translating animal research to human applications, especially considering that the potential importance of these experiences for therapeutic outcomes is still debatable in the field. While we cannot (so far) directly assess oceanic boundlessness or other subjective aspects of the psychedelic experience in animals, combinations of animal studies remain crucial for understanding the neurobiological mechanisms underlying psychedelic effects.

By addressing these areas, future research can build on the current findings and contribute to a more nuanced understanding of the factors influencing the outcomes of psilocybin treatment in mental health disorders.

6.4- Conclusion

In summary, this thesis has explored the behavioural effects of relatively high doses of psilocybin in adult female rats through three distinct experiments. The first experiment demonstrated that psilocybin significantly reduced locomotion in the open field test, with notable differences in response based on the timing between administration and test. The second experiment indicated that EMS animals did not exhibit robust depression or anxiety phenotypes, yet some effects of psilocybin were observed, particularly at 8 mg/kg. The final experiment revealed complex interactions between dose, condition, and the oestrous cycle, highlighting the nuanced effects of psilocybin on anticipatory behaviour and anxiety, with slightly better results for 4 mg/kg.

These experiments collectively contribute to the limited knowledge on high-dose psilocybin and its impact on female subjects. The insights gained from this research are essential for advancing our understanding of psilocybin's potential therapeutic applications.

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A.I. Use Statement

This thesis utilized the following artificial intelligence-based tools for search engine functionality:

1. Consensus.app. (n.d.). [Web-based application]. Available from https://consensus.app
2. Elicit.org. (n.d.). [Web-based application]. Available from https://elicit.org
3. Perplexity AI. (n.d.). [Web-based application]. Available from https://www.perplexity.ai/

And text correction and editing:

4. OpenAI. (2024). ChatGPT (Version 4). [Software]. Available from https://www.openai.com/chatgpt
5. Grammarly. (n.d.). [Text editing software]. Available from https://www.grammarly.com

Appendices

1- Means and Standard Deviations for Measures of the Experiment: Effects of Psilocybin on Locomotor Activity (Chapter 3)

Means and Standard Deviations for Central Distance Travelled

Measure	Time	Treatment	Mean	SD
Central distance travelled	3 hours	4 mg/kg	305.43	138.5
		8 mg/kg	468.91	176.89
		Saline	600.44	181.92
	24 hours	4 mg/kg	400.85	221.28
		8 mg/kg	415.38	184.59
		Saline	691.08	352.54
	8 days	4 mg/kg	646.94	215.53
		8 mg/kg	672.76	205.21
		Saline	704.71	225.94

Means and Standard Deviations for Percentage of time in Central Area

Measure	Time	Treatment	Mean	SD
Percentage of time in central area	3 hours	4 mg/kg	3.72	2.18
		8 mg/kg	4.08	2.15
		Saline	10.1	8.21
	24 hours	4 mg/kg	4.68	2.87
		8 mg/kg	4.6	1.91
		Saline	8.79	4.02
	8 days	4 mg/kg	8.52	4.9
		8 mg/kg	6.64	2.65
		Saline	9.38	4.09

Means and Standard Deviations for Central Rears

Measure	Time	Treatment	Mean	SD
Central rears	3 hours	4 mg/kg	5.27	5
		8 mg/kg	7	4.64
		Saline	18.09	10.41
	24 hours	4 mg/kg	12.25	11.66
		8 mg/kg	9.8	6.88
		Saline	15.13	8.54
	8 days	4 mg/kg	18.44	5.81
		8 mg/kg	15.1	7.28
		Saline	14.8	4.57

Means and Standard Deviations for Thigmotaxis

Measure	Time	Treatment	Mean	SD
Thigmotaxis	3 hours	4 mg/kg	96.28	2.18
		8 mg/kg	95.92	2.15
		Saline	89.9	8.21
	24 hours	4 mg/kg	95.32	2.87
		8 mg/kg	95.4	1.91
		Saline	91.2	4.02
	8 days	4 mg/kg	91.48	4.9
		8 mg/kg	93.36	2.65
		Saline	90.62	4.09

Means and Standard Deviations for Total Distance Travelled

Measure	Time	Treatment	Mean	SD
Total distance travelled	3 hours	4 mg/kg	1427.42	440.81
		8 mg/kg	1791.62	518.28
		Saline	2144.13	503.43
	24 hours	4 mg/kg	1796.57	583.64
		8 mg/kg	1731.91	544.52
		Saline	2533.66	944.83
	8 days	4 mg/kg	2343.37	673.58
		8 mg/kg	2581.07	779.74
		Saline	2616.57	721.44

Means and Standard Deviations for Total Rears

Measure	Time	Treatment	Mean	SD
Total rears	3 hours	4 mg/kg	59.27	18.4
		8 mg/kg	90.7	31.79
		Saline	135.64	24.37
	24 hours	4 mg/kg	126	46.08
		8 mg/kg	100.9	16.58
		Saline	140.25	47.22
	8 days	4 mg/kg	139.11	31.6
		8 mg/kg	135.5	27.13
		Saline	146.7	24.11

Means and Standard Deviations for Faecal Boli

Measure	Time	Treatment	Mean	SD
Faecal boli	3 hours	4 mg/kg	0	0
		8 mg/kg	0.2	0.63
		Saline	2.55	3.27
	24 hours	4 mg/kg	2.58	3.34
		8 mg/kg	3.3	3.53
		Saline	1.75	1.98
	8 days	4 mg/kg	3.44	4.8
		8 mg/kg	3.4	3.17
		Saline	2.9	2.38

Means and Standard Deviations for USVs Happy Calls

Measure	Time	Treatment	Mean	SD
USVs happy calls	3 hours	4 mg/kg	51.07	33.31
		8 mg/kg	35.72	30.52
		Saline	32.3	13.82
	24 hours	4 mg/kg	41.27	19.04
		8 mg/kg	53.94	30.45
		Saline	45.69	21.84
	8 days	4 mg/kg	36.06	20.61
		8 mg/kg	33.81	27.6
		Saline	36.98	19.03

Means and Standard Deviations for USVs Sad Calls

Measure	Time	Treatment	Mean	SD
USVs sad calls	3 hours	4 mg/kg	48.93	33.31
		8 mg/kg	54.28	33.73
		Saline	67.7	13.82
	24 hours	4 mg/kg	58.73	19.04
		8 mg/kg	46.06	30.45
		Saline	54.31	21.84
	8 days	4 mg/kg	63.94	20.61
		8 mg/kg	43.97	31.87
		Saline	63.02	19.03

2A- Means and Standard Deviations for Measures of the Experiment: Effects of Doses of 8 and 16 mg/kg of Psilocybin on Anxiety and Depression-like Symptoms (Chapter 4)

Means and Standard Deviations for Box Distance Moved

Measure - Phase	Condition	Treatment	Mean	SD
Box distance moved -baseline	Control		1313.45	401.94
	EMS		1188.82	364.54
Box distance moved -acute	Control	8 mg/Kg	1249.86	336.68
		16 mg/kg	1192.49	342.96
		Saline	1369.42	335.4
	EMS	8 mg/Kg	1680.23	503.48
		16 mg/kg	1350.66	341.55
		Saline	1185.03	360.15
Box distance moved -chronic	Control	8 mg/Kg	1637.59	472.35
		16 mg/kg	1539.59	352.42
		Saline	1871.71	702.64
	EMS	8 mg/Kg	2260.49	1029.18
		16 mg/kg	1553.24	406.43
		Saline	1715.09	614.25

Means and Standard Deviations for Box Velocity

Measure - Phase	Condition	Treatment	Mean	SD
Box velocity -baseline	Control		4.46	1.35
	EMS		4.09	1.12
Box velocity -acute	Control	8 mg/Kg	4.22	1.08
		16 mg/kg	4.08	1.11
		Saline	4.66	1.13
	EMS	8 mg/Kg	5.66	1.73
		16 mg/kg	4.64	1.05
		Saline	4.07	1.12
Box velocity -chronic	Control	8 mg/Kg	5.59	1.71
		16 mg/kg	5.26	1.2
		Saline	6.35	2.37
	EMS	8 mg/Kg	7.61	3.39
		16 mg/kg	5.4	1.21
		Saline	5.89	2.01

Means and Standard Deviations for Total Rears

Measure - Phase	Condition	Treatment	Mean	SD
Total rears -baseline	Control		39.16	13.65
	EMS		35.21	13.56
Total rears -acute	Control	8 mg/Kg	35.89	10.15
		16 mg/kg	35.85	12.6
		Saline	39.36	8.38
	EMS	8 mg/Kg	41.52	14.35
		16 mg/kg	39.44	13.1
		Saline	33.88	13.46
Total rears -chronic	Control	8 mg/Kg	41.15	9.49
		16 mg/kg	42.94	6.86
		Saline	43.85	11.2
	EMS	8 mg/Kg	40.52	13.73
		16 mg/kg	43.91	9.4
		Saline	38.26	10.53

Means and Standard Deviations for Wall Rears

Measure - Phase	Condition	Treatment	Mean	SD
Wall rears -baseline	Control		20.74	9.81
	EMS		20.64	10.32
Wall rears -acute	Control	8 mg/Kg	19.81	9.71
		16 mg/kg	19.19	7.95
		Saline	23.41	6.41
	EMS	8 mg/Kg	21.7	8.21
		16 mg/kg	23.41	9.46
		Saline	21.58	10.46
Wall rears -chronic	Control	8 mg/Kg	22.76	11.69
		16 mg/kg	21.44	8.29
		Saline	23.96	7.54
	EMS	8 mg/Kg	21.2	8.98
		16 mg/kg	24.96	7.63
		Saline	22.12	8.29

Means and Standard Deviations for Free Rears

Measure - Phase	Condition	Treatment	Mean	SD
Free rears -baseline	Control		18.29	9.58
	EMS		14.69	8.75
Free rears -acute	Control	8 mg/Kg	16.07	8.95
		16 mg/kg	16.67	9.47
		Saline	15.95	9.04
	EMS	8 mg/Kg	19.81	7.72
		16 mg/kg	16.04	6.51
		Saline	12.3	7.4
Free rears -chronic	Control	8 mg/Kg	18.39	6.57
		16 mg/kg	21.5	7.76
		Saline	19.88	8.61
	EMS	8 mg/Kg	19.31	6.68
		16 mg/kg	18.94	6.09
		Saline	16.14	8.44

Means and Standard Deviations for Alleys Distance Moved

Measure - Phase	Condition	Treatment	Mean	SD
Alleys distance moved -baseline	Control		1963.39	502.81
	EMS		1963.59	399.25
Alleys distance moved -acute	Control	8 mg/Kg	1916.52	385.1
		16 mg/kg	1997.74	562.62
		Saline	2054	380.53
	EMS	8 mg/Kg	1936.26	307.74
		16 mg/kg	2093.63	423.25
		Saline	1880.85	489.17
Alleys distance moved -chronic	Control	8 mg/Kg	1970.52	252.57
		16 mg/kg	2242.11	436.02
		Saline	2245.92	395.1
	EMS	8 mg/Kg	2022.65	204.03
		16 mg/kg	2219.94	432.18
		Saline	2084.79	422.08

Means and Standard Deviations for Alleys Velocity

Measure - Phase	Condition	Treatment	Mean	SD
Alleys velocity -baseline	Control		6.63	1.69
	EMS		6.66	1.38
Alleys velocity -acute	Control	8 mg/Kg	6.4	1.67
		16 mg/kg	7.05	1.84
		Saline	7.35	1.17
	EMS	8 mg/Kg	6.65	0.94
		16 mg/kg	7.62	2.1
		Saline	6.82	1.65
Alleys velocity -chronic	Control	8 mg/Kg	6.81	0.9
		16 mg/kg	7.88	1.6
		Saline	7.81	1.36
	EMS	8 mg/Kg	7.12	0.63
		16 mg/kg	7.84	1.52
		Saline	7.39	1.37

Means and Standard Deviations for Frequency in Alley 4

Measure - Phase	Condition	Treatment	Mean	SD
Frequency in alley 4 -baseline	Control		5.02	1.86
	EMS		5.71	2.1
Frequency in alley 4 -acute	Control	8 mg/Kg	5.37	1.49
		16 mg/kg	4.93	2.28
		Saline	5.18	1.75
	EMS	8 mg/Kg	6.15	1.58
		16 mg/kg	5.85	2.35
		Saline	5.3	1.47
Frequency in alley 4 -chronic	Control	8 mg/Kg	5.91	0.63
		16 mg/kg	6.06	2.15
		Saline	7.06	1.77
	EMS	8 mg/Kg	6.67	1.14
		16 mg/kg	6.52	1.97
		Saline	6.71	1.75

Means and Standard Deviations for Cumulative Duration in Alley 4

Measure - Phase	Condition	Treatment	Mean	SD
Cumulative duration in alley 4 -Baseline	Control		52.31	18.6
	EMS		62.54	18.75
Cumulative duration in alley 4 -acute	Control	8 mg/Kg	61.95	12.86
		16 mg/kg	44.25	21.6
		Saline	57.57	16.34
	EMS	8 mg/Kg	71.69	24.19
		16 mg/kg	64.84	24.15
		Saline	63.73	20.25
Cumulative duration in alley 4 -chronic	Control	8 mg/Kg	60.58	13.33
		16 mg/kg	48.98	13.44
		Saline	62.91	15.21
	EMS	8 mg/Kg	65.55	14.09
		16 mg/kg	59.92	13.94
		Saline	64.33	14.54

Means and Standard Deviations for Latency to eat the 1st FL

Measure - Phase	Condition	Treatment	Mean	SD
Latency to eat the 1st FL -baseline	Control		30.19	62.93
	EMS		41.38	75.54
Latency to eat the 1st FL -acute	Control	8 mg/Kg	21.33	43.34
		16 mg/kg	49.96	86.41
		Saline	6.82	12.64
	EMS	8 mg/Kg	11.07	11.14
		16 mg/kg	14.96	33.78
		Saline	30.33	48.32
Latency to eat the 1st FL -chronic	Control	8 mg/Kg	16.41	24.84
		16 mg/kg	9.87	8.76
		Saline	10.86	27.7
	EMS	8 mg/Kg	3.81	2.08
		16 mg/kg	3.81	1.11
		Saline	7.98	9.65

Means and Standard Deviations for Latency to eat FL #4

Measure - Phase	Condition	Treatment	Mean	SD
Latency to eat FL #4 -baseline	Control		126.73	77.15
	EMS		138.76	99.51
Latency to eat FL #4 -acute	Control	8 mg/Kg	121.67	85.95
		16 mg/kg	127.11	77.11
		Saline	91.56	36.65
	EMS	8 mg/Kg	108.48	61.56
		16 mg/kg	94.48	46.8
		Saline	103.45	69.52
Latency to eat FL #4 - chronic	Control	8 mg/Kg	87.41	24.02
		16 mg/kg	96.04	23.33
		Saline	87.03	41.27
	EMS	8 mg/Kg	81.28	18.33
		16 mg/kg	84.2	31.91
		Saline	85.21	28.15

Means and Standard Deviations for mSAT Score

Measure - Phase	Condition	Treatment	Mean	SD
mSAT score -baseline	Control		10.87	3.78
	EMS		12.31	6.33
mSAT score -acute	Control	8 mg/Kg	12.04	2.93
		16 mg/kg	9.85	3.57
		Saline	12.85	3.8
	EMS	8 mg/Kg	15.7	5.5
		16 mg/kg	11.78	4.29
		Saline	12.61	5.11
mSAT score -chronic	Control	8 mg/Kg	13.17	3.66
		16 mg/kg	11.24	1.23
		Saline	12.87	3.65
	EMS	8 mg/Kg	14.65	5.55
		16 mg/kg	12.7	3.76
		Saline	13.11	5.02

2B- ANOVA - Main Effects and Interactions for Measures of the Experiment: Effects of Doses of 8 and 16 mg/kg of Psilocybin on Anxiety and Depression-like Symptoms (Chapter 4)

Main Effects and Interactions for Box Distance Moved

	df	*F*	*p*
Time	(2, 108)	41.95	0.00
Time*condition	(2, 108)	3.19	0.05
Time*treatment	(4, 108)	1.50	0.21
Time*condition*treatment	(4, 108)	0.90	0.47
Condition	(1, 54)	0.33	0.57
Treatment	(2, 54)	1.41	0.25
Condition*treatment	(2, 54)	2.55	0.09

Main Effects and Interactions for Box Velocity

	df	*F*	*p*
Time	(2, 108)	44.47	0.00
Time*condition	(2, 108)	3.06	0.05
Time*treatment	(4, 108)	1.70	0.15
Time*condition*treatment	(4, 108)	0.82	0.51
Condition	(1, 54)	0.47	0.50
Treatment	(2, 54)	1.11	0.34
Condition*treatment	(2, 54)	2.47	0.09

Main Effects and Interactions for Total Rears

	df	*F*	*p*
Time	(2, 108)	6.01	0.00
Time*condition	(2, 108)	1.29	0.28
Time*treatment	(4, 108)	0.74	0.56
Time*condition*treatment	(4, 108)	0.22	0.93
Condition	(1, 54)	0.23	0.63
Treatment	(2, 54)	0.13	0.88
Condition*treatment	(2, 54)	1.20	0.31

Main Effects and Interactions for Wall Rears

	df	*F*	*p*
Time	(2, 108)	3.34	0.04
Time*condition	(2, 108)	0.40	0.67
Time*treatment	(4, 108)	0.79	0.54
Time*condition*treatment	(4, 108)	0.40	0.81
Condition	(1, 54)	0.06	0.81
Treatment	(2, 54)	0.26	0.77
Condition*treatment	(2, 54)	0.39	0.68

Main Effects and Interactions for Free Rears

	df	*F*	*p*
Time	(2, 108)	6.31	0.00
Time*condition	(2, 108)	1.66	0.20
Time*treatment	(4, 108)	1.15	0.34
Time*condition*treatment	(4, 108)	0.27	0.90
Condition	(1, 54)	0.84	0.36
Treatment	(2, 54)	1.04	0.36
Condition*treatment	(2, 54)	1.11	0.34

Main Effects and Interactions for Alleys Distance Moved

	df	*F*	*p*
Time	(2, 108)	6.92	0.00
Time*condition	(2, 108)	0.21	0.81
Time*treatment	(4, 108)	1.33	0.26
Time*condition*treatment	(4, 108)	0.32	0.87
Condition	(1, 54)	0.02	0.88
Treatment	(2, 54)	2.02	0.14
Condition*treatment	(2, 54)	0.92	0.41

Main Effects and Interactions for Alleys Velocity

	df	*F*	*p*
Time	(2, 108)	11.39	0.00
Time*condition	(2, 108)	0.11	0.90
Time*treatment	(4, 108)	0.68	0.61
Time*condition*treatment	(4, 108)	0.38	0.83
Condition	(1, 54)	0.02	0.88
Treatment	(2, 54)	2.68	0.08
Condition*treatment	(2, 54)	0.92	0.41

Main Effects and Interactions for Frequency in Alley 4

	df	*F*	*p*
Time	(2, 108)	12.00	0.00
Time*condition	(2, 108)	0.44	0.65
Time*treatment	(4, 108)	1.61	0.18
Time*condition*treatment	(4, 108)	0.09	0.98
Condition	(1, 54)	2.08	0.16
Treatment	(2, 54)	0.01	0.99
Condition*treatment	(2, 54)	0.77	0.47

Main Effects and Interactions for Cumulative Duration in Alley 4

	df	*F*	*p*
Time	(2, 108)	1.19	0.31
Time*condition	(2, 108)	0.82	0.44
Time*treatment	(4, 108)	1.17	0.33
Time*condition*treatment	(4, 108)	0.52	0.72
Condition	(1, 54)	7.16	0.01
Treatment	(2, 54)	1.62	0.21
Condition*treatment	(2, 54)	0.48	0.62

Main Effects and Interactions for Latency to eat 1^{st} FL

	df	*F*	*p*
Time	(2, 108)	8.11	0.00
Time*condition	(2, 108)	0.93	0.40
Time*treatment	(4, 108)	0.48	0.75
Time*condition*treatment	(4, 108)	1.98	0.10
Condition	(1, 54)	0.03	0.86
Treatment	(2, 54)	0.16	0.86
Condition*treatment	(2, 54)	1.65	0.20

Main Effects and Interactions for Latency to eat 1^{st} FL

	df	*F*	*p*
Time	(2, 108)	16.69	0.00
Time*condition	(2, 108)	1.02	0.36
Time*treatment	(4, 108)	0.92	0.45
Time*condition*treatment	(4, 108)	0.21	0.93
Condition	(1, 54)	0.03	0.87
Treatment	(2, 54)	0.27	0.76
Condition*treatment	(2, 54)	0.28	0.76

Main Effects and Interactions for mSAT Score

	df	*F*	*p*
Time	(2, 108)	4.60	0.01
Time*condition	(2, 108)	0.37	0.69
Time*treatment	(4, 108)	1.14	0.34
Time*condition*treatment	(4, 108)	1.40	0.24
Condition	(1, 54)	2.03	0.16
Treatment	(2, 54)	1.06	0.35
Condition*treatment	(2, 54)	0.88	0.42

3A- Means and Standard Deviations for Measures of the Experiment: Effects of Doses of 4 and 8 mg/kg of Psilocybin on Anxiety and Depression-like Symptom**s** and Impact**s** of the Oestru**s** Cycle (Chapter 5)

Means and Standard Deviations for Box Distance Moved

Measure - Phase	Condition	Treatment	Oestrus	Mean	SD
Box distance moved -baseline	SERT			749.33	490.25
	Wild Type			647.49	328.17
Box distance moved -acute	SERT	4 mg/Kg	DM	730.09	376.53
			PE	1078.15	548.89
		8 mg/kg	DM	1330.11	509.90
			PE	435.42	105.84
		Saline	DM	877.82	525.66
			PE	1276.36	605.46
	Wild Type	4 mg/Kg	DM	399.11	111.77
			PE	902.47	485.73
		8 mg/kg	DM	590.62	294.55
			PE	794.23	412.96
		Saline	DM	525.69	120.46
			PE	895.24	380.48
Box distance moved -chronic	SERT	4 mg/Kg	DM	1213.38	399.66
			PE	911.01	651.02
		8 mg/kg	DM	1598.28	299.33
			PE	833.12	157.48
		Saline	DM	1118.46	705.75
			PE	1327.95	145.76
	Wild Type	4 mg/Kg	DM	1022.13	135.36
			PE	1025.62	173.91
		8 mg/kg	DM	1031.82	393.45
			PE	999.75	443.41
		Saline	DM	496.34	450.39
			PE	1142.47	135.31

Means and Standard Deviations for Box Velocity

Measure - Phase	Condition	Treatment	Oestrus	Mean	SD
Box velocity -baseline	SERT			2.55	1.63
	Wild Type			2.19	1.1
Box velocity -acute	SERT	4 mg/Kg	DM	2.45	1.25
			PE	3.70	1.87
		8 mg/kg	DM	4.44	1.70
			PE	1.51	0.38
		Saline	DM	2.96	1.74
			PE	4.32	1.94
	Wild Type	4 mg/Kg	DM	1.35	0.37
			PE	3.07	1.57
		8 mg/kg	DM	2.18	0.98
			PE	2.74	1.36
		Saline	DM	1.94	0.50
			PE	3.01	1.29
Box velocity -chronic	SERT	4 mg/Kg	DM	4.39	1.44
			PE	3.15	2.24
		8 mg/kg	DM	5.41	1.05
			PE	3.15	0.59
		Saline	DM	3.96	2.45
			PE	4.58	0.41
	Wild Type	4 mg/Kg	DM	3.62	0.34
			PE	3.86	0.54
		8 mg/kg	DM	3.70	1.39
			PE	3.55	1.45
		Saline	DM	2.07	1.98
			PE	3.93	0.39

Means and Standard Deviations for Total Rears

Measure - Phase	Condition	Treatment	Oestrus	Mean	SD
Total rears -baseline	SERT			34.35	9.42
	Wild Type			34.72	7.9
Total rears -acute	SERT	4 mg/Kg	DM	39.95	5.78
			PE	41.20	5.87
		8 mg/kg	DM	33.56	8.68
			PE	37.68	6.27
		Saline	DM	41.15	12.31
			PE	32.00	11.31
	Wild Type	4 mg/Kg	DM	37.40	5.09
			PE	31.42	9.71
		8 mg/kg	DM	41.33	10.46
			PE	28.15	5.86
		Saline	DM	40.44	5.28
			PE	27.33	4.29
Total rears -chronic	SERT	4 mg/Kg	DM	45.05	9.10
			PE	45.35	11.00
		8 mg/kg	DM	40.38	5.76
			PE	45.04	6.42
		Saline	DM	46.40	21.52
			PE	36.19	7.17
	Wild Type	4 mg/Kg	DM	41.25	6.82
			PE	37.25	4.38
		8 mg/kg	DM	43.71	15.88
			PE	32.55	8.16
		Saline	DM	43.81	7.76
			PE	29.75	2.22

Means and Standard Deviations for Wall Rears

Measure - Phase	Condition	Treatment	Oestrus	Mean	SD
Wall rears -baseline	SERT			25.53	7.28
	Wild Type			25.4	6
Wall rears -acute	SERT	4 mg/Kg	DM	28.10	6.21
			PE	30.30	2.74
		8 mg/kg	DM	26.31	7.53
			PE	29.14	6.12
		Saline	DM	30.60	14.40
			PE	25.81	8.29
	Wild Type	4 mg/Kg	DM	29.70	5.81
			PE	22.67	4.83
		8 mg/kg	DM	28.75	5.18
			PE	22.10	4.85
		Saline	DM	27.38	5.15
			PE	20.25	2.61
Wall rears -chronic	SERT	4 mg/Kg	DM	34.85	9.40
			PE	34.15	8.62
		8 mg/kg	DM	32.13	7.50
			PE	35.46	6.66
		Saline	DM	38.15	20.49
			PE	29.25	6.13
	Wild Type	4 mg/Kg	DM	34.35	8.13
			PE	28.33	3.34
		8 mg/kg	DM	33.46	8.75
			PE	25.95	8.29
		Saline	DM	30.44	6.31
			PE	21.58	1.88

Means and Standard Deviations for Free Rears

Measure - Phase	Condition	Treatment	Oestrus	Mean	SD
Free rears -baseline	SERT			8.82	4.35
	Wild Type			9.33	3.5
Free rears -acute	SERT	4 mg/Kg	DM	11.85	3.39
			PE	10.90	3.96
		8 mg/kg	DM	7.25	3.06
			PE	8.54	2.14
		Saline	DM	10.55	4.85
			PE	6.19	3.51
	Wild Type	4 mg/Kg	DM	7.70	2.03
			PE	8.75	5.72
		8 mg/kg	DM	12.58	6.57
			PE	6.05	1.89
		Saline	DM	13.06	5.73
			PE	7.08	2.50
Free rears -chronic	SERT	4 mg/Kg	DM	10.20	2.46
			PE	11.20	2.80
		8 mg/kg	DM	8.94	4.23
			PE	9.57	3.12
		Saline	DM	8.25	2.26
			PE	7.94	2.22
	Wild Type	4 mg/Kg	DM	6.90	2.77
			PE	9.13	2.27
		8 mg/kg	DM	10.33	8.22
			PE	6.25	3.59
		Saline	DM	13.38	6.94
			PE	6.58	2.40

Means and Standard Deviations for Alleys Distance Moved

Measure - Phase	Condition	Treatment	Oestrus	Mean	SD
Alleys distance moved -baseline	SERT			2095.01	1109.86
	Wild Type			1772.57	850.62
Alleys distance moved -acute	SERT	4 mg/Kg	DM	2094.11	1250.88
			PE	2535.16	1324.88
		8 mg/kg	DM	3302.81	1568.12
			PE	1343.35	552.96
		Saline	DM	2322.47	1503.39
			PE	2152.64	1170.27
	Wild Type	4 mg/Kg	DM	1421.25	431.15
			PE	2089.48	1163.15
		8 mg/kg	DM	1554.35	623.53
			PE	2528.14	1114.43
		Saline	DM	1573.84	632.15
			PE	2661.25	823.80
Alleys distance moved -chronic	SERT	4 mg/Kg	DM	2932.47	673.69
			PE	2908.88	688.15
		8 mg/kg	DM	2936.38	667.21
			PE	2235.36	337.90
		Saline	DM	3245.35	987.55
			PE	2228.16	846.81
	Wild Type	4 mg/Kg	DM	2171.32	230.97
			PE	2789.55	569.09
		8 mg/kg	DM	2374.01	412.45
			PE	2711.52	897.74
		Saline	DM	2205.83	198.79
			PE	2520.08	477.92

Means and Standard Deviations for Alleys Velocity

Measure - Phase	Condition	Treatment	Oestrus	Mean	SD
Alleys velocity -baseline	SERT			8.92	3.98
	Wild Type			8.71	3.66
Alleys velocity -acute	SERT	4 mg/Kg	DM	10.60	2.34
			PE	9.70	4.02
		8 mg/kg	DM	11.69	6.00
			PE	8.25	3.33
		Saline	DM	9.44	4.68
			PE	8.26	4.03
	Wild Type	4 mg/Kg	DM	10.08	4.23
			PE	9.71	1.90
		8 mg/kg	DM	9.58	2.51
			PE	11.28	3.49
		Saline	DM	7.23	1.36
			PE	9.42	3.64
Alleys velocity -chronic	SERT	4 mg/Kg	DM	11.30	2.10
			PE	10.84	2.85
		8 mg/kg	DM	11.00	3.48
			PE	8.07	1.30
		Saline	DM	11.89	3.57
			PE	7.89	3.29
	Wild Type	4 mg/Kg	DM	7.74	1.19
			PE	9.46	2.00
		8 mg/kg	DM	8.32	1.24
			PE	9.75	3.48
		Saline	DM	7.87	1.22
			PE	9.25	3.07

Means and Standard Deviations for Frequency in Alley 4

Measure - Phase	Condition	Treatment	Oestrus	Mean	SD
Frequency in alley 4 -baseline	SERT			5.9	3.28
	Wild Type			5.64	3.16
Frequency in alley 4 -acute	SERT	4 mg/Kg	DM	7.25	1.94
			PE	9.45	4.74
		8 mg/kg	DM	9.69	5.54
			PE	5.93	3.26
		Saline	DM	7.85	4.40
			PE	6.75	3.18
	Wild Type	4 mg/Kg	DM	8.05	3.36
			PE	6.17	3.25
		8 mg/kg	DM	7.21	3.27
			PE	9.95	4.04
		Saline	DM	7.44	2.79
			PE	9.58	3.09
Frequency in alley 4 -chronic	SERT	4 mg/Kg	DM	8.10	2.10
			PE	8.20	1.28
		8 mg/kg	DM	9.31	2.38
			PE	6.96	2.34
		Saline	DM	8.85	1.88
			PE	6.31	1.93
	Wild Type	4 mg/Kg	DM	5.80	1.01
			PE	8.04	2.04
		8 mg/kg	DM	7.13	2.29
			PE	7.10	3.11
		Saline	DM	6.81	2.49
			PE	7.69	1.63

Means and Standard Deviations for Cumulative Duration in Alley 4

Measure - Phase	Condition	Treatment	Oestrus	Mean	SD
Cumulative duration in alley 4 -baseline	SERT			31.29	19.23
	Wild Type			31.68	19.58
Cumulative duration in alley 4 -acute	SERT	4 mg/Kg	DM	41.27	15.87
			PE	48.20	22.60
		8 mg/kg	DM	61.06	39.21
			PE	40.30	18.12
		Saline	DM	67.81	26.42
			PE	59.46	25.51
	Wild Type	4 mg/Kg	DM	32.25	10.16
			PE	40.68	29.38
		8 mg/kg	DM	58.59	27.72
			PE	56.42	21.96
		Saline	DM	45.87	23.63
			PE	63.75	21.95
Cumulative duration in alley 4 -chronic	SERT	4 mg/Kg	DM	49.68	23.11
			PE	42.00	5.14
		8 mg/kg	DM	47.38	21.27
			PE	48.32	18.42
		Saline	DM	51.41	14.88
			PE	39.52	13.14
	Wild Type	4 mg/Kg	DM	48.49	17.10
			PE	49.77	14.27
		8 mg/kg	DM	46.18	22.38
			PE	45.70	24.06
		Saline	DM	61.52	30.95
			PE	39.11	7.99

Means and Standard Deviations for Latency to eat the 1st FL

Measure - Phase	Condition	Treatment	Oestrus	Mean	SD
Latency to eat the 1st FL -baseline	SERT			10.03	8.58
	Wild Type			11.98	14.61
Latency to eat the 1st FL -acute	SERT	4 mg/Kg	DM	3.50	1.13
			PE	5.45	4.16
		8 mg/kg	DM	4.00	1.46
			PE	5.32	4.30
		Saline	DM	6.60	5.11
			PE	2.88	1.23
	Wild Type	4 mg/Kg	DM	4.35	1.78
			PE	4.38	1.08
		8 mg/kg	DM	3.71	1.65
			PE	4.75	1.70
		Saline	DM	3.13	0.52
			PE	3.67	0.14
Latency to eat the 1st FL -chronic	SERT	4 mg/Kg	DM	6.30	7.46
			PE	3.80	2.03
		8 mg/kg	DM	2.94	1.43
			PE	5.96	4.82
		Saline	DM	4.75	2.54
			PE	3.38	0.52
	Wild Type	4 mg/Kg	DM	3.35	1.89
			PE	3.92	1.53
		8 mg/kg	DM	4.17	3.85
			PE	3.65	1.23
		Saline	DM	3.63	1.11
			PE	5.00	1.09

Means and Standard Deviations for Latency to eat FL #4

Measure - Phase	Condition	Treatment	Oestrus	Mean	SD
Latency to eat FL #4 -baseline	SERT			116.77	48.68
	Wild Type			117.9	52.96
Latency to eat FL #4 -acute	SERT	4 mg/Kg	DM	89.70	31.38
			PE	68.30	8.51
		8 mg/kg	DM	77.13	7.13
			PE	89.96	41.58
		Saline	DM	76.65	10.40
			PE	80.63	11.23
	Wild Type	4 mg/Kg	DM	73.75	10.38
			PE	88.46	20.36
		8 mg/kg	DM	77.54	22.28
			PE	88.55	22.02
		Saline	DM	66.44	9.65
			PE	79.50	11.51
Latency to eat FL #4 -chronic	SERT	4 mg/Kg	DM	83.15	39.81
			PE	63.95	5.94
		8 mg/kg	DM	79.13	6.23
			PE	75.00	24.93
		Saline	DM	68.00	9.75
			PE	77.88	8.48
	Wild Type	4 mg/Kg	DM	67.05	8.86
			PE	76.71	16.38
		8 mg/kg	DM	69.71	12.13
			PE	70.85	14.38
		Saline	DM	68.31	15.16
			PE	73.50	9.58

Means and Standard Deviations for mSAT Score

Measure - Phase	Condition	Treatment	Oestrus	Mean	SD
mSAT score -baseline	SERT			10.23	2.03
	Wild Type			9.9	1.37
mSAT score -acute	SERT	4 mg/Kg	DM	10.50	0.71
			PE	10.75	1.68
		8 mg/kg	DM	11.13	1.44
			PE	9.89	0.40
		Saline	DM	10.00	0.00
			PE	10.13	0.25
	Wild Type	4 mg/Kg	DM	10.00	0.00
			PE	10.17	0.26
		8 mg/kg	DM	10.25	0.61
			PE	13.75	7.97
		Saline	DM	10.13	0.25
			PE	10.00	0.00
mSAT score -chronic	SERT	4 mg/Kg	DM	10.20	0.27
			PE	10.80	0.76
		8 mg/kg	DM	10.38	0.48
			PE	10.14	0.24
		Saline	DM	10.20	0.45
			PE	10.44	0.88
	Wild Type	4 mg/Kg	DM	10.10	0.22
			PE	10.00	0.00
		8 mg/kg	DM	10.17	0.41
			PE	10.15	0.22
		Saline	DM	10.00	0.00
			PE	10.67	1.15

Means and Standard Deviations for USVs

Measure - Phase	Condition	Treatment	Oestrus	Mean	SD
USVs -baseline 1	SERT			1.93	1.53
	Wild Type			2.05	1.43
USVs -baseline 4	SERT			2.55	2.92
	Wild Type			2.17	1.68
USVs -acute 1	SERT	4 mg/Kg	DM	6.23	1.98
			PE	1.63	1.25
		8 mg/kg	DM	1.64	0.89
			PE	1.67	0.94
		Saline	DM	2.28	1.50
			PE	1.99	1.40
	Wild Type	4 mg/Kg	DM	1.35	0.48
			PE	2.25	1.77
		8 mg/kg	DM	3.62	2.56
			PE	3.48	3.50
		Saline	DM	2.33	1.89
			PE	1.50	0.50
USVs -chronic 1	SERT	4 mg/Kg	DM	1.00	0.00
			PE	2.23	1.87
		8 mg/kg	DM	2.50	2.60
			PE	1.08	0.20
		Saline	DM	1.48	0.69
			PE	1.25	0.50
	Wild Type	4 mg/Kg	DM	1.67	0.91
			PE	1.15	0.38
		8 mg/kg	DM	1.00	0.00
			PE	1.25	0.49
		Saline	DM	1.00	0.00
			PE	1.00	0.00

3B- ANOVA - Main Effects and Interactions for Measures of the Experiment: Effects of Doses of 4 and 8 mg/kg of Psilocybin on Anxiety and Depression-like Symptom**s** and Impact**s** of the Oestru**s** Cycle – main Analysis (Chapter 5)

Main Effects and Interactions for Box Distance Moved

	df	***F***	***p***
Time	(2, 108)	19.19	0.00
Time*condition	(2, 108)	0.44	0.65
Time*treatment	(4, 108)	0.73	0.57
Time*condition*treatment	(4, 108)	0.72	0.58
Condition	(1, 54)	2.89	0.09
Treatment	(2, 54)	0.08	0.93
Condition*treatment	(2, 54)	0.76	0.47

Main Effects and Interactions for Box Velocity

	df	***F***	***p***
Time	(2, 108)	23.45	< .001
Time*condition	(2, 108)	0.22	0.802
Time*treatment	(4, 108)	0.67	0.616
Time*condition*treatment	(4, 108)	0.82	0.512
Condition	(1, 54)	2.75	0.103
Treatment	(2, 54)	0.07	0.931
Condition*treatment	(2, 54)	0.77	0.469

Main Effects and Interactions for Total Rears

	df	***F***	***p***
Time	(2, 108)	19.02	< .001
Time*condition	(2, 108)	2.59	0.08
Time*treatment	(4, 108)	0.13	0.97
Time*condition*treatment	(4, 108)	0.28	0.889
Condition	(1, 54)	1.2	0.277
Treatment	(2, 54)	0.16	0.85
Condition*treatment	(2, 54)	0.2	0.82

Main Effects and Interactions for Wall Rears

	df	***F***	***p***
Time	(2, 108)	34.65	< .001
Time*condition	(2, 108)	3.61	0.03
Time*treatment	(4, 108)	0.14	0.967
Time*condition*treatment	(4, 108)	0.39	0.815
Condition	(1, 54)	2.34	0.132
Treatment	(2, 54)	0.24	0.79
Condition*treatment	(2, 54)	0.36	0.7

Main Effects and Interactions for Free Rears

	df	*F*	*p*
Time	(2, 108)	0.09	0.918
Time*condition	(2, 108)	0.44	0.646
Time*treatment	(4, 108)	0.07	0.991
Time*condition*treatment	(4, 108)	0.37	0.827
Condition	(1, 54)	0.07	0.796
Treatment	(2, 54)	0.14	0.868
Condition*treatment	(2, 54)	2.47	0.094

Main Effects and Interactions for USVs

	df	*F*	*p*
Time	(3,69)	4.09	0.01
Time*condition	(3,69)	0.64	0.59
Time*treatment	(6,69)	0.84	0.546
Time*condition*treatment	(6,69)	1.93	0.088
Condition	(1,23)	0.01	0.93
Treatment	(2,23)	0.24	0.79
Condition*treatment	(2,23)	3.37	0.05

Main Effects and Interactions for Alleys Distance Moved

	df	*F*	*p*
Time	(2, 108)	22.52	< .001
Time*condition	(2, 108)	0.07	0.936
Time*treatment	(4, 108)	0.44	0.779
Time*condition*treatment	(4, 108)	0.4	0.805
Condition	(1, 54)	1.37	0.247
Treatment	(2, 54)	0.17	0.847
Condition*treatment	(2, 54)	0.15	0.858

Main Effects and Interactions for Alleys Velocity

	df	*F*	*p*
Time	(2, 108)	1.34	0.266
Time*condition	(2, 108)	1.36	0.26
Time*treatment	(4, 108)	0.84	0.5
Time*condition*treatment	(4, 108)	1.25	0.293
Condition	(1, 54)	0.42	0.52
Treatment	(2, 54)	0.26	0.769
Condition*treatment	(2, 54)	0.15	0.858

Main Effects and Interactions for Frequency in Alley 4

	df	*F*	*p*
Time	(2, 108)	8.79	< .001
Time*condition	(2, 108)	0.59	0.557
Time*treatment	(4, 108)	1.29	0.279
Time*condition*treatment	(4, 108)	0.47	0.76
Condition	(1, 54)	0.06	0.801
Treatment	(2, 54)	0.7	0.503
Condition*treatment	(2, 54)	0.14	0.87

Main Effects and Interactions for Cumulative Duration in Alley 4

	df	*F*	*p*
Time	(2, 108)	15.23	< .001
Time*condition	(2, 108)	0.31	0.735
Time*treatment	(4, 108)	0.98	0.419
Time*condition*treatment	(4, 108)	0.91	0.46
Condition	(1, 54)	0.02	0.9
Treatment	(2, 54)	2.1	0.133
Condition*treatment	(2, 54)	0.07	0.929

Main Effects and Interactions for Latency to eat 1st FL

	df	*F*	*p*
Time	(2, 108)	16.97	< .001
Time*condition	(2, 108)	0.51	0.604
Time*treatment	(4, 108)	0.89	0.475
Time*condition*treatment	(4, 108)	0.48	0.749
Condition	(1, 54)	0	0.987
Treatment	(2, 54)	0.72	0.493
Condition*treatment	(2, 54)	0.3	0.741

Main Effects and Interactions for Latency to Eat FL #4

	df	*F*	*p*
Time	(2, 108)	50.96	< .001
Time*condition	(2, 108)	0.12	0.885
Time*treatment	(4, 108)	0.8	0.529
Time*condition*treatment	(4, 108)	0.09	0.985
Condition	(1, 54)	0.06	0.809
Treatment	(2, 54)	0.71	0.495
Condition*treatment	(2, 54)	0.04	0.956

Main Effects and Interactions for mSAT Score

	df	*F*	*p*
Time	(2, 108)	0.82	0.443
Time*condition	(2, 108)	0.53	0.59
Time*treatment	(4, 108)	1.11	0.357
Time*condition*treatment	(4, 108)	0.61	0.657
Condition	(1, 54)	0.01	0.931
Treatment	(2, 54)	0.13	0.879
Condition*treatment	(2, 54)	2.29	0.111

3C- ANOVA - Main Effects and Interactions for measures of the experiment: Effects of Doses of 4 and 8 mg/kg of Psilocybin on Anxiety and Depression-like Symptoms and Impacts of the Oestrus Cycle (Sub Chapter 5.5)

Main Effects and Interactions for Box Distance Moved

	df	***F***	***p***
Time	(2,94)	20.8	< .001
Time*condition	(2,94)	0.72	0.488
Time*treatment	(4,94)	0.71	0.588
Time*oestrus	(2,94)	2.19	0.118
Time*condition*treatment	(4,94)	0.89	0.475
Time*condition*oestrus	(2,94)	0.09	0.918
Time*treatment*oestrus	(4,94)	3.39	0.012
Time*condition*treatment*oestrus	(4,94)	1.16	0.333
Condition	(1,47)	6	0.018
Treatment	(2,47)	0.04	0.964
Oestrus	(1,47)	1.06	0.308
Condition*treatment	(2,47)	0.39	0.682
Condition*oestrus	(1,47)	7.19	0.01
Treatment*oestrus	(2,47)	4.16	0.022
Condition*treatment*oestrus	(2,47)	2.93	0.063

Main Effects and Interactions for Box Velocity

	df	*F*	*p*
Time	(2,94)	24.58	< .001
Time*condition	(2,94)	0.4	0.672
Time*treatment	(4,94)	0.66	0.619
Time*oestrus	(2,94)	2.07	0.132
Time*condition*treatment	(4,94)	0.95	0.437
Time*condition*oestrus	(2,94)	0.13	0.875
Time*treatment*oestrus	(4,94)	2.8	0.03
Time*condition*treatment*oestrus	(4,94)	1.46	0.221
Condition	(1,47)	5.86	0.019
Treatment	(2,47)	0.02	0.979
Oestrus	(1,47)	1.19	0.28
Condition*treatment	(2,47)	0.41	0.667
Condition*oestrus	(1,47)	6.84	0.012
Treatment*oestrus	(2,47)	4.11	0.023
Condition*treatment*oestrus	(2,47)	2.79	0.072

Main Effects and Interactions for Total Rears

	df	*F*	*p*
Time	(2,94)	16.46	< .001
Time*condition	(2,94)	2.1	0.128
Time*treatment	(4,94)	0.23	0.919
Time*oestrus	(2,94)	0.05	0.955
Time*condition*treatment	(4,94)	0.2	0.939
Time*condition*oestrus	(2,94)	1.18	0.313
Time*treatment*oestrus	(4,94)	0.88	0.481
Time*condition*treatment*oestrus	(4,94)	1.61	0.177
Condition	(1,47)	1.94	0.171
Treatment	(2,47)	0.59	0.557
Oestrus	(1,47)	7.29	0.01
Condition*treatment	(2,47)	0.28	0.755
Condition*oestrus	(1,47)	2.63	0.111
Treatment*oestrus	(2,47)	0.87	0.424
Condition*treatment*oestrus	(2,47)	0.21	0.811

Main Effects and Interactions for Wall Rears

	df	*F*	*p*
Time	(2,94)	30.24	< .001
Time*condition	(2,94)	2.8	0.066
Time*treatment	(4,94)	0.23	0.92
Time*oestrus	(2,94)	0.35	0.708
Time*condition*treatment	(4,94)	0.25	0.911
Time*condition*oestrus	(2,94)	0.94	0.395
Time*treatment*oestrus	(4,94)	0.76	0.554
Time*condition*treatment*oestrus	(4,94)	1.01	0.404
Condition	(1,47)	2.9	0.095
Treatment	(2,47)	0.55	0.579
Oestrus	(1,47)	4.89	0.032
Condition*treatment	(2,47)	0.35	0.708
Condition*oestrus	(1,47)	1.87	0.178
Treatment*oestrus	(2,47)	0.35	0.71
Condition*treatment*oestrus	(2,47)	0.13	0.882

Main Effects and Interactions for Free Rears

	df	*F*	*p*
Time	(2,94)	0.04	0.964
Time*condition	(2,94)	0.5	0.611
Time*treatment	(4,94)	0.15	0.964
Time*oestrus	(2,94)	0.82	0.442
Time*condition*treatment	(4,94)	0.36	0.838
Time*condition*oestrus	(2,94)	0.81	0.447
Time*treatment*oestrus	(4,94)	1.1	0.362
Time*condition*treatment*oestrus	(4,94)	1.86	0.124
Condition	(1,47)	0	0.982
Treatment	(2,47)	0.26	0.772
Oestrus	(1,47)	3.91	0.054
Condition*treatment	(2,47)	2.48	0.095
Condition*oestrus	(1,47)	1.4	0.242
Treatment*oestrus	(2,47)	1.34	0.271
Condition*treatment*oestrus	(2,47)	0.91	0.411

Main Effects and Interactions for USVs

	df	*F*	*p*
Time	(3,51)	3.44	0.023
Time*condition	(3,51)	1.49	0.229
Time*treatment	(6,51)	1.39	0.235
Time*oestrus	(3,51)	0.43	0.733
Time*condition*treatment	(6,51)	1.68	0.144
Time*condition*oestrus	(3,51)	3.43	0.024
Time*treatment*oestrus	(6,51)	1.42	0.226
Time*condition*treatment*oestrus	(6,51)	0.61	0.723
Condition	(1,17)	0.04	0.846
Treatment	(2,17)	0.01	0.99
Oestrus	(1,17)	0.26	0.615
Condition*treatment	(2,17)	2.94	0.08
Condition*oestrus	(1,17)	0.09	0.771
Treatment*oestrus	(2,17)	0.59	0.568
Condition*treatment*oestrus	(2,17)	1.16	0.338

Main Effects and Interactions for Alleys Distance Moved

	df	*F*	*p*
Time	(2,94)	19.76	< .001
Time*condition	(2,94)	0.11	0.895
Time*treatment	(4,94)	0.68	0.604
Time*oestrus	(2,94)	0.83	0.439
Time*condition*treatment	(4,94)	0.5	0.737
Time*condition*oestrus	(2,94)	0.69	0.504
Time*treatment*oestrus	(4,94)	1.31	0.273
Time*condition*treatment*oestrus	(4,94)	2.02	0.097
Condition	(1,47)	2.41	0.127
Treatment	(2,47)	0.02	0.978
Oestrus	(1,47)	0.09	0.763
Condition*treatment	(2,47)	0.18	0.838
Condition*oestrus	(1,47)	8.38	0.006
Treatment*oestrus	(2,47)	1.41	0.255
Condition*treatment*oestrus	(2,47)	1.72	0.19

Main Effects and Interactions for Box Velocity

	df	*F*	*p*
Time	(2,94)	1.72	0.185
Time*condition	(2,94)	1.31	0.275
Time*treatment	(4,94)	0.75	0.558
Time*oestrus	(2,94)	0.1	0.901
Time*condition*treatment	(4,94)	1.21	0.313
Time*condition*oestrus	(2,94)	0.21	0.813
Time*treatment*oestrus	(4,94)	2.1	0.087
Time*condition*treatment*oestrus	(4,94)	0.53	0.711
Condition	(1,47)	0.81	0.372
Treatment	(2,47)	0.41	0.664
Oestrus	(1,47)	0.16	0.687
Condition*treatment	(2,47)	0.01	0.986
Condition*oestrus	(1,47)	6	0.018
Treatment*oestrus	(2,47)	0.77	0.468
Condition*treatment*oestrus	(2,47)	0.3	0.74

Main Effects and Interactions for Frequency in Alley 4

	df	*F*	*p*
Time	(2,94)	9.36	< .001
Time*condition	(2,94)	0.67	0.515
Time*treatment	(4,94)	1.13	0.348
Time*oestrus	(2,94)	0.23	0.794
Time*condition*treatment	(4,94)	0.43	0.786
Time*condition*oestrus	(2,94)	1.51	0.226
Time*treatment*oestrus	(4,94)	0.93	0.453
Time*condition*treatment*oestrus	(4,94)	1.47	0.218
Condition	(1,47)	0.33	0.567
Treatment	(2,47)	0.38	0.686
Oestrus	(1,47)	0.22	0.643
Condition*treatment	(2,47)	0.26	0.775
Condition*oestrus	(1,47)	1.26	0.268
Treatment*oestrus	(2,47)	1.31	0.279
Condition*treatment*oestrus	(2,47)	2.69	0.079

Main Effects and Interactions for Cumulative Duration in Alley 4

	df	*F*	*p*
Time	(2,94)		0.00
Time*condition	(2,94)	0.26	0.77
Time*treatment	(4,94)	1.15	0.34
Time*oestrus	(2,94)	0.5	0.61
Time*condition*treatment	(4,94)	0.64	0.64
Time*condition*oestrus	(2,94)	0.62	0.54
Time*treatment*oestrus	(4,94)	0.99	0.42
Time*condition*treatment*oestrus	(4,94)	0.73	0.57
Condition	(1,47)	0.02	0.90
Treatment	(2,47)	1.75	0.18
Oestrus	(1,47)	0.42	0.52
Condition*treatment	(2,47)	0.06	0.94
Condition*oestrus	(1,47)	0.68	0.41
Treatment*oestrus	(2,47)	1.18	0.32
Condition*treatment*oestrus	(2,47)	0.49	0.62

Main Effects and Interactions for Latency to eat the 1st FL

	df	***F***	***p***
Time	(2,94)	14.8	< .001
Time*condition	(2,94)	0.56	0.575
Time*treatment	(4,94)	0.89	0.476
Time*oestrus	(2,94)	0.09	0.912
Time*condition*treatment	(4,94)	0.31	0.873
Time*condition*oestrus	(2,94)	0.75	0.474
Time*treatment*oestrus	(4,94)	0.4	0.81
Time*condition*treatment*oestrus	(4,94)	1.64	0.17
Condition	(1,47)	0.02	0.881
Treatment	(2,47)	0.77	0.469
Oestrus	(1,47)	0.13	0.72
Condition*treatment	(2,47)	0.14	0.871
Condition*oestrus	(1,47)	0.99	0.325
Treatment*oestrus	(2,47)	0.84	0.439
Condition*treatment*oestrus	(2,47)	1.95	0.154

Main Effects and Interactions for Latency to eat FL #4

	df	*F*	*p*
Time	(2,94)	48.48	< .001
Time*condition	(2,94)	0.43	0.654
Time*treatment	(4,94)	0.55	0.702
Time*oestrus	(2,94)	1.57	0.213
Time*condition*treatment	(4,94)	0.13	0.971
Time*condition*oestrus	(2,94)	1.69	0.19
Time*treatment*oestrus	(4,94)	0.95	0.44
Time*condition*treatment*oestrus	(4,94)	1.93	0.111
Condition	(1,47)	0	0.995
Treatment	(2,47)	0.51	0.606
Oestrus	(1,47)	1.06	0.307
Condition*treatment	(2,47)	0.01	0.995
Condition*oestrus	(1,47)	2.24	0.141
Treatment*oestrus	(2,47)	0.45	0.642
Condition*treatment*oestrus	(2,47)	1.34	0.272

Main Effects and Interactions for mSAT Score

	df	*F*	*p*
Time	(2,94)	0.87	0.422
Time*condition	(2,94)	0.51	0.599
Time*treatment	(4,94)	1.33	0.265
Time*oestrus	(2,94)	1.35	0.264
Time*condition*treatment	(4,94)	0.61	0.655
Time*condition*oestrus	(2,94)	0.68	0.51
Time*treatment*oestrus	(4,94)	0.76	0.556
Time*condition*treatment*oestrus	(4,94)	1.13	0.349
Condition	(1,47)	0.03	0.853
Treatment	(2,47)	0.21	0.809
Oestrus	(1,47)	0.01	0.934
Condition*treatment	(2,47)	2.1	0.134
Condition*oestrus	(1,47)	1.36	0.25
Treatment*oestrus	(2,47)	0.4	0.671
Condition*treatment*oestrus	(2,47)	1.11	0.337